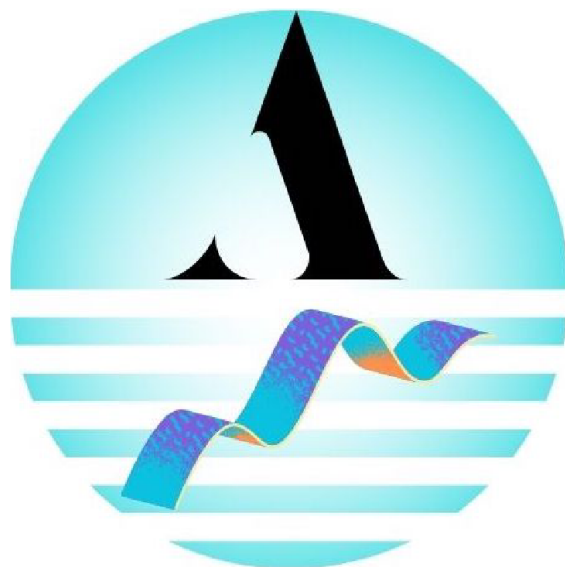




Real Analysis (MMT422J1)

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Course Objectives: To make students learn (i) basic concepts and properties of real numbers, (ii) the behavior of sequences and series in real system. Course Outcome: After the completion of this course, students shall be able to (i) apply these concepts to determine convergence and divergence of real sequences and infinite series (ii) explore new ideas in mathematical and modern analysis.	
Theory: 3 Credits	
Unit –I Real Numbers, Field and Order structure, Bounded and unbounded sets of real numbers, supremum and infimum (lub and glb), Order completeness of $\mathbb{R}$, Archimedean property, Rational and Irrational density theorems, Dedekind's property, $\mathbb{R}$ as ordered field, Incompleteness of the set of rational numbers.	
Unit –II Sequences of real numbers, bounded sequences, Convergence of sequences, Limit points of sequences, subsequences, Bolzano-Weirstrass theorem, limit inferior and limit superior, Theorems on limit and convergence of sequences, Cauchy's criteria for convergence of sequences, Cauchy sequences, Bounded monotonic sequences, Nested Interval theorem.	
Unit –III Infinite Series: Convergence and Divergence of a series, Necessary Condition for Convergence of a series, Cauchy's Criteria for Convergence of a series, Geometric Series, Series of positive terms, Test for convergence: Comparison test, Cauchy's root test, D-Alembert's ratio test, Raabe's test.	
Tutorial: 1 credit	
Unit –IV Open and Closed sets: Neighbourhood of a point, Interior of a set, Open sets, Problems based on limit point of a set, Bolzano-Weirstrass theorem property, Closed sets, Closure of a set and related results. Cardinality of a set, countable and uncountable sets, Countability of the set of Rational numbers and uncountability of reals, some simple related results.	
Books Recommended 1. S. C. Malik and S. Arora, Mathematical Analysis, New Age International (P) Ltd, Publishers, 2005. 2. T. M. Apostol, Mathematical Analysis, Narosa Publications, 2002. 3. R. Goldberg, Methods of Real Analysis, Oxford IBM Publication, 1970 4. W. Rudin, Principles of Mathematical Analysis, McGraw Hill, Indian Edition, 2017	

Chapter 1

Real Numbers and the Structure of $\mathbb{R}$

1.1 Introduction

The set of real numbers, denoted by $\mathbb{R}$, forms the bedrock of mathematical analysis. While the rational numbers $\mathbb{Q}$ allow for basic arithmetic operations, they are insufficient for the purposes of analysis (for instance, the equation $x^2 = 2$ has no solution in $\mathbb{Q}$). To study concepts like limits and continuity, we require a system that is algebraically consistent and "gap-free".

In this chapter, we formalize $\mathbb{R}$ by defining it axiomatically. We characterize the real numbers as a **Complete Ordered Field**. In this section, we focus on the first two aspects: the *Field* structure and the *Order* structure.

1.2 Field Structure of $\mathbb{R}$

A set F is called a **Field** if it is equipped with two binary operations, addition (+) and multiplication ($\cdot$), which satisfy specific properties. The set of real numbers $\mathbb{R}$ is a field.

1.2.1 The Field Axioms

For all $x, y, z \in \mathbb{R}$, the following axioms hold:

(A) Axioms of Addition 1. **Closure:** $x + y \in \mathbb{R}$.

2. **Commutativity:** $x + y = y + x$.

3. **Associativity:** $(x + y) + z = x + (y + z)$.

4. **Existence of Additive Identity:** There exists a unique element $0 \in \mathbb{R}$ such that $x + 0 = x$ for all x .

5. **Existence of Additive Inverse:** For every $x \in \mathbb{R}$, there exists a unique element $-x \in \mathbb{R}$ such that $x + (-x) = 0$.

(M) Axioms of Multiplication 1. **Closure:** $x \cdot y \in \mathbb{R}$ (usually written as xy).

2. **Commutativity:** $xy = yx$.

3. **Associativity:** $(xy)z = x(yz)$.

4. **Existence of Multiplicative Identity:** There exists a unique element $1 \in \mathbb{R}$ (with $1 \neq 0$) such that $x \cdot 1 = x$ for all x .

5. **Existence of Multiplicative Inverse:** For every $x \in \mathbb{R}$ with $x \neq 0$, there exists a unique element denoted x^{-1} (or $1/x$) such that $x \cdot x^{-1} = 1$.

(D) The Distributive Axiom 1. Multiplication distributes over addition: $x(y + z) = xy + xz$.

Remark 1.2.1. The set of Integers $\mathbb{Z}$ is **not** a field because it generally lacks multiplicative inverses (e.g., $2^{-1} \notin \mathbb{Z}$). The set of Rational numbers $\mathbb{Q}$ is a field.

1.2.2 Basic Algebraic Properties

From the axioms above, we can rigorously prove the standard rules of algebra.

Theorem 1.2.2. Let $a, b, c \in \mathbb{R}$.

1. **Cancellation Law for Addition:** If $a + c = b + c$, then $a = b$.
2. **Cancellation Law for Multiplication:** If $ac = bc$ and $c \neq 0$, then $a = b$.
3. $a \cdot 0 = 0$ for any $a \in \mathbb{R}$.
4. $(-a)b = -(ab)$ and $(-a)(-b) = ab$.

Proof. We prove (3) as an illustration of using axioms.

$$a \cdot 0 = a(0 + 0) \quad \text{(Additive Identity)}$$

$$a \cdot 0 = a \cdot 0 + a \cdot 0 \quad \text{(Distributive Axiom)}$$

Adding $-(a \cdot 0)$ to both sides:

$$0 = a \cdot 0 + (a \cdot 0 + (-(a \cdot 0)))$$

$$0 = a \cdot 0 + 0$$

$$0 = a \cdot 0$$

Thus, $a \cdot 0 = 0$. ■

Example 1.2.3. Prove that if $ab = 0$, then either $a = 0$ or $b = 0$.

Solution. Suppose $ab = 0$. If $a = 0$, the statement holds. Suppose $a \neq 0$. Then the multiplicative inverse a^{-1} exists. Multiplying the equation $ab = 0$ by a^{-1} :

$$a^{-1}(ab) = a^{-1} \cdot 0$$

Using associativity and the property that $x \cdot 0 = 0$:

$$(a^{-1}a)b = 0$$

$$1 \cdot b = 0$$

$$b = 0$$

Thus, if $a \neq 0$, then b must be 0. ■

1.3 Order Structure of $\mathbb{R}$

In addition to algebraic operations, real numbers can be compared. We formalize this by defining $\mathbb{R}$ as an **Ordered Field**.

Definition 1.3.1 (Ordered Field). An ordered field is a field F containing a non-empty subset P (called the *positive set*) satisfying the following properties:

1. **Closure under Addition:** If $x, y \in P$, then $x + y \in P$.
2. **Closure under Multiplication:** If $x, y \in P$, then $xy \in P$.
3. **Law of Trichotomy:** For every $x \in F$, exactly one of the following holds:

$$x \in P, \quad x = 0, \quad \text{or} \quad -x \in P.$$

Based on the set P , we define the inequality symbols for $x, y \in \mathbb{R}$:

- $x > 0$ means $x \in P$.
- $x < 0$ means $-x \in P$ (or $-x > 0$).
- $x > y$ means $x - y > 0$.
- $x < y$ means $y - x > 0$.
- $x \leq y$ means $x < y$ or $x = y$.

Properties of Inequalities**Theorem 1.3.2.** Let $x, y, z \in \mathbb{R}$.

1. **Transitivity:** If $x > y$ and $y > z$, then $x > z$.
2. **Compatibility with Addition:** If $x > y$, then $x + z > y + z$ for any z .
3. **Compatibility with Multiplication:**
 - If $x > y$ and $z > 0$, then $xz > yz$.
 - If $x > y$ and $z < 0$, then $xz < yz$ (the inequality reverses).
4. **Square Property:** For any $x \in \mathbb{R}$, if $x \neq 0$, then $x^2 > 0$.
5. $1 > 0$.

Proof. We provide the proof for the Square Property (4) and property (5).**Proof of (4):** Let $x \in \mathbb{R}$ and $x \neq 0$. By Trichotomy, either $x \in P$ or $-x \in P$.

- Case 1: If $x \in P$, then by closure of multiplication for P , $x \cdot x \in P$. Thus $x^2 > 0$.
- Case 2: If $-x \in P$, then $(-x) \cdot (-x) \in P$. We know $(-x)(-x) = x^2$. Thus $x^2 > 0$.

In both cases, $x^2 > 0$.**Proof of (5):** Since $1 \neq 0$ and $1 = 1^2$, by the Square Property proven above, $1^2 > 0$, which implies $1 > 0$. ■**Example 1.3.3.** Prove that if $x > 0$, then $x^{-1} > 0$.*Solution.* Given $x > 0$. Since $x \neq 0$, x^{-1} exists. By Trichotomy, x^{-1} is either positive, negative, or zero.

1. If $x^{-1} = 0$, then $x \cdot x^{-1} = x \cdot 0 = 0$. But $x \cdot x^{-1} = 1$. This implies $1 = 0$, a contradiction.
2. If $x^{-1} < 0$, then $-x^{-1} > 0$. Since $x > 0$, the product of two positive numbers must be positive:

$$x \cdot (-x^{-1}) > 0 \implies -(x \cdot x^{-1}) > 0 \implies -1 > 0 \implies 1 < 0.$$

This contradicts the theorem that $1 > 0$.Therefore, the only possibility remaining is $x^{-1} > 0$. ■

The order structure allows us to define the concept of magnitude.

Definition 1.3.4. The **absolute value** of a real number x , denoted by $|x|$, is defined as:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Theorem 1.3.5. For all $x, y \in \mathbb{R}$:

1. $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$.
2. $|-x| = |x|$.
3. $|xy| = |x||y|$.
4. **Triangle Inequality:** $|x + y| \leq |x| + |y|$.

Proof. **Proof of Triangle Inequality:** By definition, $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$. Adding these inequalities, we get:

$$-(|x| + |y|) \leq x + y \leq |x| + |y|$$

This is equivalent to the statement $|x + y| \leq |x| + |y|$. ■

Example 1.3.6. Solve for x : $|2x - 3| < 5$.

Solution. The inequality $|a| < b$ (where $b > 0$) is equivalent to $-b < a < b$.

$$-5 < 2x - 3 < 5$$

Adding 3 to all parts:

$$-2 < 2x < 8$$

Multiplying by $1/2$ (which is positive, so inequality signs are preserved):

$$-1 < x < 4$$

The solution set is $\{x \in \mathbb{R} : -1 < x < 4\}$. ■

1. Using the axioms of a field, prove that $-(-x) = x$ for all $x \in \mathbb{R}$.
2. Prove that if $x, y \in \mathbb{R}$, then $x^2 + y^2 = 0$ if and only if $x = 0$ and $y = 0$.
3. Show that if $0 < a < b$, then $a^2 < b^2$.
4. Prove the reverse triangle inequality: $||x| - |y|| \leq |x - y|$.

Definition 1.3.7 (Upper and Lower Bounds). Let S be a non-empty subset of $\mathbb{R}$.

1. The set S is said to be **bounded above** if there exists a real number M such that $x \leq M$ for all $x \in S$. The number M is called an **upper bound** of S .
2. The set S is said to be **bounded below** if there exists a real number m such that $x \geq m$ for all $x \in S$. The number m is called a **lower bound** of S .
3. A set is **bounded** if it is both bounded above and bounded below.

Example 1.3.8. Let $S = \{\frac{1}{n} : n \in \mathbb{N}\} = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$. Determine if S is bounded.

Solution. **Upper Bound:** Since $n \geq 1$ for all $n \in \mathbb{N}$, we have $\frac{1}{n} \leq 1$. Thus, 1 is an upper bound. Any number greater than 1 (e.g., 5) is also an upper bound.

Lower Bound: Since $n > 0$, $\frac{1}{n} > 0$. Thus 0 is a lower bound.

Since S has both an upper and a lower bound, S is a bounded set. ■

1.4 Bounded and Unbounded Sets

The order structure of $\mathbb{R}$ allows us to discuss the geometric notion of the "extent" of a set. A fundamental concept in analysis is determining whether a set of real numbers is confined within a finite interval or extends indefinitely.

1.4.1 Upper and Lower Bounds

Definition 1.4.1 (Bounded Above and Upper Bounds). Let S be a subset of $\mathbb{R}$. The set S is said to be **bounded above** if there exists a real number M such that

$$x \leq M \quad \text{for all } x \in S.$$

The number M is called an **upper bound** of S .

Definition 1.4.2 (Bounded Below and Lower Bounds). The set S is said to be **bounded below** if there exists a real number m such that

$$m \leq x \quad \text{for all } x \in S.$$

The number m is called a **lower bound** of S .

Definition 1.4.3 (Bounded Set). A set S is said to be **bounded** if it is both bounded above and bounded below. Equivalently, S is bounded if there exists a real number $K > 0$ such that

$$|x| \leq K \quad \text{for all } x \in S.$$

Definition 1.4.4 (Unbounded Set). A set is **unbounded** if it is not bounded.

- It is **unbounded above** if for every real number M , there exists some $x \in S$ such that $x > M$.
- It is **unbounded below** if for every real number m , there exists some $x \in S$ such that $x < m$.

Example 1.4.5. Consider the following subsets of $\mathbb{R}$:

1. The set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ is bounded below (by 1, 0, -5, etc.) but is unbounded above.
2. The interval $S = (-2, 5)$ is bounded. Any number $M \geq 5$ is an upper bound, and any number $m \leq -2$ is a lower bound.
3. The set $S = \{x \in \mathbb{R} : x^2 < 4\}$ is bounded, as it corresponds to the interval $(-2, 2)$.

1.5 Supremum and Infimum

If a set is bounded above, it has infinitely many upper bounds. For example, if $S = [0, 1]$, then 1, 2, 100, and π are all upper bounds. In mathematical analysis, we are interested in the *smallest* of these upper bounds, as it provides the sharpest boundary of the set.

1.5.1 The Least Upper Bound (Supremum)

Definition 1.5.1 (Supremum). Let S be a subset of $\mathbb{R}$ that is bounded above. A real number u is called the **Supremum** (or **Least Upper Bound**) of S if it satisfies the following two conditions:

1. u is an upper bound of S (i.e., $x \leq u$ for all $x \in S$).
2. If v is any other upper bound of S , then $u \leq v$.

We write $u = \sup S$ or $u = \text{lub } S$.

It is often difficult to use the second condition directly in proofs. The following theorem provides an analytical criterion involving ϵ (epsilon) which is central to real analysis.

Theorem 1.5.2 (ϵ -Characterization of Supremum). Let S be a non-empty subset of $\mathbb{R}$ bounded above, and let $u \in \mathbb{R}$. Then $u = \sup S$ if and only if:

1. u is an upper bound of S .
2. For every $\epsilon > 0$, there exists an element $x_\epsilon \in S$ such that $u - \epsilon < x_\epsilon$.

Proof. ($\Rightarrow$) Suppose $u = \sup S$. By definition, u is an upper bound. Let $\epsilon > 0$ be given. Then $u - \epsilon < u$. Since u is the *least* upper bound, $u - \epsilon$ cannot be an upper bound of S . Therefore, there must be some element in S greater than $u - \epsilon$.

($\Leftarrow$) Suppose conditions (1) and (2) hold. Condition (1) states u is an upper bound. We must show it is the least. Let v be any other upper bound. We want to show $u \leq v$. Suppose, for the sake of contradiction, that $u > v$. Let $\epsilon = u - v$. Then $\epsilon > 0$. By condition (2), there exists $x \in S$ such that

$$x > u - \epsilon = u - (u - v) = v.$$

So $x > v$. But this contradicts the assumption that v is an upper bound of S . Thus, we must have $u \leq v$. Hence, u is the supremum. ■

1.5.2 The Greatest Lower Bound (Infimum)

The concept of infimum is the dual to the supremum.

Definition 1.5.3 (Infimum). Let S be a subset of $\mathbb{R}$ that is bounded below. A real number w is called the **Infimum** (or **Greatest Lower Bound**) of S if:

1. w is a lower bound of S (i.e., $w \leq x$ for all $x \in S$).
2. If t is any other lower bound of S , then $t \leq w$.

We write $w = \inf S$ or $w = \text{glb } S$.

Theorem 1.5.4 (ϵ -Characterization of Infimum). Let S be a non-empty subset of $\mathbb{R}$ bounded below, and let $w \in \mathbb{R}$. Then $w = \inf S$ if and only if:

1. w is a lower bound of S .
2. For every $\epsilon > 0$, there exists an element $x_\epsilon \in S$ such that $x_\epsilon < w + \epsilon$.

1.5.3 Relationship Between Max/Min and Sup/Inf

It is crucial to distinguish between the maximum of a set and its supremum.

- If S contains its supremum (i.e., $\sup S \in S$), then $\sup S$ is called the **maximum** of S , denoted $\max S$.
- If S contains its infimum (i.e., $\inf S \in S$), then $\inf S$ is called the **minimum** of S , denoted $\min S$.

Many sets have a supremum but no maximum.

Example 1.5.5. Find the supremum and infimum of the set $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\}$.

Solution. The elements of the set are $\{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\}$.

Infimum: The smallest element is clearly 0 (for $n = 1$). Since $0 \in S$ and $0 \leq x$ for all $x \in S$, $\inf S = \min S = 0$.

Supremum: Observe that $1 - \frac{1}{n} < 1$ for all $n \in \mathbb{N}$. Thus, 1 is an upper bound. We claim $\sup S = 1$. Let $\epsilon > 0$. We must find $n \in \mathbb{N}$ such that $1 - \frac{1}{n} > 1 - \epsilon$. Rearranging the inequality:

$$1 - \frac{1}{n} > 1 - \epsilon \iff -\frac{1}{n} > -\epsilon \iff \frac{1}{n} < \epsilon \iff n > \frac{1}{\epsilon}.$$

By the Archimedean property (which ensures such an integer exists), we can choose an integer $n > 1/\epsilon$. For this n , the element $x = 1 - 1/n$ satisfies the condition. Thus, $\sup S = 1$. Note that $1 \notin S$, so S has no maximum. ■

Example 1.5.6. Let A and B be non-empty bounded sets of real numbers. Let $C = \{x + y : x \in A, y \in B\}$. Prove that $\sup C = \sup A + \sup B$.

Solution. Let $u = \sup A$ and $v = \sup B$. First, we show $u + v$ is an upper bound for C . For any $z \in C$, we have $z = x + y$ where $x \in A$ and $y \in B$. Since $x \leq u$ and $y \leq v$, it follows that $x + y \leq u + v$. Thus, $u + v$ is an upper bound of C . Hence, $\sup C \leq u + v$.

Now, we show that $u + v$ is the *least* upper bound. Let $\epsilon > 0$. We need to find an element in C greater than $(u + v) - \epsilon$. Since $u = \sup A$, there exists $x_0 \in A$ such that $x_0 > u - \epsilon/2$. Since $v = \sup B$, there exists $y_0 \in B$ such that $y_0 > v - \epsilon/2$. Consider the element $z_0 = x_0 + y_0 \in C$.

$$z_0 = x_0 + y_0 > (u - \epsilon/2) + (v - \epsilon/2) = u + v - \epsilon.$$

This satisfies the ϵ -criterion for the supremum. Therefore, $\sup C = u + v$. ■

1.6 Exercises

Exercise 1.6.1. Find the supremum and infimum of the set $S = \left\{ \frac{(-1)^n}{n} : n \in \mathbb{N} \right\}$.

Exercise 1.6.2. Let $S \subset \mathbb{R}$ be bounded below. Prove that $\inf S = -\sup\{-s : s \in S\}$.

Exercise 1.6.3. Give an example of a set $S \subset \mathbb{Q}$ (rational numbers) that is bounded above but has no supremum in $\mathbb{Q}$. (Note: This motivates the Completeness Axiom of $\mathbb{R}$).

In the previous sections, we established that $\mathbb{R}$ is an Ordered Field. However, the set of rational numbers $\mathbb{Q}$ is also an ordered field, yet it is insufficient for mathematical analysis. The deficiency of $\mathbb{Q}$ lies in the existence of "gaps" on the number line—for instance, there is no rational number whose square is 2.

The property that fills these gaps and distinguishes $\mathbb{R}$ from $\mathbb{Q}$ is called **Completeness**. This property guarantees that "limits" exist within the system.

1.7 Order Completeness of $\mathbb{R}$

The completeness of the real number system is stated as an axiom concerning the existence of suprema for bounded sets.

Axiom 1.7.1 (The Least Upper Bound Axiom / Completeness Axiom). *Every non-empty subset of real numbers that is bounded above has a supremum (least upper bound) in $\mathbb{R}$.*

That is, if $S \subset \mathbb{R}$ is non-empty and bounded above, then there exists a real number $u = \sup S$ such that:

1. u is an upper bound of S .
2. If v is any upper bound of S , then $u \leq v$.

Remark 1.7.2. This axiom fails for $\mathbb{Q}$. Consider the set $S = \{q \in \mathbb{Q} : q > 0, q^2 < 2\}$. This set is bounded above in $\mathbb{Q}$ (e.g., by 2, 1.5, etc.). However, it has no least upper bound in $\mathbb{Q}$ because $\sqrt{2}$ is irrational. In $\mathbb{R}$, $\sup S = \sqrt{2}$.

From the Least Upper Bound (LUB) Axiom, we can deduce the corresponding property for lower bounds.

Theorem 1.7.3 (The Greatest Lower Bound Property). *Every non-empty subset of real numbers that is bounded below has an infimum in $\mathbb{R}$.*

Proof. Let S be a non-empty subset of $\mathbb{R}$ bounded below. Let L be the set of all lower bounds of S . Since S is bounded below, L is non-empty. Since S is non-empty, fix some $s_0 \in S$. For any $y \in L$, $y \leq s$ for all $s \in S$, so specifically $y \leq s_0$. Thus, L is bounded above by any element of S . By the Completeness Axiom, L has a supremum in $\mathbb{R}$. Let $w = \sup L$. We claim $w = \inf S$.

1. Since w is the supremum of the set of lower bounds, w is larger than any other lower bound (Condition 2 of Infimum).
2. We must show w is itself a lower bound. If not, there exists $s \in S$ such that $s < w$. Since $s < w$, w is not the *least* upper bound of L unless there is an element of L greater than s . But elements of L are lower bounds of S , so no element of L can be greater than s . This contradiction implies w is a lower bound.

Thus, the infimum exists. ■

1.7.1 Application: Existence of n -th Roots

The most immediate consequence of Order Completeness is the existence of roots for positive real numbers.

Theorem 1.7.4. *For every real number $x > 0$ and every integer $n \geq 1$, there exists a unique positive real number y such that $y^n = x$.*

Proof Sketch. (For the case $n = 2, x = 2$). Let $S = \{t \in \mathbb{R} : t > 0 \text{ and } t^2 < 2\}$. S is non-empty ($1 \in S$) and bounded above (by 2). By the Completeness Axiom, let $y = \sup S$. By the Trichotomy law, $y^2 < 2$, $y^2 > 2$, or $y^2 = 2$. Using the order properties, one can show that if $y^2 < 2$, we can find a slightly larger number $y + \epsilon$ still in S (contradicting that y is an upper bound). Similarly, if $y^2 > 2$, we can find a slightly smaller number that is still an upper bound (contradicting that y is the least upper bound). Therefore, we must have $y^2 = 2$. ■

1.8 The Archimedean Property

The Archimedean property describes the relationship between real numbers and the natural numbers. Intuitively, it states that $\mathbb{N}$ is not bounded in $\mathbb{R}$, or equivalently, no matter how small a positive real number is, adding it to itself enough times will eventually exceed any given large number.

Theorem 1.8.1 (Archimedean Property of $\mathbb{R}$). *If $x \in \mathbb{R}$ and $y \in \mathbb{R}$ with $x > 0$, then there exists a positive integer $n \in \mathbb{N}$ such that*

$$nx > y.$$

Proof. We prove this by contradiction. Suppose the statement is false. Then there exist $x > 0$ and $y \in \mathbb{R}$ such that for all $n \in \mathbb{N}$, $nx \leq y$. Consider the set $A = \{nx : n \in \mathbb{N}\}$. By our assumption, y is an upper bound of A . Thus, A is a non-empty set bounded above. By the **Completeness Axiom**, A must have a supremum. Let $\alpha = \sup A$.

Since $x > 0$, we have $\alpha - x < \alpha$. Because α is the least upper bound, $\alpha - x$ cannot be an upper bound of A . Therefore, there exists some element in A , say mx (where $m \in \mathbb{N}$), such that:

$$mx > \alpha - x$$

Rearranging the inequality:

$$mx + x > \alpha \implies (m + 1)x > \alpha.$$

However, since $m \in \mathbb{N}$, then $m + 1 \in \mathbb{N}$, which implies $(m + 1)x \in A$. The statement $(m + 1)x > \alpha$ contradicts the fact that α is the supremum (upper bound) of A . Thus, our initial assumption is false. There must exist an n such that $nx > y$. ■

1.8.1 Corollaries of the Archimedean Property

The Archimedean property yields several critical results used in convergence proofs.

Corollary 1.8.2. *The set of natural numbers $\mathbb{N}$ is unbounded above.*

Proof. Apply the Archimedean property with $x = 1$ and $y = M$ for any arbitrary M . There exists $n \cdot 1 > M$, so $n > M$. ■

Corollary 1.8.3. *For any $\epsilon > 0$, there exists a natural number $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < \epsilon$.*

Proof. Let $x = \epsilon$ and $y = 1$. By the Archimedean property, there exists $n \in \mathbb{N}$ such that $n\epsilon > 1$. Dividing by n (since $n \geq 1 > 0$), we get $\epsilon > \frac{1}{n}$. Since $n > 0$, $\frac{1}{n} > 0$. Thus $0 < \frac{1}{n} < \epsilon$. ■

Corollary 1.8.4. *If $y > 0$, there exists a unique $n \in \mathbb{N}$ such that $n - 1 \leq y < n$.*

Remark 1.8.5. This justifies the existence of the "floor function" or integer part of a real number.

1.9 Examples

Example 1.9.1. Let $S = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$. Prove that $\inf S = 0$.

Solution. Clearly, 0 is a lower bound for S since $\frac{1}{n} > 0$ for all n . We must show that 0 is the *greatest* lower bound. Let t be any number such that $t > 0$. We want to show that t cannot be a lower bound. Since $t > 0$, by the Corollary to the Archimedean Property, there exists an $n_0 \in \mathbb{N}$ such that $\frac{1}{n_0} < t$. Since $\frac{1}{n_0} \in S$ and $\frac{1}{n_0} < t$, t is not a lower bound of S . Thus, any lower bound must be ≤ 0 . Therefore, $\inf S = 0$. ■

Example 1.9.2. Prove that if $x \in \mathbb{R}$ and $x \geq 0$, and if $x < \epsilon$ for every $\epsilon > 0$, then $x = 0$.

Solution. Suppose $x > 0$. By the Archimedean property, there exists an $n \in \mathbb{N}$ such that $\frac{1}{n} < x$. Let $\epsilon_0 = \frac{1}{n}$. Then $\epsilon_0 > 0$. By hypothesis, we should have $x < \epsilon_0$. But we found $\epsilon_0 < x$, which implies $x < x$, a contradiction. Thus, the assumption $x > 0$ is false. Since $x \geq 0$, we must have $x = 0$. ■

- Exercise 1.9.3.** 1. Let S be a non-empty subset of $\mathbb{R}$ bounded above. Let $u = \sup S$. Show that for every $n \in \mathbb{N}$, there exists an element $s_n \in S$ such that $u - \frac{1}{n} < s_n \leq u$.
2. Prove that the set of integers $\mathbb{Z}$ is neither bounded above nor bounded below.
3. If $S \subseteq \mathbb{R}$ contains a non-empty open interval, show that S is uncountable. (Hint: Relate to the Archimedean property and density).

1.10 Density of Rational and Irrational Numbers

One of the most profound consequences of the Archimedean Property is the "denseness" of the rational and irrational numbers within the real line. Geometrically, this implies that between any two points on the line, no matter how close together, there is an infinite "cloud" of both rational and irrational numbers.

1.10.1 The Density of Rational Numbers

Theorem 1.10.1 (The Density Theorem). *If x and y are distinct real numbers with $x < y$, then there exists a rational number $q \in \mathbb{Q}$ such that*

$$x < q < y.$$

Proof. Since $x < y$, we have $y - x > 0$. By the **Archimedean Property**, there exists a positive integer $n \in \mathbb{N}$ such that

$$n(y - x) > 1, \quad \text{or equivalently,} \quad ny - nx > 1.$$

This inequality implies that the gap between nx and ny is greater than 1. Therefore, there must contain an integer between them.

To formalize this, we apply the corollary of the Archimedean property to the real number nx . There exists a unique integer $m \in \mathbb{Z}$ such that

$$m - 1 \leq nx < m.$$

From this inequality, we deduce two things:

1. $nx < m$, which implies $x < \frac{m}{n}$.
2. $m \leq nx + 1$.

Combining the second deduction with our initial Archimedean scaling ($ny > nx + 1$):

$$m \leq nx + 1 < ny \implies m < ny \implies \frac{m}{n} < y.$$

Thus, we have:

$$x < \frac{m}{n} < y.$$

Let $q = \frac{m}{n}$. Since $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, q is a rational number satisfying $x < q < y$. ■

Corollary 1.10.2. *Between any two distinct real numbers, there are infinitely many rational numbers.*

Reasoning. By the theorem, there is one rational q_1 such that $x < q_1 < y$. Now apply the theorem to the interval (x, q_1) to find q_2 , and so on. ■

1.10.2 The Density of Irrational Numbers

The set of irrational numbers, denoted by $\mathbb{R} \setminus \mathbb{Q}$ (or sometimes $\mathbb{I}$), is also dense in $\mathbb{R}$.

Theorem 1.10.3. *If x and y are real numbers with $x < y$, then there exists an irrational number ξ such that*

$$x < \xi < y.$$

Proof. We utilize the known irrational number $\sqrt{2}$ and the density of rationals. Consider the real numbers $\frac{x}{\sqrt{2}}$ and $\frac{y}{\sqrt{2}}$. Since $x < y$ and $\sqrt{2} > 0$, we have

$$\frac{x}{\sqrt{2}} < \frac{y}{\sqrt{2}}.$$

By the **Rational Density Theorem**, there exists a rational number $r \in \mathbb{Q}$ (where $r \neq 0$) such that

$$\frac{x}{\sqrt{2}} < r < \frac{y}{\sqrt{2}}.$$

Multiplying through by $\sqrt{2}$, we obtain:

$$x < r\sqrt{2} < y.$$

Let $\xi = r\sqrt{2}$. Since r is a non-zero rational and $\sqrt{2}$ is irrational, their product ξ is irrational. Thus, we have found an irrational number strictly between x and y . ■

Example 1.10.4. Find a rational number and an irrational number between 2 and 2.5.

Solution. **Rational:** The midpoint $\frac{2+2.5}{2} = 2.25 = \frac{9}{4}$ is rational. **Irrational:** We need a number ξ such that $2 < \xi < 2.5$. Consider $\sqrt{5}$. We know $2^2 = 4$ and $2.5^2 = 6.25$. Since $4 < 5 < 6.25$, it follows that $2 < \sqrt{5} < 2.5$. Alternatively, using the theorem construction: $2 < 2 + \frac{\pi}{100} < 2.5$. ■

1.11 Dedekind's Property

While the supremum principle is the standard analytic way to describe the completeness of $\mathbb{R}$, Richard Dedekind provided a geometric-algebraic characterization known as *Dedekind Cuts* (or *Sections*). This concept formalizes the idea that the real line has "no gaps."

1.11.1 Dedekind Cuts

We imagine "cutting" the real line into two distinct pieces.

Definition 1.11.1 (Dedekind Cut). A **Dedekind Cut** (or Partition) of the set of real numbers $\mathbb{R}$ is a pair of non-empty subsets (L, U) (Lower and Upper classes) such that:

1. $L \cup U = \mathbb{R}$.
2. $L \cap U = \emptyset$.
3. Every element in L is strictly less than every element in U . That is, if $x \in L$ and $y \in U$, then $x < y$.

1.11.2 Dedekind's Fundamental Theorem

The crucial property of the real numbers is that whenever such a cut is made, the "knife" must hit exactly one point on the line.

Theorem 1.11.2 (Dedekind's Property). *For every Dedekind cut (L, U) of $\mathbb{R}$, there exists a unique real number α such that one of the following holds:*

1. α is the greatest element of L (in which case U has no smallest element), OR
2. α is the smallest element of U (in which case L has no greatest element).

Proof. Since the cut (L, U) partitions $\mathbb{R}$, L is non-empty. Furthermore, any element $u \in U$ serves as an upper bound for L (since every element in L is strictly less than every element in U).

By the **Completeness Axiom**, the set L must have a supremum in $\mathbb{R}$. Let

$$\alpha = \sup L.$$

Since $\mathbb{R} = L \cup U$ and $L \cap U = \emptyset$, the number α must belong to either L or U . We analyze these two cases:

Case 1: $\alpha \in L$.

Since α is the supremum of L and $\alpha \in L$, it is the greatest element of L (i.e., $\alpha = \max L$). Consequently, every $y \in U$ must satisfy $y > \alpha$. By the density of real numbers, U cannot have a smallest element (for any $y \in U$, there exists z such that $\alpha < z < y$, and z must be in U).

Case 2: $\alpha \in U$.

Since α is the *least* upper bound of L , no number smaller than α can be an upper bound of L . Consequently, no number smaller than α can belong to U (as elements of U are upper bounds for L). Thus, for all $y \in U$, we have $y \geq \alpha$. Since $\alpha \in U$, it follows that $\alpha = \min U$. In this case, L has no greatest element (since $\alpha \notin L$).

Uniqueness:

Suppose there are two distinct numbers $\alpha < \beta$ satisfying the property. By the density of $\mathbb{R}$, there exists a number γ such that $\alpha < \gamma < \beta$. Based on the cut properties relative to α and β , γ would have to belong to both L and U (or neither), which is a contradiction. Thus, α is unique. ■

In simpler terms, the cut is "produced" by a unique number α . The number α acts as the boundary between the lower class L and the upper class U .

Remark 1.11.3. Dedekind's Property is logically equivalent to the **Completeness Axiom** (Supremum Property).

- If we assume Dedekind's Property, we can prove every bounded set has a supremum.
- If we assume the Supremum Property, we can prove Dedekind's Property.

1.11.3 Comparison with Rational Numbers

To understand why this property is special to $\mathbb{R}$, consider a cut in the rational numbers $\mathbb{Q}$.

Example 1.11.4 (Failure of Dedekind's Property in $\mathbb{Q}$). Consider the set of rational numbers $\mathbb{Q}$. Let us define a cut (L, U) of $\mathbb{Q}$ defined by $\sqrt{2}$:

$$L = \{x \in \mathbb{Q} : x \leq 0 \text{ or } x^2 < 2\}$$

$$U = \{x \in \mathbb{Q} : x > 0 \text{ and } x^2 > 2\}$$

1. Every element of L is less than every element of U .
2. $L \cup U = \mathbb{Q}$ (since $x^2 = 2$ has no solution in $\mathbb{Q}$).
3. Does L have a greatest element? No. If $x \in L$ with $x > 0$, we can find a rational slightly larger than x whose square is still less than 2.

4. Does U have a smallest element? No. If $y \in U$, we can find a rational slightly smaller than y whose square is still greater than 2.

In the system $\mathbb{Q}$, this cut defines a "gap". The number required to fill this gap ($\sqrt{2}$) is missing. In $\mathbb{R}$, Dedekind's property guarantees this gap is filled.

- Exercise 1.11.5.** 1. Prove that between any two distinct rational numbers, there exists an irrational number.
2. Prove that between any two distinct irrational numbers, there exists a rational number.
3. Let (L, U) be a section (Dedekind cut) of $\mathbb{R}$. Prove that U is bounded below and L is bounded above.

Exercise 1.11.6. Construct a Dedekind Cut (L, U) of $\mathbb{Q}$ corresponding to the rational number 3. Identify the greatest element of L or the smallest element of U .

1.12 The Structure of $\mathbb{R}$ as an Ordered Field

Throughout this unit, we have established various properties of the real numbers. To summarize, the system of real numbers $\mathbb{R}$ is characterized by three distinct layers of structure:

1. **Algebraic Structure:** $\mathbb{R}$ is a *Field*. It possesses two binary operations ($+$ and $\cdot$) that satisfy the laws of closure, commutativity, associativity, identity, inverses, and distributivity.
2. **Order Structure:** $\mathbb{R}$ is an *Ordered* set. There exists a strict linear ordering ($<$) compatible with the field operations.
3. **Topological Structure (Completeness):** $\mathbb{R}$ is *Order Complete*. Every non-empty subset bounded above has a supremum within the set.

Combining the first two, we arrive at the definition of $\mathbb{R}$ as an Ordered Field.

Definition 1.12.1 ($\mathbb{R}$ as an Ordered Field). The set $\mathbb{R}$ is an Ordered Field. This implies that for all $x, y, z \in \mathbb{R}$:

- (Field Properties) $x + y$, xy , $-x$, and x^{-1} (for $x \neq 0$) exist and behave according to standard arithmetic axioms.
- (Trichotomy) Exactly one of $x < y$, $x = y$, or $x > y$ holds.
- (Monotonicity) $x < y \implies x + z < y + z$, and if $z > 0$, then $xz < yz$.

While the set of rational numbers $\mathbb{Q}$ also satisfies every axiom of an Ordered Field, it fails the third structural requirement: Completeness. This failure is what necessitates the extension from $\mathbb{Q}$ to $\mathbb{R}$.

1.13 Incompleteness of the Set of Rational Numbers

The "Incompleteness" of $\mathbb{Q}$ refers to the existence of "gaps" or "holes" in the rational number line. While the rationals are dense (there is a rational between any two numbers), they do not form a continuum.

To prove $\mathbb{Q}$ is incomplete, we must demonstrate that there exists a subset of $\mathbb{Q}$ that is bounded above but has no least upper bound (supremum) within $\mathbb{Q}$.

1.13.1 The Existence of Gaps

The classic example of a "gap" corresponds to the number $\sqrt{2}$. We first prove that this number cannot be rational.

Theorem 1.13.1 (Irrationality of $\sqrt{2}$). *There is no rational number r such that $r^2 = 2$.*

Proof. We proceed by contradiction. Suppose there exists a rational number r such that $r^2 = 2$. Since $r \in \mathbb{Q}$, it can be written in the form

$$r = \frac{p}{q},$$

where $p, q \in \mathbb{Z}$, $q \neq 0$, and we assume that p and q are *coprime* (they have no common factors other than 1). Squaring both sides:

$$\left(\frac{p}{q}\right)^2 = 2 \implies p^2 = 2q^2.$$

This equation implies that p^2 is an even number. Since the square of an odd integer is odd, p itself must be even. Thus, we can write $p = 2k$ for some integer k . Substituting this back into the equation:

$$(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies 2k^2 = q^2.$$

This equation implies that q^2 is even, and consequently, q must be even.

We have concluded that both p and q are even. This means they share a common factor of 2. This contradicts our initial assumption that p and q were coprime (in lowest terms). Therefore, no such rational number r exists. ■

We now formally prove that the Ordered Field $\mathbb{Q}$ is not complete.

Theorem 1.13.2 (Incompleteness of $\mathbb{Q}$). *The set of rational numbers $\mathbb{Q}$ is not order complete. Specifically, let*

$$S = \{x \in \mathbb{Q} : x > 0 \text{ and } x^2 < 2\}.$$

Then S is bounded above in $\mathbb{Q}$, but S has no supremum in $\mathbb{Q}$.

Proof. 1. Boundedness: The set S is non-empty (since $1 \in S$ as $1^2 = 1 < 2$). The set S is bounded above by 2, because if $x > 2$, then $x^2 > 4 > 2$, so $x \notin S$. Thus, 2 is an upper bound in $\mathbb{Q}$.

2. Non-existence of Supremum in $\mathbb{Q}$: Suppose, for the sake of contradiction, that S has a supremum in $\mathbb{Q}$. Let $p \in \mathbb{Q}$ be the supremum of S . By the Trichotomy law, since $p^2 \neq 2$ (proven in the previous theorem), we have two cases: $p^2 < 2$ or $p^2 > 2$.

Case A: Suppose $p^2 < 2$. We aim to show that p is not an upper bound by finding a rational $q > p$ such that $q \in S$. Let us construct q using the formula:

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}.$$

Then q is clearly rational. Consider $q^2 - 2$:

$$q^2 - 2 = \left(\frac{2p + 2}{p + 2}\right)^2 - 2 = \frac{4p^2 + 8p + 4 - 2(p^2 + 4p + 4)}{(p + 2)^2} = \frac{2(p^2 - 2)}{(p + 2)^2}.$$

Since we assumed $p^2 < 2$, the numerator $2(p^2 - 2)$ is negative, while the denominator is positive. Thus, $q^2 - 2 < 0$, which means $q^2 < 2$. So $q \in S$. Also, note that since $p^2 < 2$, $q - p = \frac{2 - p^2}{p + 2} > 0$, so $q > p$. We have found a member of S greater than p . Thus, p is not an upper bound. Contradiction.

Case B: Suppose $p^2 > 2$. We aim to show that p is not the *least* upper bound by finding a smaller upper bound $q < p$. Using the same formula $q = \frac{2p + 2}{p + 2}$, we analyze $q^2 - 2$:

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}.$$

Since we assumed $p^2 > 2$, the numerator is positive. Thus $q^2 > 2$. This implies q is an upper bound for S (since any $x \in S$ has $x^2 < 2 < q^2$). Furthermore, $p - q = \frac{p^2 - 2}{p + 2} > 0$, so $q < p$. We have found an upper bound q smaller than p . Thus, p is not the supremum. Contradiction.

Conclusion: Since p^2 cannot be less than 2, cannot be greater than 2, and cannot equal 2 (in $\mathbb{Q}$), no such rational p exists. Therefore, $\mathbb{Q}$ lacks the Least Upper Bound property. ■

Summary of Unit I

The journey through Unit I leads to this fundamental realization:

- $\mathbb{Q}$ allows for arithmetic (Field).
- $\mathbb{Q}$ allows for comparison (Order).
- However, $\mathbb{Q}$ fails to support limiting processes (Incompleteness).

The Real Number system $\mathbb{R}$ is explicitly constructed to preserve the Field and Order properties of $\mathbb{Q}$ while adding the **Completeness Axiom**, thereby filling the gaps and providing a solid foundation for Calculus.

Exercise 1.13.3. 1. Prove that if $x^2 = 3$, then x is not a rational number.

2. Let $A = \{x \in \mathbb{Q} : x^2 < 3\}$. Prove that A is bounded above in $\mathbb{Q}$ but has no rational least upper bound.

3. Explain why the set of Integers $\mathbb{Z}$ is an ordered ring but not an ordered field.

Chapter 2

Sequences of Real Numbers

In the previous unit, we studied the static properties of the real number system $\mathbb{R}$. We now transition to the dynamic aspect of analysis: limiting processes. The most fundamental tool for studying limits is the *sequence*.

A sequence can be thought of as a list of numbers written in a definite order:

$$x_1, x_2, x_3, \dots, x_n, \dots$$

Unlike a set, order matters in a sequence, and elements can repeat.

2.1 Sequences

Definition 2.1.1 (Sequence). A **sequence** of real numbers is a function $f : \mathbb{N} \rightarrow \mathbb{R}$, where $\mathbb{N}$ is the set of natural numbers.

Usually, we do not use the functional notation $f(n)$. Instead, we denote the value $f(n)$ by x_n (or a_n , u_n , etc.).

- x_n is called the **n -th term** of the sequence.
- The sequence itself is denoted by $\{x_n\}_{n=1}^{\infty}$, $\{x_n\}$, or simply (x_n) .

Remark 2.1.2. It is important to distinguish between the *sequence* $\{x_n\}$ (which has an infinite number of terms ordered by index) and the *range set* $\{x_n : n \in \mathbb{N}\}$ (which is merely the set of values taken by the sequence). For example, the sequence defined by $x_n = (-1)^n$ is $-1, 1, -1, 1, \dots$. It has infinitely many terms, but its range set is finite: $\{-1, 1\}$.

Example 2.1.3. 1. **Identity Sequence:** $x_n = n$. The sequence is $1, 2, 3, 4, \dots$

2. **Constant Sequence:** $x_n = c$ for some $c \in \mathbb{R}$. The sequence is $c, c, c, \dots$

3. **Harmonic Sequence:** $x_n = \frac{1}{n}$. The sequence is $1, \frac{1}{2}, \frac{1}{3}, \dots$

4. **Alternating Sequence:** $x_n = (-1)^{n+1}$. The sequence is $1, -1, 1, -1, \dots$

5. **Recursive Sequence:** Defined by $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$.

2.1.1 Bounded Sequences

Just as we classified sets of real numbers as bounded or unbounded, we can classify sequences based on the range of values they take.

Definition 2.1.4 (Bounded Above). A sequence $\{x_n\}$ is said to be **bounded above** if there exists a real number M such that

$$x_n \leq M \quad \text{for all } n \in \mathbb{N}.$$

The number M is called an **upper bound** of the sequence.

Definition 2.1.5 (Bounded Below). A sequence $\{x_n\}$ is said to be **bounded below** if there exists a real number m such that

$$x_n \geq m \quad \text{for all } n \in \mathbb{N}.$$

The number m is called a **lower bound** of the sequence.

Definition 2.1.6 (Bounded Sequence). A sequence $\{x_n\}$ is said to be **bounded** if it is both bounded above and bounded below. Equivalently, a sequence is bounded if there exists a positive real number $K > 0$ such that

$$|x_n| \leq K \quad \text{for all } n \in \mathbb{N}.$$

Definition 2.1.7 (Unbounded Sequence). A sequence is **unbounded** if it is not bounded. This means that for any $K > 0$, there exists at least one term x_n such that $|x_n| > K$.

Example 2.1.8. Show that the sequence $x_n = \frac{2n+1}{n+3}$ is bounded.

Solution. We examine the magnitude of the n -th term:

$$x_n = \frac{2n+1}{n+3}.$$

Since $n \geq 1$, both numerator and denominator are positive, so $x_n > 0$. Thus, 0 is a lower bound. For the upper bound, we can rewrite the term:

$$\frac{2n+1}{n+3} = \frac{2(n+3)-5}{n+3} = 2 - \frac{5}{n+3}.$$

Since $\frac{5}{n+3} > 0$ for all $n \in \mathbb{N}$, it follows that

$$2 - \frac{5}{n+3} < 2.$$

Thus, $x_n < 2$ for all n . Since $0 < x_n < 2$ for all n , the sequence is bounded (we may take $K = 2$). ■

Example 2.1.9. Consider the sequence defined by $x_n = (-1)^n n$. Is this sequence bounded?

Solution. The terms of the sequence are $-1, 2, -3, 4, -5, \dots$. To check for boundedness, consider the absolute value $|x_n| = |(-1)^n n| = n$. The set of values $\{|x_n| : n \in \mathbb{N}\} = \{1, 2, 3, \dots\}$ is the set of natural numbers $\mathbb{N}$. By the Archimedean property, $\mathbb{N}$ is not bounded above. Therefore, there is no K such that $|x_n| \leq K$ for all n . The sequence is unbounded. ■

Theorem 2.1.10 (Algebra of Bounded Sequences). Let $\{x_n\}$ and $\{y_n\}$ be two bounded sequences of real numbers, and let $c \in \mathbb{R}$. Then:

1. The sum sequence $\{x_n + y_n\}$ is bounded.
2. The difference sequence $\{x_n - y_n\}$ is bounded.
3. The product sequence $\{x_n \cdot y_n\}$ is bounded.
4. The scalar multiple sequence $\{c \cdot x_n\}$ is bounded.

Proof. Since $\{x_n\}$ and $\{y_n\}$ are bounded, there exist positive constants B_x and B_y such that $|x_n| \leq B_x$ and $|y_n| \leq B_y$ for all n .

1. **Sum and Difference:** By the Triangle Inequality,

$$|x_n \pm y_n| \leq |x_n| + |y_n| \leq B_x + B_y.$$

Thus, both sequences are bounded by $B_x + B_y$.

2. **Product:** Using the multiplicative property of absolute values,

$$|x_n y_n| = |x_n| |y_n| \leq B_x B_y.$$

Thus, the product sequence is bounded by $B_x B_y$.

3. **Scalar Multiple:** For any constant $c \in \mathbb{R}$,

$$|cx_n| = |c| |x_n| \leq |c| B_x.$$

Thus, the sequence is bounded by $|c| B_x$. ■

2.1.2 Supremum and Infimum of a Sequence

Since the range of a sequence is a subset of $\mathbb{R}$, if the sequence is bounded, we can define its supremum and infimum.

Definition 2.1.11. • The **supremum** of a sequence $\{x_n\}$, denoted $\sup x_n$, is the supremum of its range set $\{x_n : n \in \mathbb{N}\}$.

• The **infimum** of a sequence $\{x_n\}$, denoted $\inf x_n$, is the infimum of its range set $\{x_n : n \in \mathbb{N}\}$.

Example 2.1.12. Find the supremum and infimum of $x_n = 1 + \frac{(-1)^n}{n}$.

Solution. Let us list the first few terms to understand the behavior:

- $n = 1 : x_1 = 1 - 1 = 0$
- $n = 2 : x_2 = 1 + 1/2 = 1.5$
- $n = 3 : x_3 = 1 - 1/3 = 0.66 \dots$
- $n = 4 : x_4 = 1 + 1/4 = 1.25$

The terms oscillate around 1. The "even" terms are $1 + 1/2, 1 + 1/4, 1 + 1/6, \dots$. This subsequence decreases towards 1. The maximum value is at $n = 2$, which is 1.5. Thus, $\sup x_n = 1.5$. The "odd" terms are $1 - 1, 1 - 1/3, 1 - 1/5, \dots$. This subsequence increases towards 1. The minimum value is at $n = 1$, which is 0. Thus, $\inf x_n = 0$. The sequence is bounded between 0 and 1.5. ■

Theorem 2.1.13 (Existence of Bounds). *Every sequence $\{x_n\}$ that is bounded above has a supremum in $\mathbb{R}$. Similarly, every sequence that is bounded below has an infimum in $\mathbb{R}$.*

Proof. Let $S = \{x_n : n \in \mathbb{N}\}$ be the range set of the sequence. Since the sequence is bounded above, the set S is bounded above in $\mathbb{R}$. By the **Completeness Axiom** (Least Upper Bound Axiom) of real numbers, every non-empty subset of $\mathbb{R}$ that is bounded above has a supremum. Thus, $\sup x_n$ exists. The proof for the existence of the infimum is analogous, relying on the Greatest Lower Bound property of $\mathbb{R}$. ■

Theorem 2.1.14 (Algebraic Inequalities of Supremum and Infimum). *Let $\{x_n\}$ and $\{y_n\}$ be bounded sequences. Then:*

1. $\sup(x_n + y_n) \leq \sup x_n + \sup y_n$.
2. $\inf(x_n + y_n) \geq \inf x_n + \inf y_n$.
3. $\inf x_n + \sup y_n \leq \sup(x_n + y_n)$.
4. $\sup(c \cdot x_n) = c \cdot \sup x_n$ if $c \geq 0$.
5. $\sup(c \cdot x_n) = c \cdot \inf x_n$ if $c < 0$.

Proof. Let $u = \sup x_n$, $v = \sup y_n$, $a = \inf x_n$, and $b = \inf y_n$.

1. **Sum Supremum:** For all n , $x_n \leq u$ and $y_n \leq v$. Adding these inequalities yields $x_n + y_n \leq u + v$. Since $u + v$ is an upper bound for the sequence $\{x_n + y_n\}$, the least upper bound must satisfy:

$$\sup(x_n + y_n) \leq u + v = \sup x_n + \sup y_n.$$

2. **Sum Infimum:** Similarly, $x_n \geq a$ and $y_n \geq b$ implies $x_n + y_n \geq a + b$. Thus, $a + b$ is a lower bound, implying:

$$\inf(x_n + y_n) \geq a + b = \inf x_n + \inf y_n.$$

3. **Mixed Inequality:** We know $x_n + y_n \leq \sup(x_n + y_n)$. Rearranging gives $y_n \leq \sup(x_n + y_n) - x_n$. Since this holds for all n , taking the supremum of the left side (and noting that $\sup(-x_n) = -\inf x_n$) yields:

$$\sup y_n \leq \sup(x_n + y_n) + \sup(-x_n) = \sup(x_n + y_n) - \inf x_n.$$

Rearranging gives $\inf x_n + \sup y_n \leq \sup(x_n + y_n)$.

4. **Scalar ($c \geq 0$):** If $c \geq 0$, the inequality $x_n \leq u$ implies $cx_n \leq cu$. Thus cu is an upper bound, so $\sup(cx_n) \leq cu$. Conversely, for any $\epsilon > 0$, there exists $x_k > u - \epsilon/c$, implying $cx_k > cu - \epsilon$. Thus $\sup(cx_n) = c \sup x_n$.
5. **Scalar ($c < 0$):** If $c < 0$, multiplying $x_n \geq a$ by c reverses the inequality: $cx_n \leq ca$. Thus ca (which is $c \inf x_n$) is an upper bound for $\{cx_n\}$. By symmetry, it is the least upper bound. ■

Theorem 2.1.15 (Comparison Principle). Let $\{x_n\}$ and $\{y_n\}$ be sequences such that $x_n \leq y_n$ for all $n \in \mathbb{N}$.

1. If both are bounded above, then $\sup x_n \leq \sup y_n$.
2. If both are bounded below, then $\inf x_n \leq \inf y_n$.

Proof. 1. Since y_n is bounded above, let $M = \sup y_n$. By definition, $y_n \leq M$ for all n . Given $x_n \leq y_n$, by transitivity we have $x_n \leq M$ for all n . Thus, M is an upper bound for the sequence $\{x_n\}$. Since $\sup x_n$ is the *least* upper bound, it follows that $\sup x_n \leq M = \sup y_n$.

2. Similarly, let $m = \inf x_n$. Then $m \leq x_n$ for all n . Given $x_n \leq y_n$, we have $m \leq y_n$ for all n . Thus, m is a lower bound for $\{y_n\}$. Since $\inf y_n$ is the *greatest* lower bound, we conclude $\inf x_n = m \leq \inf y_n$. ■

Exercise 2.1.16. 1. Write out the first five terms of the sequence defined by $x_1 = 1$ and $x_{n+1} = \frac{x_n}{1+x_n}$. Prove by induction that $x_n = \frac{1}{n}$.

2. Let $\{x_n\}$ and $\{y_n\}$ be two bounded sequences. Let $z_n = x_n + y_n$. Prove that $\{z_n\}$ is a bounded sequence and that

$$\sup(x_n + y_n) \leq \sup x_n + \sup y_n.$$

3. Give an example of a sequence that is bounded below but not bounded above.

2.2 Convergence of Sequences

The central concept of analysis is the limit. A sequence converges if its terms get arbitrarily close to a specific number as the index increases.

Definition 2.2.1 (Convergence). A sequence of real numbers $\{x_n\}$ is said to **converge** to a real number l if for every $\epsilon > 0$, there exists a natural number N (depending on ϵ) such that

$$|x_n - l| < \epsilon \quad \text{for all } n \geq N.$$

The number l is called the **limit** of the sequence. We write:

$$\lim_{n \rightarrow \infty} x_n = l \quad \text{or} \quad x_n \rightarrow l.$$

Definition 2.2.2 (Divergence). A sequence is said to **diverge** if it does not converge to any real number.

Example 2.2.3. Using the definition of convergence, prove that

$$\lim_{n \rightarrow \infty} \frac{2n^2 - 3}{n^2 + 1} = 2.$$

Solution. Let $\epsilon > 0$ be arbitrary. We seek a natural number N such that for all $n \geq N$:

$$\left| \frac{2n^2 - 3}{n^2 + 1} - 2 \right| < \epsilon.$$

We simplify the expression inside the absolute value:

$$\left| \frac{2n^2 - 3 - 2(n^2 + 1)}{n^2 + 1} \right| = \left| \frac{2n^2 - 3 - 2n^2 - 2}{n^2 + 1} \right| = \left| \frac{-5}{n^2 + 1} \right| = \frac{5}{n^2 + 1}.$$

To satisfy the condition $< \epsilon$, we require:

$$\frac{5}{n^2 + 1} < \frac{5}{n^2} < \epsilon.$$

(Note: We used the inequality $n^2 + 1 > n^2$ to simplify the fraction). Solving for n :

$$\frac{5}{n^2} < \epsilon \implies n^2 > \frac{5}{\epsilon} \implies n > \sqrt{\frac{5}{\epsilon}}.$$

By the Archimedean property, there exists a natural number N such that $N > \sqrt{\frac{5}{\epsilon}}$. Thus, for all $n \geq N$, the inequality holds. Therefore, the limit is 2. ■

2.2.1 Fundamental Theorems on Convergence

Theorem 2.2.4 (Uniqueness of Limit). A sequence cannot converge to more than one limit. If $\lim_{n \rightarrow \infty} x_n = l$ and $\lim_{n \rightarrow \infty} x_n = m$, then $l = m$.

Proof. Suppose, for the sake of contradiction, that $l \neq m$. Let $\epsilon = \frac{|l-m|}{2} > 0$. By the definition of convergence, there exist integers N_1 and N_2 such that:

$$|x_n - l| < \epsilon \quad \forall n \geq N_1 \quad \text{and} \quad |x_n - m| < \epsilon \quad \forall n \geq N_2.$$

Let $N = \max(N_1, N_2)$. For any $n \geq N$, the Triangle Inequality yields:

$$|l - m| = |(l - x_n) + (x_n - m)| \leq |x_n - l| + |x_n - m| < \epsilon + \epsilon = 2\epsilon.$$

Substituting ϵ , we obtain $|l - m| < |l - m|$, a contradiction. Thus, $l = m$. ■

Theorem 2.2.5 (Boundedness of Convergent Sequences). Every convergent sequence is bounded.

Proof. Let $\{x_n\}$ converge to l . Choosing $\epsilon = 1$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - l| < 1$. By the Triangle Inequality, $|x_n| - |l| \leq |x_n - l| < 1$, which implies $|x_n| < |l| + 1$ for all $n \geq N$. To bound the entire sequence, let

$$M = \max\{|x_1|, |x_2|, \dots, |x_{N-1}|, |l| + 1\}.$$

Then $|x_n| \leq M$ for all $n \in \mathbb{N}$. Hence, the sequence is bounded. ■

Theorem 2.2.6 (Algebra of Limits). Let $\{x_n\}$ and $\{y_n\}$ be sequences converging to l and m respectively. Then:

$$1. \lim_{n \rightarrow \infty} (x_n + y_n) = l + m.$$

2. $\lim_{n \rightarrow \infty} (x_n - y_n) = l - m$.
3. $\lim_{n \rightarrow \infty} (c \cdot x_n) = c \cdot l$ for any constant $c \in \mathbb{R}$.
4. $\lim_{n \rightarrow \infty} (x_n \cdot y_n) = l \cdot m$.
5. $\lim_{n \rightarrow \infty} \left(\frac{x_n}{y_n} \right) = \frac{l}{m}$, provided $m \neq 0$ and $y_n \neq 0$ for all n .
6. $\lim_{n \rightarrow \infty} |x_n| = |l|$.

Proof. We prove selected parts; others follow similarly. Let $\epsilon > 0$.

1. **Sum:** Choose N such that $|x_n - l| < \epsilon/2$ and $|y_n - m| < \epsilon/2$ for $n \geq N$. Then:

$$|(x_n + y_n) - (l + m)| \leq |x_n - l| + |y_n - m| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

2. **Scalar Multiple:** If $c = 0$, the result is trivial. If $c \neq 0$, choose N such that $|x_n - l| < \epsilon/|c|$. Then $|cx_n - cl| = |c||x_n - l| < \epsilon$.

3. **Product:** Since $\{x_n\}$ is convergent, it is bounded; let $|x_n| \leq K$ for all n . Consider the decomposition:

$$x_n y_n - lm = x_n(y_n - m) + m(x_n - l).$$

Choose N such that $|x_n - l| < \frac{\epsilon}{2(|m|+1)}$ and $|y_n - m| < \frac{\epsilon}{2K}$. Then:

$$|x_n y_n - lm| \leq |x_n||y_n - m| + |m||x_n - l| < K \cdot \frac{\epsilon}{2K} + |m| \cdot \frac{\epsilon}{2(|m|+1)} < \epsilon.$$

4. **Quotient:** It suffices to show $1/y_n \rightarrow 1/m$ and use the Product rule. Since $y_n \rightarrow m \neq 0$, eventually $|y_n| > |m|/2$. Thus:

$$\left| \frac{1}{y_n} - \frac{1}{m} \right| = \frac{|m - y_n|}{|y_n||m|} < \frac{|m - y_n|}{|m|^2/2} = \frac{2}{|m|^2} |y_n - m|.$$

The result follows as $|y_n - m| \rightarrow 0$.

5. **Absolute Value:** By the Reverse Triangle Inequality, $||x_n| - |l|| \leq |x_n - l| < \epsilon$.

■

Example 2.2.7. Let a sequence be defined recursively by $x_1 = 1$ and $x_{n+1} = \sqrt{2x_n + 3}$. Prove that the sequence converges and find its limit.

Solution. We utilize the **Monotone Convergence Theorem**. We must show the sequence is monotonic and bounded.

Boundedness: We claim that $x_n < 3$ for all n . *Base Case:* $x_1 = 1 < 3$. *Inductive Step:* Assume $x_k < 3$. Then:

$$x_{k+1} = \sqrt{2x_k + 3} < \sqrt{2(3) + 3} = \sqrt{9} = 3.$$

Thus, the sequence is bounded above by 3. Since $x_1 > 0$ and the recurrence involves positive roots, it is bounded below by 0.

Monotonicity: We consider the difference $x_{n+1} - x_n$. It is easier to compare squares since terms are positive:

$$x_{n+1}^2 - x_n^2 = (2x_n + 3) - x_n^2 = -(x_n^2 - 2x_n - 3) = -(x_n - 3)(x_n + 1).$$

Since $0 < x_n < 3$, the term $(x_n - 3)$ is negative and $(x_n + 1)$ is positive. Therefore, $x_{n+1}^2 - x_n^2 > 0$, which implies $x_{n+1} > x_n$. The sequence is strictly monotonically increasing.

Limit: Since $\{x_n\}$ is increasing and bounded above, it converges to a limit l . Taking the limit of the relation $x_{n+1} = \sqrt{2x_n + 3}$:

$$l = \sqrt{2l + 3} \implies l^2 = 2l + 3 \implies l^2 - 2l - 3 = 0.$$

Factoring gives $(l - 3)(l + 1) = 0$. Since $x_n > 0$, the limit must be $l = 3$.

■

Theorem 2.2.8 (Sandwich / Squeeze Theorem). *Let $\{x_n\}$, $\{y_n\}$, and $\{z_n\}$ be sequences such that:*

$$x_n \leq y_n \leq z_n \quad \text{for all } n \geq N_0.$$

If $\lim_{n \rightarrow \infty} x_n = l$ and $\lim_{n \rightarrow \infty} z_n = l$, then $\{y_n\}$ also converges and $\lim_{n \rightarrow \infty} y_n = l$.

Proof. Let $\epsilon > 0$. Since $x_n \rightarrow l$ and $z_n \rightarrow l$, there exists an integer $K \geq N_0$ such that for all $n \geq K$:

$$l - \epsilon < x_n \quad \text{and} \quad z_n < l + \epsilon.$$

Using the hypothesis $x_n \leq y_n \leq z_n$, we combine the inequalities:

$$l - \epsilon < x_n \leq y_n \leq z_n < l + \epsilon.$$

Consequently, $l - \epsilon < y_n < l + \epsilon$, or equivalently $|y_n - l| < \epsilon$. Thus, $y_n \rightarrow l$. ■

Example 2.2.9. Evaluate the limit of the sequence given by:

$$x_n = (3^n + 4^n)^{1/n}.$$

Solution. We apply the **Sandwich (Squeeze) Theorem**. Observe that $4^n \leq 3^n + 4^n \leq 4^n + 4^n = 2 \cdot 4^n$. Taking the n -th root of the inequality:

$$(4^n)^{1/n} \leq (3^n + 4^n)^{1/n} \leq (2 \cdot 4^n)^{1/n}$$

$$4 \leq x_n \leq 2^{1/n} \cdot 4.$$

We know that $\lim_{n \rightarrow \infty} 4 = 4$. We also know that $\lim_{n \rightarrow \infty} 2^{1/n} = 1$ (standard limit $\lim a^{1/n} = 1$ for $a > 0$). Therefore, $\lim_{n \rightarrow \infty} (2^{1/n} \cdot 4) = 1 \cdot 4 = 4$. Since the bounding sequences both converge to 4, by the Sandwich Theorem:

$$\lim_{n \rightarrow \infty} x_n = 4. \quad \text{■}$$

2.2.2 Subsequences

A subsequence is obtained by deleting some terms of the original sequence while maintaining the relative order of the remaining terms.

Definition 2.2.10 (Subsequence). Let $\{x_n\}$ be a sequence. Let $\{n_k\}_{k=1}^{\infty}$ be a strictly increasing sequence of natural numbers (i.e., $n_1 < n_2 < n_3 < \dots$). Then the sequence $\{x_{n_k}\}$ is called a **subsequence** of $\{x_n\}$.

Theorem 2.2.11 (Convergence of Subsequences). *If a sequence $\{x_n\}$ converges to l , then every subsequence $\{x_{n_k}\}$ of $\{x_n\}$ also converges to l .*

Proof. Let $\epsilon > 0$. Since $\lim_{n \rightarrow \infty} x_n = l$, there exists a natural number N such that $|x_n - l| < \epsilon$ for all $n \geq N$. For any subsequence, the strictly increasing indices satisfy $n_k \geq k$ for all k . Therefore, if we choose $k \geq N$, it follows that $n_k \geq N$. Consequently:

$$|x_{n_k} - l| < \epsilon \quad \text{for all } k \geq N.$$

Thus, $\{x_{n_k}\}$ converges to l . ■

Corollary 2.2.12 (Divergence Criteria). *If a sequence $\{x_n\}$ contains two subsequences that converge to distinct limits, then $\{x_n\}$ is divergent.*

Proof. Suppose, for the sake of contradiction, that $\{x_n\}$ converges to a limit L . By the preceding theorem, every subsequence of $\{x_n\}$ must also converge to L . Let the two subsequences converge to l_1 and l_2 . Then we must have $l_1 = L$ and $l_2 = L$, which implies $l_1 = l_2$. This contradicts the hypothesis that the limits are distinct ($l_1 \neq l_2$). Therefore, the original sequence $\{x_n\}$ does not converge. ■

2.2.3 Limit Points (Cluster Points)

It is important to distinguish between the *limit* of a sequence and a *limit point*. A sequence has at most one limit, but it may have multiple limit points.

Definition 2.2.13 (Limit Point). A real number ξ is called a **limit point** (or cluster point) of a sequence $\{x_n\}$ if, for every $\epsilon > 0$, the interval $(\xi - \epsilon, \xi + \epsilon)$ contains infinitely many terms of the sequence.

Theorem 2.2.14 (Characterization of Limit Points). A number ξ is a limit point of $\{x_n\}$ if and only if there exists a subsequence of $\{x_n\}$ that converges to ξ .

Example 2.2.15. Consider the sequence $x_n = (-1)^n$. The terms are $-1, 1, -1, 1, \dots$

- The number 1 is a limit point because the subsequence $x_{2k} \rightarrow 1$.
- The number -1 is a limit point because the subsequence $x_{2k-1} \rightarrow -1$.

This sequence diverges, but it has two limit points.

The Bolzano-Weierstrass Theorem

This is one of the most significant theorems in real analysis, establishing a link between boundedness and sequential compactness.

Theorem 2.2.16 (Bolzano-Weierstrass Theorem). Every bounded sequence of real numbers has a convergent subsequence.

Proof. Let $\{x_n\}$ be a bounded sequence contained in the closed interval $I_1 = [a, b]$. We construct a sequence of nested intervals using the bisection method. Divide I_1 into two halves; at least one half must contain infinitely many terms of the sequence. Call this closed interval I_2 . Repeating this process, we obtain a sequence of nested closed intervals $I_1 \supseteq I_2 \supseteq I_3 \dots$ such that the length of I_k is $(b - a)/2^{k-1}$ and each I_k contains infinitely many terms of $\{x_n\}$.

By the **Nested Interval Theorem**, the intersection $\bigcap_{k=1}^{\infty} I_k$ contains a unique point ξ . We construct a subsequence $\{x_{n_k}\}$ as follows: choose n_1 such that $x_{n_1} \in I_1$. Generally, choose $n_k > n_{k-1}$ such that $x_{n_k} \in I_k$. Since both x_{n_k} and ξ lie in I_k , we have $|x_{n_k} - \xi| \leq \text{length}(I_k) \rightarrow 0$ as $k \rightarrow \infty$. Thus, the subsequence converges to ξ . ■

Remark 2.2.17. This theorem guarantees that if terms of a sequence are confined within a finite interval, they must "cluster" around at least one point.

Theorem 2.2.18 (Existence of Limit Points). Every bounded sequence of real numbers has at least one limit point.

Proof. Let $\{x_n\}$ be a bounded sequence. By the **Bolzano-Weierstrass Theorem**, every bounded sequence possesses a convergent subsequence. Let $\{x_{n_k}\}$ be such a subsequence converging to a limit ξ . By the definition of a limit point, the limit of a convergent subsequence is a limit point of the original sequence. Thus, ξ is a limit point of $\{x_n\}$. ■

Theorem 2.2.19 (Relationship to Convergence). A bounded sequence $\{x_n\}$ is convergent if and only if it has exactly one limit point.

Proof. ($\Rightarrow$) Suppose $\{x_n\}$ converges to l . Then every subsequence of $\{x_n\}$ also converges to l . Thus, l is a limit point. Furthermore, no number $\xi \neq l$ can be a limit point, because any neighborhood of ξ disjoint from a neighborhood of l would contain at most finitely many terms. Thus, the limit point is unique.

($\Leftarrow$) Conversely, suppose $\{x_n\}$ is bounded and has a unique limit point l . We proceed by contradiction. Assume $\{x_n\}$ does not converge to l . Then there exists an $\epsilon > 0$ such that infinitely many terms of the sequence satisfy $|x_n - l| \geq \epsilon$. Let these terms form a subsequence $\{y_k\}$. Since $\{x_n\}$ is bounded, $\{y_k\}$ is also bounded. By the Bolzano-Weierstrass Theorem, $\{y_k\}$ must contain a convergent subsequence $\{z_m\}$ with limit l' . Since $|z_m - l| \geq \epsilon$ for all terms, the limit satisfies $|l' - l| \geq \epsilon$, which implies $l' \neq l$. This means $\{x_n\}$ has a second limit point l' , contradicting the hypothesis of uniqueness. Therefore, $\{x_n\}$ must converge to l . ■

2.3 Limit Superior and Limit Inferior

For a bounded sequence that does not necessarily converge, we can define upper and lower limits based on the set of its limit points.

Definition 2.3.1. Let $\{x_n\}$ be a bounded sequence. Let E be the set of all limit points of $\{x_n\}$.

1. The **Limit Superior** ($\limsup x_n$ or $\overline{\lim} x_n$) is the supremum of the set E .

$$\limsup_{n \rightarrow \infty} x_n = \sup E.$$

2. The **Limit Inferior** ($\liminf x_n$ or $\underline{\lim} x_n$) is the infimum of the set E .

$$\liminf_{n \rightarrow \infty} x_n = \inf E.$$

Theorem 2.3.2 (Analytical Definition of LimSup and LimInf). Let $\{x_n\}$ be a bounded sequence. Then the limits of the supremum and infimum tails exist:

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} (\sup\{x_k : k \geq n\})$$

$$\liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} (\inf\{x_k : k \geq n\})$$

Proof. We provide the proof for the limit superior; the limit inferior follows analogously. Let $S_n = \{x_k : k \geq n\}$. Since the sequence $\{x_n\}$ is bounded, S_n is bounded. Define $u_n = \sup S_n$. Observe that $S_{n+1} \subset S_n$ (we are removing x_n from the set). The supremum of a subset is less than or equal to the supremum of the set, so $u_{n+1} \leq u_n$. Thus, the sequence $\{u_n\}$ is monotonically decreasing. Furthermore, since $\{x_n\}$ is bounded below, $\{u_n\}$ is bounded below. By the **Monotone Convergence Theorem**, the sequence $\{u_n\}$ converges. Therefore, $\lim_{n \rightarrow \infty} u_n$ exists. ■

Theorem 2.3.3 (Convergence via LimSup/LimInf). A bounded sequence $\{x_n\}$ converges to l if and only if:

$$\limsup_{n \rightarrow \infty} x_n = \liminf_{n \rightarrow \infty} x_n = l.$$

Proof. Let $u_n = \sup\{x_k : k \geq n\}$ and $v_n = \inf\{x_k : k \geq n\}$. By definition, for all $k \geq n$, we have $v_n \leq x_k \leq u_n$. Specifically:

$$v_n \leq x_n \leq u_n.$$

($\Leftarrow$) Suppose $\lim v_n = l$ and $\lim u_n = l$. By the **Squeeze Theorem** applied to the inequality above, it follows immediately that $\lim_{n \rightarrow \infty} x_n = l$.

($\Rightarrow$) Suppose $\lim_{n \rightarrow \infty} x_n = l$. For any $\epsilon > 0$, there exists N such that for all $k \geq N$, $l - \epsilon < x_k < l + \epsilon$. This implies that for $n \geq N$:

$$l - \epsilon \leq v_n \leq u_n \leq l + \epsilon.$$

Since ϵ is arbitrary, both sequences u_n and v_n must converge to l . ■

Example 2.3.4. Find the limit points, $\limsup x_n$, and $\liminf x_n$ for the sequence:

$$x_n = (-1)^n \left(1 + \frac{1}{n}\right) + \sin\left(\frac{n\pi}{2}\right).$$

Solution. We analyze the behavior of the sequence by considering subsequences based on $n \pmod{4}$.

- If $n = 4k$: $(-1)^{4k} = 1$, $\sin(2k\pi) = 0$.

$$x_{4k} = 1\left(1 + \frac{1}{4k}\right) + 0 \rightarrow 1.$$

- If $n = 4k + 1$: $(-1)^{4k+1} = -1$, $\sin(2k\pi + \pi/2) = 1$.

$$x_{4k+1} = -1\left(1 + \frac{1}{4k+1}\right) + 1 = -1 - \frac{1}{4k+1} + 1 = -\frac{1}{4k+1} \rightarrow 0.$$

- If $n = 4k + 2$: $(-1)^{4k+2} = 1$, $\sin(2k\pi + \pi) = 0$.

$$x_{4k+2} = 1\left(1 + \frac{1}{4k+2}\right) + 0 \rightarrow 1.$$

- If $n = 4k + 3$: $(-1)^{4k+3} = -1$, $\sin(2k\pi + 3\pi/2) = -1$.

$$x_{4k+3} = -1\left(1 + \frac{1}{4k+3}\right) - 1 = -2 - \frac{1}{4k+3} \rightarrow -2.$$

The set of limit points (subsequential limits) is $E = \{-2, 0, 1\}$. From this set:

- Limit Superior: $\limsup_{n \rightarrow \infty} x_n = \sup E = 1$.
- Limit Inferior: $\liminf_{n \rightarrow \infty} x_n = \inf E = -2$.

Since $\limsup \neq \liminf$, the sequence diverges. ■

Example 2.3.5. Evaluate the limit of the sequence defined by

$$x_n = \sqrt{n^2 + n} - n.$$

Solution. Direct substitution yields the indeterminate form $\infty - \infty$. We proceed by rationalizing the expression (multiplying by the conjugate).

$$\begin{aligned} x_n &= \frac{(\sqrt{n^2 + n} - n)(\sqrt{n^2 + n} + n)}{\sqrt{n^2 + n} + n} \\ &= \frac{(n^2 + n) - n^2}{\sqrt{n^2 + n} + n} \\ &= \frac{n}{\sqrt{n^2(1 + \frac{1}{n})} + n}. \end{aligned}$$

Factor out n from the denominator (since $n > 0$, $\sqrt{n^2} = n$):

$$x_n = \frac{n}{n\sqrt{1 + \frac{1}{n}} + n} = \frac{n}{n\left(\sqrt{1 + \frac{1}{n}} + 1\right)}.$$

Using the Algebra of Limits and the fact that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$:

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} = \frac{1}{\sqrt{1 + 0} + 1} = \frac{1}{2}.$$
■

Example 2.3.6. Prove that $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$.

Solution. We utilize the **Ratio Test for Sequences**: If $\{x_n\}$ is a sequence of positive terms such that $\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = L < 1$, then $\lim_{n \rightarrow \infty} x_n = 0$.

Let $x_n = \frac{n!}{n^n}$. Consider the ratio of consecutive terms:

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} \\ &= \frac{(n+1) \cdot n!}{(n+1) \cdot (n+1)^n} \cdot \frac{n^n}{n!} \\ &= \frac{n^n}{(n+1)^n} \\ &= \frac{1}{\left(\frac{n+1}{n}\right)^n} \\ &= \frac{1}{\left(1 + \frac{1}{n}\right)^n}. \end{aligned}$$

We recall the standard limit $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$. Therefore:

$$\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \frac{1}{e}.$$

Since $e \approx 2.718 > 1$, the limit $L = \frac{1}{e} < 1$. Thus, by the Ratio Test, $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$. ■

Example 2.3.7. Find the limit of the sequence

$$x_n = \frac{1}{n^2 + 1} + \frac{1}{n^2 + 2} + \cdots + \frac{1}{n^2 + n}.$$

Solution. We cannot simply apply the limit to each term because the number of terms (n) approaches infinity. Instead, we use the **Sandwich (Squeeze) Theorem**.

Let us bound the sequence x_n . The sum consists of n terms. The *smallest* term in the sum is the last one: $\frac{1}{n^2+n}$. The *largest* term in the sum is the first one: $\frac{1}{n^2+1}$.

Therefore, we can establish the inequality:

$$n \cdot \left(\frac{1}{n^2 + n}\right) \leq x_n \leq n \cdot \left(\frac{1}{n^2 + 1}\right).$$

Simplifying the bounds:

$$\frac{n}{n^2 + n} \leq x_n \leq \frac{n}{n^2 + 1}.$$

Now, we evaluate the limits of the bounding sequences:

- Lower Bound Limit: $\lim_{n \rightarrow \infty} \frac{n}{n^2 + n} = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$.
- Upper Bound Limit: $\lim_{n \rightarrow \infty} \frac{n}{n^2 + 1} = \lim_{n \rightarrow \infty} \frac{1/n}{1+1/n^2} = \frac{0}{1} = 0$.

Since both the lower and upper bounds converge to 0, by the Sandwich Theorem:

$$\lim_{n \rightarrow \infty} x_n = 0. \quad \blacksquare$$

In the previous sections, determining whether a sequence $\{x_n\}$ converges required a candidate for the limit l . We then verified the limit using the definition $|x_n - l| < \epsilon$. However, in many practical situations, the limit is unknown.

Augustin-Louis Cauchy introduced an intrinsic criterion for convergence that depends solely on the internal behavior of the sequence terms. The fundamental idea is that if the terms of a sequence eventually "cluster" infinitely close to each other, the sequence must be converging to some limit.

2.4 Cauchy Sequences

Definition 2.4.1 (Cauchy Sequence). A sequence of real numbers $\{x_n\}$ is called a **Cauchy sequence** (or a fundamental sequence) if for every $\epsilon > 0$, there exists a positive integer N such that

$$|x_n - x_m| < \epsilon \quad \text{for all } n, m \geq N.$$

Remark 2.4.2. The inequality $|x_n - x_m| < \epsilon$ implies that the distance between *any* two terms beyond the index N becomes arbitrarily small. It is not sufficient for consecutive terms merely to get closer (e.g., $|x_{n+1} - x_n| \rightarrow 0$ does not guarantee a sequence is Cauchy, as seen in the harmonic series partial sums).

Example 2.4.3. Show that the sequence $\{\frac{1}{n}\}$ is a Cauchy sequence.

Solution. Let $\epsilon > 0$ be given. We look for N such that for $n, m \geq N$, $|\frac{1}{n} - \frac{1}{m}| < \epsilon$. By the Triangle Inequality:

$$\left| \frac{1}{n} - \frac{1}{m} \right| \leq \frac{1}{n} + \frac{1}{m}.$$

If we choose $n, m \geq N$, then $\frac{1}{n} \leq \frac{1}{N}$ and $\frac{1}{m} \leq \frac{1}{N}$. Thus:

$$\left| \frac{1}{n} - \frac{1}{m} \right| \leq \frac{1}{N} + \frac{1}{N} = \frac{2}{N}.$$

To satisfy the condition, we require $\frac{2}{N} < \epsilon$, or $N > \frac{2}{\epsilon}$. By the Archimedean property, such an integer N exists. Thus, the sequence is Cauchy. ■

2.4.1 Cauchy's General Principle of Convergence

The power of Real Analysis lies in the equivalence between Cauchy sequences and convergent sequences. We establish this in two steps.

Theorem 2.4.4. *Every convergent sequence of real numbers is a Cauchy sequence.*

Proof. Let $\{x_n\}$ converge to a limit l . Let $\epsilon > 0$ be given. By the definition of convergence, there exists an integer N such that for all $k \geq N$:

$$|x_k - l| < \frac{\epsilon}{2}.$$

Now, for any $n, m \geq N$, we use the Triangle Inequality:

$$\begin{aligned} |x_n - x_m| &= |(x_n - l) - (x_m - l)| \\ &\leq |x_n - l| + |x_m - l| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Thus, $\{x_n\}$ is a Cauchy sequence. ■

Example 2.4.5. Let $\{x_n\}$ be the sequence of partial sums of the Harmonic series, i.e., $x_n = \sum_{k=1}^n \frac{1}{k}$. Use Cauchy's General Principle to prove that $\{x_n\}$ diverges.

Solution. We assume, for the sake of contradiction, that $\{x_n\}$ is convergent. By Cauchy's Criterion, it must be a Cauchy sequence. Therefore, for $\epsilon = \frac{1}{2}$, there must exist an N such that $|x_m - x_n| < \frac{1}{2}$ for all $m, n \geq N$. Let $n \geq N$ and choose $m = 2n$. Consider the difference:

$$|x_{2n} - x_n| = \sum_{k=1}^{2n} \frac{1}{k} - \sum_{k=1}^n \frac{1}{k} = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}.$$

There are exactly n terms in this sum. The smallest term is $\frac{1}{2n}$. Replacing each term with the smallest term $\frac{1}{2n}$, we get:

$$|x_{2n} - x_n| \geq \underbrace{\frac{1}{2n} + \cdots + \frac{1}{2n}}_{n \text{ times}} = n \cdot \frac{1}{2n} = \frac{1}{2}.$$

This contradicts the condition that $|x_{2n} - x_n| < \frac{1}{2}$ for sufficiently large n . Thus, the sequence is not Cauchy, and consequently, it diverges. ■

Boundedness of Cauchy Sequences

To prove the converse, we first need to establish that Cauchy sequences behave like convergent sequences in terms of boundedness.

Theorem 2.4.6. *Every Cauchy sequence of real numbers is bounded.*

Proof. Let $\{x_n\}$ be a Cauchy sequence. Let $\epsilon = 1$. There exists a natural number N such that for all $n \geq N$:

$$|x_n - x_N| < 1.$$

By the Triangle Inequality, $|x_n| - |x_N| \leq |x_n - x_N| < 1$, which implies:

$$|x_n| < |x_N| + 1 \quad \text{for all } n \geq N.$$

Let $M = \max\{|x_1|, |x_2|, \dots, |x_{N-1}|, |x_N| + 1\}$. Then $|x_n| \leq M$ for all $n \in \mathbb{N}$. Hence, the sequence is bounded. ■

This theorem is the cornerstone of the completeness of $\mathbb{R}$. It asserts that in the real number system, if terms get close together, they must converge to a limit existing within the system.

Example 2.4.7. Consider the sequence $x_n = \sin(n)$ (where n is in radians). Prove that this sequence has a convergent subsequence.

Solution. The Sine function is bounded for all real arguments. Specifically:

$$-1 \leq \sin(n) \leq 1 \quad \text{for all } n \in \mathbb{N}.$$

Thus, $\{x_n\}$ is a bounded sequence with upper bound $M = 1$ and lower bound $m = -1$. By the **Bolzano-Weierstrass Theorem**, every bounded sequence of real numbers must possess at least one convergent subsequence. Therefore, there exists a strictly increasing sequence of integers $\{n_k\}$ and a real number $\alpha \in [-1, 1]$ such that:

$$\lim_{k \rightarrow \infty} \sin(n_k) = \alpha.$$

(Note: Proving that the set of limit points is actually the entire interval $[-1, 1]$ requires the fact that $\{n \pmod{2\pi}\}$ is dense in $[0, 2\pi]$, but the B-W theorem provides immediate existence of convergence). ■

Theorem 2.4.8 (Cauchy's Convergence Criterion). *A sequence of real numbers is convergent if and only if it is a Cauchy sequence.*

Proof. ($\Rightarrow$) Already proved (Every convergent sequence is Cauchy).

($\Leftarrow$) Let $\{x_n\}$ be a Cauchy sequence.

By the previous theorem, $\{x_n\}$ is bounded. Since $\{x_n\}$ is bounded, by the **Bolzano-Weierstrass Theorem**, it has a subsequence $\{x_{n_k}\}$ that converges to some real number l .

We now show that the entire sequence $\{x_n\}$ converges to l . Let $\epsilon > 0$. Since $\{x_n\}$ is Cauchy, there exists N_1 such that:

$$|x_n - x_m| < \frac{\epsilon}{2} \quad \text{for all } n, m \geq N_1.$$

Since the subsequence $x_{n_k} \rightarrow l$, there exists N_2 such that:

$$|x_{n_k} - l| < \frac{\epsilon}{2} \quad \text{for all } n_k \geq N_2.$$

Let $N = \max(N_1, N_2)$. Choose any index k such that $n_k \geq N$. For any $n \geq N$, we have:

$$\begin{aligned} |x_n - l| &= |x_n - x_{n_k} + x_{n_k} - l| \\ &\leq |x_n - x_{n_k}| + |x_{n_k} - l|. \end{aligned}$$

Since $n \geq N$ and $n_k \geq N \geq N_1$, the Cauchy condition gives $|x_n - x_{n_k}| < \frac{\epsilon}{2}$. Since $n_k \geq N_2$, the limit condition gives $|x_{n_k} - l| < \frac{\epsilon}{2}$. Therefore:

$$|x_n - l| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \quad \text{for all } n \geq N.$$

Thus, $\{x_n\}$ converges to l . ■

Example 2.4.9. Let $\{x_n\}$ be defined by $x_n = 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}$. Use Cauchy's Criterion to prove that $\{x_n\}$ is convergent.

Solution. We must show that for any $\epsilon > 0$, there exists an N such that $|x_m - x_n| < \epsilon$ for $m > n \geq N$. Let $m > n$. Then:

$$|x_m - x_n| = \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \cdots + \frac{1}{m!}.$$

We observe that for $k \geq 1$, $k! \geq 2^{k-1}$. Specifically, $(n+k)! \geq (n+1)! \cdot (n+2) \cdots \geq 2^{n+k-1}$ is a loose bound, but a tighter, standard comparison is sufficient:

$$\frac{1}{(n+k)!} \leq \frac{1}{2^{n+k-1}}.$$

Alternatively, notice $\frac{1}{k!} \leq \frac{1}{2^{k-1}}$ for $k \geq 1$. Let's use a simpler inequality: for $k > n$, $\frac{1}{k!} \leq \frac{1}{2^{k-1}}$. Then:

$$\begin{aligned} |x_m - x_n| &< \frac{1}{2^n} + \frac{1}{2^{n+1}} + \cdots + \frac{1}{2^{m-1}} \\ &= \frac{1}{2^n} \left(1 + \frac{1}{2} + \cdots + \frac{1}{2^{m-n-1}} \right). \end{aligned}$$

The term in the parenthesis is a geometric sum bounded by its infinite sum: $\sum_{k=0}^{\infty} (\frac{1}{2})^k = 2$. Thus:

$$|x_m - x_n| < \frac{1}{2^n} \cdot 2 = \frac{1}{2^{n-1}}.$$

To satisfy the condition $< \epsilon$, we set $\frac{1}{2^{n-1}} < \epsilon$, which implies $2^{n-1} > 1/\epsilon$. By the Archimedean property (applied to powers of 2), such an N exists. Thus, $\{x_n\}$ is a Cauchy sequence and therefore convergent. ■

Example 2.4.10 (Contractive Sequence). Let $\{x_n\}$ be a sequence such that $|x_{n+1} - x_n| \leq r|x_n - x_{n-1}|$ for all $n \geq 2$, where $0 < r < 1$. Prove that $\{x_n\}$ is a Cauchy sequence.

Solution. First, we establish a bound for the difference of consecutive terms. By repeated application of the hypothesis:

$$|x_{n+1} - x_n| \leq r|x_n - x_{n-1}| \leq r^2|x_{n-1} - x_{n-2}| \leq \cdots \leq r^{n-1}|x_2 - x_1|.$$

Let $C = |x_2 - x_1|$. Then $|x_{n+1} - x_n| \leq Cr^{n-1}$.

Now, let $m > n$. We estimate $|x_m - x_n|$ using the Triangle Inequality:

$$\begin{aligned} |x_m - x_n| &= |(x_m - x_{m-1}) + (x_{m-1} - x_{m-2}) + \cdots + (x_{n+1} - x_n)| \\ &\leq |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \cdots + |x_{n+1} - x_n| \\ &\leq Cr^{m-2} + Cr^{m-3} + \cdots + Cr^{n-1} \\ &= Cr^{n-1}(1 + r + r^2 + \cdots + r^{m-n-1}). \end{aligned}$$

The geometric sum is strictly less than the sum of the infinite series $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ (since $0 < r < 1$). Thus:

$$|x_m - x_n| < Cr^{n-1} \cdot \frac{1}{1-r}.$$

Since $0 < r < 1$, $\lim_{n \rightarrow \infty} r^{n-1} = 0$. Therefore, for any $\epsilon > 0$, we can choose N sufficiently large such that $\frac{C}{1-r}r^{N-1} < \epsilon$. For all $m > n \geq N$, $|x_m - x_n| < \epsilon$. The sequence is Cauchy and hence convergent. ■

Exercise 2.4.11. 1. Prove directly from the definition that the sequence $x_n = \frac{n}{n+1}$ is a Cauchy sequence.

2. Show that the sequence $x_n = (-1)^n$ is **not** a Cauchy sequence by taking $\epsilon = 1$.

3. Consider the sequence defined by $x_n = \sum_{k=1}^n \frac{1}{k}$. Use Cauchy's Criterion to prove that this sequence diverges. (Hint: Consider the difference $|x_{2n} - x_n|$).

2.5 Monotonic Sequences

While the general definition of convergence involves arbitrary oscillations around a limit, a special class of sequences exhibits a much simpler behavior: they move in only one direction. These are known as monotonic sequences. Their study provides one of the most powerful tools in analysis for establishing convergence.

Definition 2.5.1 (Monotonic Sequences). A sequence $\{x_n\}$ is said to be **monotonic** if it is either monotonically increasing or monotonically decreasing.

1. A sequence is **monotonically increasing** (or non-decreasing) if

$$x_{n+1} \geq x_n \quad \text{for all } n \in \mathbb{N}.$$

2. A sequence is **strictly increasing** if $x_{n+1} > x_n$ for all n .

3. A sequence is **monotonically decreasing** (or non-increasing) if

$$x_{n+1} \leq x_n \quad \text{for all } n \in \mathbb{N}.$$

4. A sequence is **strictly decreasing** if $x_{n+1} < x_n$ for all n .

2.5.1 The Monotone Convergence Theorem

Intuitively, if a sequence constantly moves upwards but hits a ceiling (an upper bound), it must crowd together against that ceiling. This intuition is formalized in the Monotone Convergence Theorem, which relies heavily on the **Completeness Axiom** (Supremum Property) of $\mathbb{R}$.

Theorem 2.5.2 (Monotone Convergence Theorem). *A monotonic sequence of real numbers is convergent if and only if it is bounded.*

1. If $\{x_n\}$ is non-decreasing and bounded above, it converges to its supremum.
2. If $\{x_n\}$ is non-increasing and bounded below, it converges to its infimum.

Proof. We prove the first case. Let $\{x_n\}$ be a non-decreasing sequence that is bounded above. Let $S = \{x_n : n \in \mathbb{N}\}$ be the range set of the sequence. Since the sequence is bounded above, the set S is bounded above. By the **Completeness Axiom**, S has a least upper bound in $\mathbb{R}$. Let

$$u = \sup S.$$

We claim that $\lim_{n \rightarrow \infty} x_n = u$.

Let $\epsilon > 0$ be given. Since u is the supremum, $u - \epsilon$ is not an upper bound for S . Therefore, there exists at least one term in the sequence, say x_N , such that

$$x_N > u - \epsilon.$$

Since the sequence is non-decreasing, for all $n \geq N$, we have $x_n \geq x_N$. Combining inequalities:

$$u - \epsilon < x_N \leq x_n \leq u < u + \epsilon.$$

Note that $x_n \leq u$ because u is an upper bound. Thus, for all $n \geq N$:

$$u - \epsilon < x_n < u + \epsilon \implies |x_n - u| < \epsilon.$$

Therefore, the sequence converges to u .

The proof for the non-increasing case is analogous, utilizing the infimum. ■

Corollary 2.5.3. *A monotonic sequence diverges if and only if it is unbounded.*

- If $\{x_n\}$ is non-decreasing and unbounded, $\lim_{n \rightarrow \infty} x_n = +\infty$.
- If $\{x_n\}$ is non-increasing and unbounded, $\lim_{n \rightarrow \infty} x_n = -\infty$.

2.5.2 Application: Recursive Sequences

Monotonicity is particularly useful for sequences defined recursively where a direct formula for the limit is difficult to deduce without first proving existence.

Example 2.5.4. Let $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$. Prove that $\{x_n\}$ converges and find its limit.

Solution. **1. Boundedness:** We claim $x_n < 2$ for all n . Base case: $x_1 = 1 < 2$. Inductive step: Assume $x_k < 2$. Then

$$x_{k+1} = \sqrt{2 + x_k} < \sqrt{2 + 2} = \sqrt{4} = 2.$$

Thus, by induction, the sequence is bounded above by 2.

2. Monotonicity: We examine $x_{n+1}^2 - x_n^2$:

$$x_{n+1}^2 - x_n^2 = (2 + x_n) - x_n^2 = -(x_n^2 - x_n - 2) = -(x_n - 2)(x_n + 1).$$

Since $1 \leq x_n < 2$, the term $(x_n - 2)$ is negative and $(x_n + 1)$ is positive. Thus the product $-(x_n - 2)(x_n + 1)$ is positive. Therefore, $x_{n+1}^2 > x_n^2 \implies x_{n+1} > x_n$. The sequence is strictly increasing.

3. Convergence: Since $\{x_n\}$ is increasing and bounded above, by the Monotone Convergence Theorem, it converges to a limit l .

4. Finding the Limit: Taking the limit of the recursion relation:

$$\lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} \sqrt{2 + x_n}$$

$$l = \sqrt{2 + l} \implies l^2 = 2 + l \implies l^2 - l - 2 = 0.$$

Factoring yields $(l - 2)(l + 1) = 0$. The roots are $l = 2$ and $l = -1$. Since $x_n > 0$, the limit must be positive. Thus, $\lim_{n \rightarrow \infty} x_n = 2$. ■

2.6 The Nested Interval Theorem

The geometric manifestation of the Completeness of $\mathbb{R}$ is found in the behavior of nested intervals. This concept essentially states that if you "zoom in" infinitely on the real line with closed intervals, you will not fall through a hole; you will capture a point.

Definition 2.6.1 (Nested Sequence of Intervals). A sequence of intervals $\{I_n\}$ is called **nested** if

$$I_{n+1} \subseteq I_n \quad \text{for all } n \in \mathbb{N}.$$

That is, $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$.

Cantor's Intersection Theorem

Theorem 2.6.2 (Nested Interval Theorem). Let $\{I_n\}$ be a sequence of closed and bounded intervals, where $I_n = [a_n, b_n]$. If the sequence is nested ($I_{n+1} \subseteq I_n$), then the intersection

$$\bigcap_{n=1}^{\infty} I_n$$

is non-empty. Furthermore, if the lengths of the intervals converge to zero (i.e., $\lim_{n \rightarrow \infty} (b_n - a_n) = 0$), then the intersection consists of exactly one point.

Proof. Since the intervals are nested, we have the containment $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$. This implies inequality relations for the endpoints:

$$a_1 \leq a_2 \leq a_3 \leq \dots \leq a_n \leq \dots \leq b_n \leq \dots \leq b_2 \leq b_1.$$

Analysis of Endpoints: The sequence of left endpoints $\{a_n\}$ is monotonically increasing. Also, every b_m is an upper bound for the set of all a_n 's. Specifically, $a_n \leq b_1$ for all n . Thus, $\{a_n\}$ is

increasing and bounded above. By the Monotone Convergence Theorem, it converges to a limit $\alpha = \sup\{a_n\}$.

Similarly, the sequence of right endpoints $\{b_n\}$ is monotonically decreasing and bounded below (by a_1). Thus, it converges to a limit $\beta = \inf\{b_n\}$.

Non-Empty Intersection: Since $a_n \leq b_n$ for all n , taking limits yields $\alpha \leq \beta$. For any n , we have $a_n \leq \alpha$ (since α is sup) and $\beta \leq b_n$ (since β is inf). Therefore, for any x such that $\alpha \leq x \leq \beta$, we have

$$a_n \leq \alpha \leq x \leq \beta \leq b_n.$$

This implies $x \in [a_n, b_n]$ for all n . Thus, the interval $[\alpha, \beta]$ lies in the intersection $\cap I_n$. Since $\alpha \leq \beta$, this set is non-empty.

Uniqueness: Suppose $\lim_{n \rightarrow \infty} (b_n - a_n) = 0$. Since $\lim a_n = \alpha$ and $\lim b_n = \beta$, we have:

$$\beta - \alpha = \lim_{n \rightarrow \infty} b_n - \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (b_n - a_n) = 0.$$

Therefore, $\alpha = \beta$. The intersection contains exactly one point $\xi = \alpha = \beta$. ■

Remark 2.6.3. The condition that the intervals are **closed** is essential. Consider the nested open intervals $I_n = (0, \frac{1}{n})$. The intersection $\bigcap_{n=1}^{\infty} (0, \frac{1}{n})$ is empty. For any $x > 0$, by the Archimedean property, there exists n such that $\frac{1}{n} < x$, so $x \notin I_n$. Similarly, the condition of **boundedness** is necessary. Consider $I_n = [n, \infty)$. The intersection is empty.

Example 2.6.4. Let $I_n = [0, \frac{1}{n}]$. Prove that $\bigcap_{n=1}^{\infty} I_n = \{0\}$.

Solution. We define the intervals $I_n = [a_n, b_n]$ where $a_n = 0$ and $b_n = \frac{1}{n}$.

1. **Nested Condition:** Since $n < n+1$, we have $\frac{1}{n+1} < \frac{1}{n}$. Also $0 = 0$. Thus $[0, \frac{1}{n+1}] \subset [0, \frac{1}{n}]$. The intervals are nested.
2. **Length limit:** The length of I_n is $b_n - a_n = \frac{1}{n}$. Clearly $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

By the **Nested Interval Theorem**, the intersection contains exactly one point. Since $0 \in I_n$ for every n , the point 0 is in the intersection. Suppose there is another point $x > 0$ in the intersection. By the Archimedean property, there exists $k \in \mathbb{N}$ such that $\frac{1}{k} < x$. This implies $x \notin [0, \frac{1}{k}] = I_k$. Thus $x \notin \cap I_n$. Therefore, the intersection is the singleton set $\{0\}$. ■

- Exercise 2.6.5.**
1. Consider the sequence defined by $x_n = \left(1 + \frac{1}{n}\right)^n$. Show that this sequence is monotonically increasing and bounded above. (The limit of this sequence is Euler's number e).
 2. Let $\{x_n\}$ be a bounded sequence. Let $S_n = \sup\{x_k : k \geq n\}$. Prove that the sequence $\{S_n\}$ is monotonically decreasing and converges to $\limsup x_n$.
 3. Give an example of a nested sequence of closed intervals whose intersection is empty. (Hint: Consider unbounded intervals).

Chapter 3

Infinite Series: Convergence and Divergence

In the previous unit, we studied sequences, which are ordered lists of numbers. In this unit, we study the sum of the terms of a sequence. An infinite series represents the attempt to add infinitely many numbers together.

3.1 Infinite Series

Definition 3.1.1 (Infinite Series). Let $\{u_n\}$ be a sequence of real numbers. An expression of the form

$$u_1 + u_2 + u_3 + \cdots + u_n + \cdots$$

is called an **infinite series**. It is denoted by the symbol

$$\sum_{n=1}^{\infty} u_n \quad \text{or simply} \quad \sum u_n.$$

The numbers $u_1, u_2, \dots$ are called the **terms** of the series.

Sequence of Partial Sums

Since it is impossible to physically add infinite numbers, we define the "sum" of a series through the concept of partial sums.

Definition 3.1.2 (Partial Sums). Consider the series $\sum u_n$. We construct a new sequence $\{S_n\}$ defined as follows:

$$\begin{aligned} S_1 &= u_1 \\ S_2 &= u_1 + u_2 \\ S_3 &= u_1 + u_2 + u_3 \\ &\vdots \\ S_n &= u_1 + u_2 + \cdots + u_n = \sum_{k=1}^n u_k \end{aligned}$$

The number S_n is called the **n -th partial sum** of the series. The sequence $\{S_n\}$ is called the **sequence of partial sums** of the series.

Convergence and Divergence

The behavior of the infinite series $\sum u_n$ is defined entirely by the behavior of the sequence $\{S_n\}$.

Definition 3.1.3 (Convergence of a Series). The series $\sum u_n$ is said to **converge** (or be convergent) if the sequence of partial sums $\{S_n\}$ converges to a finite real number S . If

$$\lim_{n \rightarrow \infty} S_n = S,$$

then we say that S is the **sum** of the infinite series, and we write:

$$\sum_{n=1}^{\infty} u_n = S.$$

Definition 3.1.4 (Divergence of a Series). The series $\sum u_n$ is said to **diverge** (or be divergent) if the sequence of partial sums $\{S_n\}$ does not converge to a finite limit. There are two main ways a series can diverge:

1. **Proper Divergence:** If $\lim_{n \rightarrow \infty} S_n = +\infty$ or $\lim_{n \rightarrow \infty} S_n = -\infty$.
2. **Oscillation:** If the sequence $\{S_n\}$ is bounded or unbounded but does not tend to a unique limit (e.g., S_n oscillates between two values).

In the following examples, we determine the nature of the series by explicitly finding a formula for S_n and taking the limit $n \rightarrow \infty$.

3.1.1 Geometric Series

Example 3.1.5. Discuss the convergence of the geometric series:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^{n-1}} + \cdots$$

Solution. Here, the terms constitute a geometric progression with first term $a = 1$ and common ratio $r = \frac{1}{2}$. We form the n -th partial sum:

$$S_n = 1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}}.$$

Using the standard algebraic formula for the sum of a finite geometric progression ($S_n = \frac{a(1-r^n)}{1-r}$ for $r \neq 1$):

$$S_n = \frac{1 \cdot \left(1 - \left(\frac{1}{2}\right)^n\right)}{1 - \frac{1}{2}} = \frac{1 - \frac{1}{2^n}}{\frac{1}{2}} = 2 \left(1 - \frac{1}{2^n}\right).$$

Now we take the limit as $n \rightarrow \infty$. Since $\frac{1}{2^n} \rightarrow 0$:

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} 2 \left(1 - \frac{1}{2^n}\right) = 2(1 - 0) = 2.$$

Since the limit of partial sums is finite (2), the series **converges**, and its sum is 2. ■

Example 3.1.6. Discuss the series $\sum_{n=0}^{\infty} (-1)^n = 1 - 1 + 1 - 1 + \cdots$

Solution. We examine the sequence of partial sums $\{S_n\}$:

- $S_1 = 1$
- $S_2 = 1 - 1 = 0$
- $S_3 = 1 - 1 + 1 = 1$
- $S_4 = 1 - 1 + 1 - 1 = 0$

In general:

$$S_n = \begin{cases} 1 & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

As $n \rightarrow \infty$, S_n does not approach a unique limit; it oscillates finitely between 1 and 0. Therefore, the series **diverges** (specifically, it oscillates finitely). ■

Telescoping Series

A Telescoping series is one where the partial sums can be simplified by cancelling middle terms, allowing us to find S_n directly.

Example 3.1.7. Find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

Solution. The general term is $u_n = \frac{1}{n(n+1)}$. We use the method of partial fractions to rewrite u_n :

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Now, we write out the partial sum S_n :

$$\begin{aligned} S_n &= u_1 + u_2 + u_3 + \cdots + u_n \\ S_n &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right). \end{aligned}$$

Observe the pattern of cancellation: the second part of each parenthesis cancels with the first part of the next parenthesis.

$$S_n = 1 + \underbrace{\left(-\frac{1}{2} + \frac{1}{2}\right)}_0 + \underbrace{\left(-\frac{1}{3} + \frac{1}{3}\right)}_0 + \cdots + \underbrace{\left(-\frac{1}{n} + \frac{1}{n}\right)}_0 - \frac{1}{n+1}.$$

The sum collapses to the first term and the very last component:

$$S_n = 1 - \frac{1}{n+1}.$$

Now we determine convergence by taking the limit:

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1 - 0 = 1.$$

The series **converges** to 1. ■

Example 3.1.8. Show that the series $\sum_{n=1}^{\infty} \log\left(\frac{n}{n+1}\right)$ diverges.

Solution. The general term is $u_n = \log\left(\frac{n}{n+1}\right) = \log n - \log(n+1)$. We form the partial sum S_n :

$$\begin{aligned} S_n &= (\log 1 - \log 2) + (\log 2 - \log 3) + \cdots + (\log n - \log(n+1)) \\ &= \log 1 - \log(n+1) \\ &= 0 - \log(n+1) = -\log(n+1). \end{aligned}$$

Taking the limit:

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} [-\log(n+1)] = -\infty.$$

Since the limit of partial sums is infinite, the series **diverges** to $-\infty$. ■

3.2 General Theorems on Convergence of Series

3.2.1 The Necessary Condition for Convergence

One of the most common misconceptions in the study of series is that if the terms get smaller, the series must converge. While the terms *must* approach zero for convergence to occur, this condition alone is not sufficient.

Theorem 3.2.1 (Necessary Condition for Convergence). *If an infinite series $\sum_{n=1}^{\infty} u_n$ is convergent, then the limit of its n -th term as n approaches infinity is zero. That is:*

$$\lim_{n \rightarrow \infty} u_n = 0.$$

Proof. Let $\{S_n\}$ denote the sequence of partial sums of the series $\sum u_n$. By definition,

$$S_n = u_1 + u_2 + \cdots + u_n$$

and

$$S_{n-1} = u_1 + u_2 + \cdots + u_{n-1}.$$

Therefore, for $n \geq 2$, we can express the n -th term as:

$$u_n = S_n - S_{n-1}.$$

Since the series $\sum u_n$ is convergent, the sequence of partial sums $\{S_n\}$ converges to a finite limit, say S . Consequently, the shifted sequence $\{S_{n-1}\}$ also converges to S as $n \rightarrow \infty$.

Taking the limit of the expression for u_n :

$$\begin{aligned} \lim_{n \rightarrow \infty} u_n &= \lim_{n \rightarrow \infty} (S_n - S_{n-1}) \\ &= \lim_{n \rightarrow \infty} S_n - \lim_{n \rightarrow \infty} S_{n-1} \\ &= S - S \\ &= 0. \end{aligned}$$

Thus, if the series converges, the n -th term must tend to zero. ■

Remark 3.2.2 (Important Warning). The converse of this theorem is **false**. If $\lim_{n \rightarrow \infty} u_n = 0$, the series $\sum u_n$ may or may not converge.

- **Example:** The Harmonic Series $\sum \frac{1}{n}$. Here, $u_n = \frac{1}{n} \rightarrow 0$, yet the series diverges to infinity.

Solution. By definition, the series converges or diverges based on the behavior of its sequence of partial sums, denoted by $\{S_n\}$, where:

$$S_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

Since every term $\frac{1}{n}$ is positive, the sequence $\{S_n\}$ is strictly monotonically increasing. Therefore, to prove divergence, it suffices to show that $\{S_n\}$ is unbounded.

We consider the subsequence of partial sums with indices that are powers of 2, specifically S_{2^k} .

$$\begin{aligned} S_2 &= 1 + \frac{1}{2} \\ S_4 &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) \\ &> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) = 1 + \frac{2}{2} \\ S_8 &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) \\ &> 1 + \frac{1}{2} + \frac{1}{2} + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) = 1 + \frac{3}{2}. \end{aligned}$$

By induction, for any integer k :

$$S_{2^k} > 1 + \frac{k}{2}.$$

As $n \rightarrow \infty$, we can choose k such that $2^k \leq n$, and since k grows arbitrarily large, the expression $1 + \frac{k}{2}$ tends to infinity. Consequently, $\lim_{n \rightarrow \infty} S_n = \infty$.

Since the limit of the partial sums is infinite, the series diverges. ■

This theorem is most useful in its contrapositive form as a **Test for Divergence**:

Corollary 3.2.3 (The Divergence Test). *If $\lim_{n \rightarrow \infty} u_n \neq 0$ (or if the limit does not exist), then the series $\sum u_n$ is divergent.*

3.3 Cauchy's General Principle of Convergence for Series

Just as with sequences, Cauchy's criterion allows us to establish the convergence of a series without knowing the value of its sum. It translates the behavior of the "tail" of the series.

Theorem 3.3.1 (Cauchy's Criterion for Series). *The infinite series $\sum_{n=1}^{\infty} u_n$ converges if and only if for every $\epsilon > 0$, there exists a positive integer N such that for all $m > n \geq N$:*

$$|S_m - S_n| = |u_{n+1} + u_{n+2} + \cdots + u_m| < \epsilon.$$

Proof. Let $S_n = \sum_{k=1}^n u_k$ denote the n -th partial sum. The series $\sum u_n$ converges if and only if the sequence $\{S_n\}$ converges. By Cauchy's criterion for sequences, convergence is equivalent to the condition that for every $\epsilon > 0$, there exists an integer N such that $|S_m - S_n| < \epsilon$ for all $m > n \geq N$.

Since $S_m - S_n = u_{n+1} + u_{n+2} + \cdots + u_m$, this condition is identical to requiring

$$|u_{n+1} + u_{n+2} + \cdots + u_m| < \epsilon$$

for all $m > n \geq N$, which completes the proof. ■

Example 3.3.2 (Divergence of the Harmonic Series). Use Cauchy's General Principle to prove that the series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Solution. We proceed by contradiction. Assume the series converges. By Cauchy's General Principle, for any $\epsilon > 0$, there exists an integer N such that for all $m > n \geq N$,

$$\left| \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{m} \right| < \epsilon.$$

Let us choose a specific value for ϵ , say $\epsilon = \frac{1}{2}$. Then there must exist a specific N satisfying the condition. Since the condition holds for *any* $m > n \geq N$, let us choose $n = N$ and $m = 2N$. The sum of terms becomes:

$$S_{2N} - S_N = \frac{1}{N+1} + \frac{1}{N+2} + \cdots + \frac{1}{2N}.$$

There are exactly N terms in this sum. Notice that the smallest term is the last one, $\frac{1}{2N}$. Therefore, replacing every term with $\frac{1}{2N}$ yields a smaller value:

$$S_{2N} - S_N > \underbrace{\frac{1}{2N} + \frac{1}{2N} + \cdots + \frac{1}{2N}}_{N \text{ times}} = N \cdot \frac{1}{2N} = \frac{1}{2}.$$

Thus, $|S_{2N} - S_N| > \frac{1}{2}$. This contradicts the Cauchy requirement that the difference must be less than $\frac{1}{2}$. Therefore, the series diverges. ■

Example 3.3.3 (Convergence of the Inverse Squares). Show that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

Solution. Let $\epsilon > 0$ be given. We examine the partial sum difference for $m > n$:

$$|S_m - S_n| = \frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \cdots + \frac{1}{m^2}.$$

We utilize the inequality $\frac{1}{k^2} < \frac{1}{k(k-1)}$ for $k \geq 2$. Applying this to each term:

$$\begin{aligned} |S_m - S_n| &< \frac{1}{n(n+1)} + \frac{1}{(n+1)(n+2)} + \cdots + \frac{1}{(m-1)m} \\ &= \left(\frac{1}{n} - \frac{1}{n+1}\right) + \left(\frac{1}{n+1} - \frac{1}{n+2}\right) + \cdots + \left(\frac{1}{m-1} - \frac{1}{m}\right). \end{aligned}$$

This is a telescoping sum. All intermediate terms cancel:

$$|S_m - S_n| < \frac{1}{n} - \frac{1}{m}.$$

Since $\frac{1}{m} > 0$, we have $|S_m - S_n| < \frac{1}{n}$. To satisfy the Cauchy condition, we need $\frac{1}{n} < \epsilon$, which implies $n > \frac{1}{\epsilon}$. Choose an integer $N > \frac{1}{\epsilon}$. Then for all $m > n \geq N$:

$$|S_m - S_n| < \frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

Thus, the series converges. ■

Example 3.3.4 (Divergence of $1/\sqrt{n}$). Prove that the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ diverges.

Solution. We assume the series converges. Then, for $\epsilon = \frac{1}{2}$, there must exist an N such that for all $m > n \geq N$, the sum of the terms is less than $\frac{1}{2}$. Let $n = N$ and $m = 2N$. Consider the sum:

$$\Sigma = \frac{1}{\sqrt{N+1}} + \frac{1}{\sqrt{N+2}} + \cdots + \frac{1}{\sqrt{2N}}.$$

In this sum of N terms, the smallest term is $\frac{1}{\sqrt{2N}}$. Thus:

$$\Sigma > N \cdot \frac{1}{\sqrt{2N}} = \frac{N}{\sqrt{2}\sqrt{N}} = \frac{\sqrt{N}}{\sqrt{2}}.$$

Since N is a natural number ($N \geq 1$), $\frac{\sqrt{N}}{\sqrt{2}} \geq \frac{1}{\sqrt{2}} \approx 0.707$. Consequently, $\Sigma > 0.707 > \frac{1}{2}$. This contradicts the requirement that the sum must be less than $\frac{1}{2}$. Therefore, the series diverges. ■

Example 3.3.5 (Convergence of a Geometric Series). Show that $\sum_{n=1}^{\infty} \frac{1}{2^n}$ converges using Cauchy's Criterion.

Solution. Let $\epsilon > 0$. Consider the sum of terms from $n+1$ to m :

$$|S_m - S_n| = \frac{1}{2^{n+1}} + \frac{1}{2^{n+2}} + \cdots + \frac{1}{2^m}.$$

We factor out the first term, $\frac{1}{2^{n+1}}$:

$$|S_m - S_n| = \frac{1}{2^{n+1}} \left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{m-n-1}}\right).$$

The expression in the parentheses is a finite geometric series with ratio $1/2$. The sum of this finite series is strictly less than the sum of the infinite series $\sum_{k=0}^{\infty} (\frac{1}{2})^k$, which equals 2. Therefore:

$$|S_m - S_n| < \frac{1}{2^{n+1}} \cdot 2 = \frac{1}{2^n}.$$

To satisfy $|S_m - S_n| < \epsilon$, it suffices to have $\frac{1}{2^n} < \epsilon$, or $2^n > \frac{1}{\epsilon}$. By the Archimedean property, there exists N such that $2^N > \frac{1}{\epsilon}$. Thus, for all $m > n \geq N$, the condition is met, and the series converges. ■

The following theorems rely on the linearity of limits applied to the partial sums.

Theorem 3.3.6 (Algebra of Convergent Series). *Let $\sum u_n$ and $\sum v_n$ be two convergent series with sums S and T respectively, and let $c \in \mathbb{R}$. Then:*

1. **Summation:** *The series $\sum(u_n + v_n)$ converges and its sum is $S + T$.*
2. **Difference:** *The series $\sum(u_n - v_n)$ converges and its sum is $S - T$.*
3. **Scalar Multiplication:** *The series $\sum(c \cdot u_n)$ converges and its sum is $c \cdot S$.*

Proof. Let $\{S_n\}$ and $\{T_n\}$ denote the sequences of partial sums for $\sum u_n$ and $\sum v_n$ respectively. By hypothesis, $\lim_{n \rightarrow \infty} S_n = S$ and $\lim_{n \rightarrow \infty} T_n = T$.

Summation: Due to the linearity of finite sums, the n -th partial sum of the series $\sum(u_n + v_n)$ is $S_n + T_n$. Applying the algebra of limits for sequences:

$$\lim_{n \rightarrow \infty} (S_n + T_n) = \lim_{n \rightarrow \infty} S_n + \lim_{n \rightarrow \infty} T_n = S + T.$$

Thus, the series converges to $S + T$. The proof for the difference series follows identically.

Scalar Multiplication: The n -th partial sum of $\sum(c \cdot u_n)$ is $\sum_{k=1}^n cu_k = c \sum_{k=1}^n u_k = cS_n$. By the scalar property of limits:

$$\lim_{n \rightarrow \infty} (cS_n) = c \lim_{n \rightarrow \infty} S_n = cS.$$

Thus, the series converges to cS . ■

Theorem 3.3.7 (Effect of Removing Terms). *The convergence or divergence of an infinite series is not affected by the addition, removal, or alteration of a finite number of terms.*

Proof. Let $\sum u_n$ be the original series and $\sum v_n$ be the modified series such that $u_n = v_n$ for all $n > k$. Let $\{S_n\}$ and $\{T_n\}$ denote their respective partial sums. For any $n > k$, the difference between the partial sums is a constant finite value C determined by the sum of the differences of the first k terms:

$$T_n = S_n + \sum_{i=1}^k (v_i - u_i) = S_n + C.$$

Since C is a constant, the limit $\lim_{n \rightarrow \infty} T_n$ exists if and only if $\lim_{n \rightarrow \infty} S_n$ exists. Thus, the convergence behavior is identical. ■

Theorem 3.3.8 (Grouping of Terms). *If a series $\sum u_n$ converges to S , then any series obtained by inserting parentheses (grouping terms) also converges to S .*

Proof. Let $\{S_n\}$ be the sequence of partial sums of $\sum u_n$. By hypothesis, $\lim_{n \rightarrow \infty} S_n = S$. When parentheses are inserted to group terms, the k -th partial sum of the new "grouped" series, denoted T_k , is identical to the partial sum S_{n_k} of the original series for some index n_k (where $n_1 < n_2 < \dots$ corresponds to the positions of the closing parentheses). Thus, the sequence $\{T_k\}$ is simply a subsequence of $\{S_n\}$. Since the original sequence converges to S , every subsequence must also converge to S . ■

Theorem 3.3.9 (Boundedness of Positive Term Series). *If $u_n \geq 0$ for all n , then the series $\sum u_n$ converges if and only if its sequence of partial sums $\{S_n\}$ is bounded above.*

Proof. Since each term u_n is non-negative, the sequence of partial sums is non-decreasing:

$$S_{n+1} = S_n + u_{n+1} \geq S_n.$$

The sequence $\{S_n\}$ is therefore monotonic. By the **Monotone Convergence Theorem**, a monotonic sequence converges if and only if it is bounded. Since it is bounded below by u_1 , convergence is equivalent to being bounded above. ■

Example 3.3.10. Prove that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

Solution. Since every term $u_n = \frac{1}{n^2}$ is positive, the series converges if and only if its sequence of partial sums $\{S_n\}$ is bounded above.

We attempt to bound S_n by comparing it to a known convergent sum. We observe that for $n \geq 2$:

$$\frac{1}{n^2} < \frac{1}{n(n-1)}.$$

Using partial fractions, $\frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$. Now, consider the partial sum S_n :

$$\begin{aligned} S_n &= 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \\ &< 1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(n-1)n} \\ &= 1 + \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right). \end{aligned}$$

The terms in the parentheses telescope:

$$S_n < 1 + 1 - \frac{1}{n} = 2 - \frac{1}{n} < 2.$$

Thus, $S_n < 2$ for all n . Since $\{S_n\}$ is monotonic (increasing) and bounded above by 2, the series converges. ■

Example 3.3.11. Prove that the series $\sum_{n=1}^{\infty} \frac{1}{n!}$ converges.

Solution. The terms are positive. We observe that for $n \geq 2$:

$$n! = 1 \cdot 2 \cdot 3 \cdots n \geq 1 \cdot 2 \cdot 2 \cdots 2 = 2^{n-1}.$$

Therefore, $\frac{1}{n!} \leq \frac{1}{2^{n-1}}$. The partial sum S_n satisfies:

$$S_n = 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} \leq 1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}}.$$

The right-hand side is a finite geometric series with sum less than 2. Thus, $\{S_n\}$ is bounded above, implies the series converges. ■

Example 3.3.12. Find the sum of the series $\sum_{n=1}^{\infty} \left(\frac{3}{4^n} - \frac{2}{5^n}\right)$.

Solution. We treat this as the difference of two geometric series. Let $u_n = \frac{1}{4^n}$ and $v_n = \frac{1}{5^n}$. Both are geometric series with $|r| < 1$, so they converge.

1. First Series: $\sum \frac{3}{4^n} = 3 \sum (\frac{1}{4})^n$. Using the sum formula $\frac{a}{1-r}$ where first term $a = 1/4$ and $r = 1/4$:

$$\sum_{n=1}^{\infty} \frac{1}{4^n} = \frac{1/4}{1 - 1/4} = \frac{1/4}{3/4} = \frac{1}{3}.$$

Thus, $\sum \frac{3}{4^n} = 3 \cdot \frac{1}{3} = 1$.

2. Second Series: $\sum \frac{2}{5^n} = 2 \sum (\frac{1}{5})^n$. First term $a = 1/5$, $r = 1/5$:

$$\sum_{n=1}^{\infty} \frac{1}{5^n} = \frac{1/5}{1 - 1/5} = \frac{1/5}{4/5} = \frac{1}{4}.$$

Thus, $\sum \frac{2}{5^n} = 2 \cdot \frac{1}{4} = \frac{1}{2}$.

By the theorem on the **Algebra of Convergent Series**, the series converges to the difference of the sums:

$$S = 1 - \frac{1}{2} = \frac{1}{2}.$$

Example 3.3.13. Discuss the convergence of $\sum_{n=10}^{\infty} \frac{1}{n(n+1)}$.

Solution. The convergence of a series depends only on the "tail" of the series. We know from previous examples that the telescoping series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges to 1. According to the theorem on the **Effect of Removing Terms**, removing the first 9 terms does not alter the convergence. Thus, this series converges.

To find the *value*, we subtract the first 9 terms from the total sum:

$$\sum_{n=10}^{\infty} u_n = \left(\sum_{n=1}^{\infty} u_n \right) - \left(\sum_{n=1}^9 u_n \right).$$

The sum of the first 9 terms is $1 - \frac{1}{10} = \frac{9}{10}$. Thus, the sum is $1 - \frac{9}{10} = \frac{1}{10}$. ■

Example 3.3.14. Consider the series $\sum_{n=1}^{\infty} (-1)^{n+1} = 1 - 1 + 1 - 1 + \dots$. Show that grouping terms leads to an apparent contradiction if the series is not convergent.

Solution. Let S be the sum.

1. Grouping Case A:

$$(1 - 1) + (1 - 1) + (1 - 1) + \dots = 0 + 0 + 0 + \dots = 0.$$

This suggests $S = 0$.

2. Grouping Case B:

$$1 - (1 - 1) - (1 - 1) - \dots = 1 - 0 - 0 - \dots = 1.$$

This suggests $S = 1$.

Since the "sum" depends on how we group the terms, the sequence of partial sums does not converge to a unique limit. This illustrates the remark: **Grouping preserves the sum only if the series is already known to be convergent**. Since this series diverges (oscillates), arbitrary grouping is invalid. ■

3.4 Series of Positive Terms

Theorem 3.4.1 (Fundamental Theorem for Positive Term Series). *If $\sum u_n$ is a series of positive terms (i.e., $u_n > 0$ for all n), then the series converges if and only if its sequence of partial sums $\{S_n\}$ is bounded above.*

Proof. Let $S_n = u_1 + u_2 + \dots + u_n$ be the n -th partial sum. Since $u_n > 0$ for all n , we have:

$$S_{n+1} = S_n + u_{n+1} > S_n.$$

Thus, $\{S_n\}$ is a strictly monotonically increasing sequence. By the **Monotone Convergence Theorem** (from Unit II), a monotonically increasing sequence converges if and only if it is bounded above. Consequently, the series $\sum u_n$ converges if and only if $\{S_n\}$ is bounded. ■

Theorem 3.4.2 (Comparison Test (First Type / Direct Comparison)). *Let $\sum u_n$ and $\sum v_n$ be two series of positive real numbers such that*

$$u_n \leq k \cdot v_n \quad \text{for all } n \geq m,$$

where k is a fixed positive number and m is a fixed natural number.

1. *If $\sum v_n$ converges, then $\sum u_n$ also converges.*

2. *If $\sum u_n$ diverges, then $\sum v_n$ also diverges.*

Proof. Let $S_n = \sum_{i=1}^n u_i$ and $T_n = \sum_{i=1}^n v_i$ denote the partial sums of the two series. Since convergence is unaffected by a finite number of terms, we may assume without loss of generality that the inequality holds for all $n \geq 1$ (i.e., $m = 1$). Thus, $u_n \leq kv_n$ for all n . Summing this inequality from 1 to n :

$$S_n \leq k \cdot T_n.$$

1. **Proof of Convergence:** Assume $\sum v_n$ converges. Then $\{T_n\}$ converges to a limit T , and since terms are positive, $T_n \leq T$ for all n . The inequality becomes $S_n \leq kT$. This shows that the sequence $\{S_n\}$ is bounded above by kT . Since $\sum u_n$ is a positive term series, by the **Fundamental Theorem**, it must converge.
2. **Proof of Divergence:** Assume $\sum u_n$ diverges. Since it is a positive term series, it must diverge to $+\infty$. Thus, $\lim_{n \rightarrow \infty} S_n = \infty$. From the inequality $S_n \leq kT_n$, we deduce $T_n \geq \frac{1}{k}S_n$. Taking the limit as $n \rightarrow \infty$, it follows that $T_n \rightarrow \infty$. Hence, $\sum v_n$ diverges. ■

Theorem 3.4.3 (Limit Form Comparison Test). *Let $\sum u_n$ and $\sum v_n$ be two series of positive terms. If*

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = l,$$

where l is a finite and non-zero number ($l \neq 0$), then both series converge or diverge together.

Proof. Since $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = l$ and $l > 0$, we can bound the ratio within a strictly positive interval. Let $\epsilon = \frac{l}{2} > 0$. By the definition of the limit, there exists a natural number N such that for all $n \geq N$:

$$\left| \frac{u_n}{v_n} - l \right| < \frac{l}{2}.$$

Opening the absolute value, we get:

$$l - \frac{l}{2} < \frac{u_n}{v_n} < l + \frac{l}{2} \implies \frac{l}{2} < \frac{u_n}{v_n} < \frac{3l}{2}.$$

Since $v_n > 0$, we can multiply through to obtain a double inequality:

$$\frac{l}{2}v_n < u_n < \frac{3l}{2}v_n \quad \text{for all } n \geq N.$$

We now apply the **Direct Comparison Test**:

1. **If $\sum v_n$ converges:** The right-hand inequality $u_n < (\frac{3l}{2})v_n$ implies that $\sum u_n$ converges.
2. **If $\sum v_n$ diverges:** The left-hand inequality $u_n > (\frac{l}{2})v_n$ (or equivalently $v_n < \frac{2}{l}u_n$) implies that $\sum u_n$ diverges.

The converse holds symmetrically by considering the limit of v_n/u_n which is $1/l$. Thus, the series behave identically. ■

3.4.1 Cauchy's Root Test

Theorem 3.4.4 (Cauchy's Root Test). *Let $\sum u_n$ be a series of positive terms. Let*

$$l = \lim_{n \rightarrow \infty} (u_n)^{1/n}.$$

1. *If $l < 1$, the series converges.*
2. *If $l > 1$, the series diverges.*
3. *If $l = 1$, the test is inconclusive (the series may converge or diverge).*

Proof of Cauchy's Root Test. Let $l = \lim_{n \rightarrow \infty} (u_n)^{1/n}$.

1. **Case $l < 1$ (Convergence):** Choose a real number r such that $l < r < 1$. By the definition of the limit, there exists an integer N such that for all $n \geq N$:

$$(u_n)^{1/n} < r \implies u_n < r^n.$$

Since $0 < r < 1$, the geometric series $\sum r^n$ converges. By the **Direct Comparison Test**, $\sum u_n$ converges.

2. **Case $l > 1$ (Divergence):** Since the limit is strictly greater than 1, there exists an integer N such that for all $n \geq N$:

$$(u_n)^{1/n} > 1 \implies u_n > 1.$$

Consequently, $\lim_{n \rightarrow \infty} u_n \neq 0$. By the **Divergence Test**, the series diverges. ■

Example 3.4.5 (Basic Algebraic Function). Test the convergence of the series:

$$\sum_{n=1}^{\infty} \left(\frac{3n+2}{5n-1} \right)^n$$

Solution. Let the general term be $u_n = \left(\frac{3n+2}{5n-1} \right)^n$. Applying the Root Test, we evaluate $\lim_{n \rightarrow \infty} (u_n)^{1/n}$:

$$L = \lim_{n \rightarrow \infty} \left[\left(\frac{3n+2}{5n-1} \right)^n \right]^{1/n} = \lim_{n \rightarrow \infty} \frac{3n+2}{5n-1}.$$

Dividing numerator and denominator by n :

$$L = \lim_{n \rightarrow \infty} \frac{3 + 2/n}{5 - 1/n} = \frac{3 + 0}{5 - 0} = \frac{3}{5}.$$

Since $L = \frac{3}{5} < 1$, the series **converges**. ■

Example 3.4.6 (Involving Standard Limits). Discuss the convergence of the series:

$$\sum_{n=1}^{\infty} \left(1 + \frac{1}{n} \right)^{-n^2}$$

Solution. Let $u_n = \left(1 + \frac{1}{n} \right)^{-n^2}$. We evaluate the n -th root:

$$(u_n)^{1/n} = \left[\left(1 + \frac{1}{n} \right)^{-n^2} \right]^{1/n} = \left(1 + \frac{1}{n} \right)^{-n}.$$

We can rewrite this expression as:

$$\left(1 + \frac{1}{n} \right)^{-n} = \frac{1}{\left(1 + 1/n \right)^n}.$$

Now, taking the limit as $n \rightarrow \infty$:

$$L = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + 1/n \right)^n} = \frac{1}{e}.$$

Since $e \approx 2.718$, we have $L = \frac{1}{e} < 1$. Therefore, by Cauchy's Root Test, the series **converges**. ■

Example 3.4.7 (Logarithmic Functions). Test the series for convergence:

$$\sum_{n=2}^{\infty} \frac{1}{(\ln n)^{n/2}}$$

Solution. Let $u_n = \frac{1}{(\ln n)^{n/2}}$. Applying the Root Test:

$$L = \lim_{n \rightarrow \infty} \left(\frac{1}{(\ln n)^{n/2}} \right)^{1/n} = \lim_{n \rightarrow \infty} \frac{1}{(\ln n)^{1/2}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{\ln n}}.$$

As $n \rightarrow \infty$, $\ln n \rightarrow \infty$, and consequently $\sqrt{\ln n} \rightarrow \infty$. Thus:

$$L = 0.$$

Since $L = 0 < 1$, the series **converges**. (Note: The Ratio Test would have been very difficult to apply here due to the $(\ln n)$ term). ■

Example 3.4.8 (Parameter Dependent Series). Test the convergence of the series for $x > 0$:

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2} x^n$$

Solution. Let $u_n = \left(\frac{n}{n+1} \right)^{n^2} x^n$. Calculating the limit of the n -th root:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} (u_n)^{1/n} \\ &= \lim_{n \rightarrow \infty} \left[\left(\frac{n}{n+1} \right)^{n^2} x^n \right]^{1/n} \\ &= \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n \cdot x \\ &= x \cdot \lim_{n \rightarrow \infty} \frac{1}{\left(\frac{n+1}{n} \right)^n} \\ &= x \cdot \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n} \right)^n}. \end{aligned}$$

Using the standard limit $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$:

$$L = \frac{x}{e}.$$

Now we analyze the convergence based on L :

1. **Converges** if $L < 1 \implies \frac{x}{e} < 1 \implies x < e$.
2. **Diverges** if $L > 1 \implies \frac{x}{e} > 1 \implies x > e$.
3. **Test Fails** if $L = 1 \implies x = e$.

Supplementary check for $x = e$: The general term becomes $u_n = \left(\frac{n}{n+1} \right)^{n^2} e^n$. Consider the limit of the term u_n .

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{e^n}{\left(1 + \frac{1}{n} \right)^{n^2}}.$$

This limit is non-zero (in fact, it tends to $\sqrt{e}$ using more advanced asymptotic analysis like Taylor series). Since $u_n \not\rightarrow 0$, the series diverges for $x = e$.

Conclusion: Converges for $x < e$, diverges for $x \geq e$. ■

3.4.2 D'Alembert's Ratio Test

Theorem 3.4.9 (D'Alembert's Ratio Test). Let $\sum u_n$ be a series of positive terms. Let

$$l = \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n}.$$

1. If $l < 1$, the series converges.
2. If $l > 1$, the series diverges.
3. If $l = 1$, the test is inconclusive (the series may converge or diverge).

Proof of D'Alembert's Ratio Test. Let $l = \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n}$.

1. **Case $l < 1$ (Convergence):** Choose r such that $l < r < 1$. There exists N such that for all $n \geq N$, $\frac{u_{n+1}}{u_n} < r$. For any $n > N$, we can write:

$$u_n = \frac{u_n}{u_{n-1}} \cdot \frac{u_{n-1}}{u_{n-2}} \cdots \frac{u_{N+1}}{u_N} \cdot u_N < r \cdot r \cdots r \cdot u_N = u_N \cdot r^{n-N}.$$

Thus, $u_n < \left(\frac{u_N}{r^N}\right) r^n$. Since $\sum r^n$ converges (geometric series with $r < 1$), $\sum u_n$ converges by comparison.

2. **Case $l > 1$ (Divergence):** There exists N such that for all $n \geq N$, $\frac{u_{n+1}}{u_n} > 1$. This implies $u_{n+1} > u_n$. Thus, the sequence of terms $\{u_n\}$ is increasing for $n \geq N$ and cannot approach 0. The series diverges.

■

Example 3.4.10 (Basic Application involving Factorials). Test the convergence of the series:

$$\sum_{n=1}^{\infty} \frac{2^n}{n!}$$

Solution. Let the general term be $u_n = \frac{2^n}{n!}$. Then the next term is $u_{n+1} = \frac{2^{n+1}}{(n+1)!}$.

We form the ratio $\frac{u_{n+1}}{u_n}$:

$$\begin{aligned} \frac{u_{n+1}}{u_n} &= \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} \\ &= \frac{2^n \cdot 2}{(n+1) \cdot n!} \cdot \frac{n!}{2^n} \\ &= \frac{2}{n+1}. \end{aligned}$$

Now, we take the limit as $n \rightarrow \infty$:

$$L = \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0.$$

Since $L = 0 < 1$, by D'Alembert's Ratio Test, the series **converges**.

■

Example 3.4.11 (Polynomials and Exponentials). Discuss the convergence of the series:

$$\sum_{n=1}^{\infty} \frac{n^2 + 1}{3^n}$$

Solution. Let $u_n = \frac{n^2 + 1}{3^n}$. Then $u_{n+1} = \frac{(n+1)^2 + 1}{3^{n+1}}$.

Consider the limit of the ratio:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} \\ &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)^2 + 1}{3^{n+1}} \cdot \frac{3^n}{n^2 + 1} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{3} \cdot \frac{n^2 + 2n + 2}{n^2 + 1} \right) \\ &= \frac{1}{3} \lim_{n \rightarrow \infty} \left(\frac{1 + \frac{2}{n} + \frac{2}{n^2}}{1 + \frac{1}{n^2}} \right). \end{aligned}$$

As $n \rightarrow \infty$, the terms with n in the denominator vanish.

$$L = \frac{1}{3} \cdot 1 = \frac{1}{3}.$$

Since $L = \frac{1}{3} < 1$, the series **converges**. ■

Example 3.4.12 (Complex Factorial Expressions). Test for convergence:

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$$

Solution. Let $u_n = \frac{(n!)^2}{(2n)!} = \frac{n! \cdot n!}{(2n)!}$. The next term is $u_{n+1} = \frac{((n+1)!)^2}{(2(n+1))!} = \frac{((n+1)!)^2}{(2n+2)!}$.

We simplify the ratio. Note that $(2n+2)! = (2n+2)(2n+1)(2n)!$.

$$\begin{aligned} \frac{u_{n+1}}{u_n} &= \frac{((n+1)!)^2}{(2n+2)!} \cdot \frac{(2n)!}{(n!)^2} \\ &= \frac{(n+1)!}{(n!)} \cdot \frac{(n+1)!}{(n!)} \cdot \frac{(2n)!}{(2n+2)!} \\ &= (n+1) \cdot (n+1) \cdot \frac{1}{(2n+2)(2n+1)} \\ &= \frac{(n+1)^2}{2(n+1)(2n+1)} \\ &= \frac{n+1}{2(2n+1)} = \frac{n+1}{4n+2}. \end{aligned}$$

Taking the limit:

$$L = \lim_{n \rightarrow \infty} \frac{n(1 + 1/n)}{n(4 + 2/n)} = \frac{1}{4}.$$

Since $L = \frac{1}{4} < 1$, the series **converges**. ■

Example 3.4.13 (Parameter Dependent Series). Test the convergence of the series for all positive values of x :

$$\sum_{n=1}^{\infty} \frac{x^n}{n} \quad (x > 0).$$

Solution. Let $u_n = \frac{x^n}{n}$. Then $u_{n+1} = \frac{x^{n+1}}{n+1}$.

Calculate the limit of the ratio:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| \quad (\text{Since } x > 0, \text{ absolute value is redundant}) \\ &= \lim_{n \rightarrow \infty} \left(\frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n} \right) \\ &= \lim_{n \rightarrow \infty} \left(x \cdot \frac{n}{n+1} \right) \\ &= x \cdot \lim_{n \rightarrow \infty} \frac{1}{1 + 1/n} = x. \end{aligned}$$

We must analyze the result based on the value of $L = x$:

1. **Case** $x < 1$: Since $L < 1$, the series **converges**.
2. **Case** $x > 1$: Since $L > 1$, the series **diverges**.
3. **Case** $x = 1$: Here $L = 1$, so the Ratio Test is **inconclusive**. We must substitute $x = 1$ into the original series to test separately. The series becomes $\sum_{n=1}^{\infty} \frac{1^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n}$. This is the Harmonic Series, which is known to **diverge**.

Conclusion: The series converges if $0 < x < 1$ and diverges if $x \geq 1$. ■

3.4.3 Raabe's Test

Theorem 3.4.14 (Raabe's Test). Let $\sum u_n$ be a series of positive terms. Let

$$l = \lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right).$$

1. If $l > 1$, the series converges.
2. If $l < 1$, the series diverges.
3. If $l = 1$, the test is inconclusive.

Proof of Raabe's Test. Raabe's test determines convergence by comparing $\sum u_n$ to the p -series $\sum \frac{1}{n^p}$.

Let $L_n = n \left(\frac{u_n}{u_{n+1}} - 1 \right)$.

1. **Case $l > 1$ (Convergence):** Choose a real number p such that $1 < p < l$. Since $\lim L_n = l > p$, eventually $n \left(\frac{u_n}{u_{n+1}} - 1 \right) > p$. Rearranging, we get $\frac{u_n}{u_{n+1}} > 1 + \frac{p}{n}$. Using Bernoulli's Inequality (or Taylor expansion), observe that for large n :

$$\left(1 + \frac{1}{n} \right)^p < 1 + \frac{p}{n} < \frac{u_n}{u_{n+1}}.$$

Let $v_n = \frac{1}{n^p}$. Then $\frac{v_n}{v_{n+1}} = \frac{(n+1)^p}{n^p} = \left(1 + \frac{1}{n} \right)^p$. Thus, $\frac{u_n}{u_{n+1}} > \frac{v_n}{v_{n+1}}$, or $\frac{u_{n+1}}{v_{n+1}} < \frac{u_n}{v_n}$. The sequence $\left\{ \frac{u_n}{v_n} \right\}$ is monotonically decreasing and bounded. Thus $u_n < k v_n$ for some constant k . Since $\sum v_n = \sum \frac{1}{n^p}$ converges (as $p > 1$), $\sum u_n$ converges.

2. **Case $l < 1$ (Divergence):** Eventually $n \left(\frac{u_n}{u_{n+1}} - 1 \right) < 1$, which implies $\frac{u_n}{u_{n+1}} < 1 + \frac{1}{n} = \frac{n+1}{n}$. Rearranging yields $\frac{u_{n+1}}{u_n} > \frac{1/(n+1)}{1/n}$. Comparing u_n with the harmonic series $v_n = \frac{1}{n}$, we see that the ratio of terms of u_n decreases slower than that of the divergent harmonic series. Hence, $\sum u_n$ diverges.

■

Example 3.4.15 (Ratio Test Failure). Test the convergence of the series:

$$\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}$$

Solution. Let the general term be $u_n = \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)}$. The ratio of consecutive terms is

$$\frac{u_{n+1}}{u_n} = \frac{2n+1}{2n+2}.$$

As $n \rightarrow \infty$, this ratio approaches 1, rendering D'Alembert's Ratio Test inconclusive. We therefore apply Raabe's Test by examining the limit of $n \left(\frac{u_n}{u_{n+1}} - 1 \right)$.

Substituting the reciprocal ratio $\frac{u_n}{u_{n+1}} = \frac{2n+2}{2n+1}$, we obtain:

$$L = \lim_{n \rightarrow \infty} n \left(\frac{2n+2}{2n+1} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{(2n+2) - (2n+1)}{2n+1} \right).$$

Simplifying the expression yields:

$$L = \lim_{n \rightarrow \infty} \frac{n}{2n+1} = \frac{1}{2}.$$

Since the limit $L = \frac{1}{2} < 1$, Raabe's Test indicates that the series **diverges**. ■

Example 3.4.16 (Product Series). Discuss the convergence of the series:

$$\sum_{n=1}^{\infty} \frac{3 \cdot 6 \cdot 9 \cdots (3n)}{7 \cdot 10 \cdot 13 \cdots (3n+4)}$$

Solution. We define the n -th term as $u_n = \frac{3 \cdot 6 \cdots (3n)}{7 \cdot 10 \cdots (3n+4)}$. The subsequent term u_{n+1} introduces factors of $3(n+1)$ in the numerator and $3(n+1)+4$ in the denominator. Thus, the ratio is:

$$\frac{u_{n+1}}{u_n} = \frac{3n+3}{3n+7}.$$

Since $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = 1$, the Ratio Test fails to provide a conclusion. We proceed with Raabe's Test, calculating the limit $L = \lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right)$.

Using the inverted ratio $\frac{u_n}{u_{n+1}} = \frac{3n+7}{3n+3}$:

$$L = \lim_{n \rightarrow \infty} n \left(\frac{3n+7}{3n+3} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{4}{3n+3} \right).$$

This simplifies to $L = \lim_{n \rightarrow \infty} \frac{4n}{3n+3} = \frac{4}{3}$. Since the limit exceeds 1, the series **converges**. ■

Example 3.4.17 (Variable Parameter x). Test the convergence of the series for $x > 0$:

$$\sum_{n=1}^{\infty} \frac{n!}{(x+1)(x+2) \cdots (x+n)}$$

Solution. Let $u_n = \frac{n!}{(x+1) \cdots (x+n)}$. The ratio of successive terms is

$$\frac{u_{n+1}}{u_n} = \frac{n+1}{n+x+1}.$$

Regardless of the value of x , the limit of this ratio is 1, so we require the more sensitive Raabe's Test. We evaluate the limit L :

$$L = \lim_{n \rightarrow \infty} n \left(\frac{n+x+1}{n+1} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{x}{n+1} \right).$$

The limit is clearly $L = x$. According to Raabe's Test, the series **converges** if $x > 1$ and **diverges** if $x < 1$.

In the boundary case where $x = 1$, the limit is 1, and the test is technically inconclusive. However, substituting $x = 1$ directly into the general term yields $u_n = \frac{n!}{(n+1)!} = \frac{1}{n+1}$. This results in the harmonic series, which is known to diverge. Thus, the series diverges for all $x \leq 1$. ■

Example 3.4.18 (Complex Powers and Factorials). Examine the convergence of the series:

$$\sum_{n=1}^{\infty} \left[\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \right] \cdot \frac{1}{n}$$

Solution. Let $u_n = \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)} \cdot \frac{1}{n}$. The ratio of consecutive terms involves both the factorial-like components and the $\frac{1}{n}$ factor:

$$\frac{u_{n+1}}{u_n} = \frac{2n+1}{2n+2} \cdot \frac{n}{n+1}.$$

The limit of this product is $1 \cdot 1 = 1$, necessitating Raabe's Test. We compute the test limit L using the reciprocal $\frac{u_n}{u_{n+1}} = \frac{2n+2}{2n+1} \cdot \frac{n+1}{n}$:

$$L = \lim_{n \rightarrow \infty} n \left(\frac{(2n+2)(n+1)}{(2n+1)n} - 1 \right).$$

Finding a common denominator and expanding the numerator:

$$L = \lim_{n \rightarrow \infty} n \left(\frac{2n^2 + 4n + 2 - (2n^2 + n)}{2n^2 + n} \right) = \lim_{n \rightarrow \infty} n \left(\frac{3n + 2}{2n^2 + n} \right).$$

This simplifies to $L = \lim_{n \rightarrow \infty} \frac{3n^2 + 2n}{2n^2 + n} = \frac{3}{2}$. Since $L = \frac{3}{2} > 1$, the series **converges**. ■

Exercise 3.4.19. Determine the convergence or divergence of the following series. For series involving a variable x , assume $x > 0$ unless otherwise stated.

1. $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$
2. $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2}$
3. $\sum_{n=1}^{\infty} \frac{1}{(\ln n)^n} \quad (n \geq 2)$
4. $\sum_{n=1}^{\infty} \left(\frac{3n+1}{4n+3}\right)^n$
5. $\sum_{n=1}^{\infty} \frac{n^n x^n}{(n+1)^n}$
6. $\sum_{n=1}^{\infty} \frac{n!}{n^n}$
7. $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$
8. $\sum_{n=1}^{\infty} \frac{n^2 + 1}{3^n}$
9. $\sum_{n=1}^{\infty} \frac{x^n}{n!}$
10. $\sum_{n=1}^{\infty} \frac{n^p}{n!} \quad (p \in \mathbb{R})$
11. $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} x^n$
12. $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} x^n$
13. $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \frac{1}{2n+1}$
14. $\sum_{n=1}^{\infty} \frac{3 \cdot 6 \cdot 9 \cdots (3n)}{7 \cdot 10 \cdot 13 \cdots (3n+4)} x^n$
15. $\sum_{n=1}^{\infty} \left[\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \right]^p \quad (p > 0)$
16. $\sum_{n=1}^{\infty} \frac{n!}{(x+1)(x+2) \cdots (x+n)}$
17. $\sum_{n=1}^{\infty} \frac{4 \cdot 7 \cdot 10 \cdots (3n+1)}{1 \cdot 2 \cdot 3 \cdots n} x^n$
18. $\sum_{n=1}^{\infty} \frac{\alpha(\alpha+1) \cdots (\alpha+n-1)}{\beta(\beta+1) \cdots (\beta+n-1)} \quad (0 < \alpha < \beta)$
19. $\sum_{n=1}^{\infty} x^{\ln n}$
20. $\sum_{n=1}^{\infty} \left(\frac{an+b}{cn+d} \right)^n \quad (a, c > 0)$

Answers

1. Converges (Root Test limit $1/e$).
2. Converges (Root Test limit $1/e$).
3. Converges (Root Test limit 0).
4. Converges (Root Test limit $3/4$).
5. Converges if $x < e$, Diverges if $x \geq e$.
6. Converges (Ratio Test limit $1/e$).
7. Converges (Ratio Test limit $2/e$).
8. Converges (Ratio Test limit $1/3$).
9. Converges for all $x > 0$.
10. Converges for all p .

- | | |
|---|---|
| 11. Converges if $x < 4$, Diverges if $x \geq 4$. | 16. Converges if $x > 1$, Diverges if $x \leq 1$. |
| 12. Converges if $x < 1$, Diverges if $x \geq 1$. | 17. Converges if $x < 1/3$, Diverges if $x \geq 1/3$. |
| 13. Converges (Raabe's Test limit $3/2$). | 18. Converges if $\beta - \alpha > 1$, Diverges if $\beta - \alpha \leq 1$. |
| 14. Converges if $x \leq 1$, Diverges if $x > 1$. | 19. Converges if $x < 1/e$, Diverges if $x \geq 1/e$. |
| 15. Converges if $p > 2$, Diverges if $p \leq 2$. | 20. Converges if $a < c$, Diverges if $a \geq c$. |

3.5 SOME MORE USEFUL TESTS

3.5.1 The Logarithmic Test

The Logarithmic test is often employed when the Ratio test is inconclusive ($L = 1$) and the presence of logarithms makes Raabe's test cumbersome to apply directly.

Theorem 3.5.1 (Logarithmic Test). *Let $\sum u_n$ be a series of positive terms such that*

$$\lim_{n \rightarrow \infty} \left(n \ln \frac{u_n}{u_{n+1}} \right) = l.$$

1. If $l > 1$, the series converges.
2. If $l < 1$, the series diverges.

Proof. **Case 1: Convergence** ($l > 1$)

Since $l > 1$, we can choose a real number p such that $1 < p < l$. Based on the definition of the limit, there exists an integer N such that for all $n \geq N$:

$$n \ln \frac{u_n}{u_{n+1}} > p.$$

Dividing by n and exponentiating both sides, we obtain:

$$\frac{u_n}{u_{n+1}} > e^{p/n}.$$

Recall the expansion $e^x = 1 + x + \frac{x^2}{2!} + \dots$. For $x > 0$, it follows that $e^x > 1 + x$. However, to compare with the auxiliary series $v_n = \frac{1}{n^p}$, we require a slightly sharper inequality. Note that for large n ,

$$e^{p/n} > \left(1 + \frac{1}{n} \right)^p = \frac{(n+1)^p}{n^p} = \frac{v_n}{v_{n+1}}.$$

Thus, for $n \geq N$, we have the inequality $\frac{u_n}{u_{n+1}} > \frac{v_n}{v_{n+1}}$. Since the auxiliary series $\sum v_n = \sum \frac{1}{n^p}$ converges (as $p > 1$), it follows by the Ratio Comparison Test that $\sum u_n$ converges.

Case 2: Divergence ($l < 1$)

Since $l < 1$, we choose a real number p such that $l < p < 1$. For all $n \geq N$, we have:

$$n \ln \frac{u_n}{u_{n+1}} < p \implies \frac{u_n}{u_{n+1}} < e^{p/n}.$$

We compare this with the divergent series $v_n = \frac{1}{n^p}$ (where $p \leq 1$). For sufficiently large n , the inequality $e^{p/n} < (1 + \frac{1}{n})^p$ holds. Consequently, $\frac{u_n}{u_{n+1}} < \frac{v_n}{v_{n+1}}$. Since $\sum v_n$ diverges, the series $\sum u_n$ diverges. ■

3.5.2 Gauss's Test

Gauss's test is considered one of the most powerful convergence tests, often succeeding where the Ratio, Raabe, and Logarithmic tests fail. It relies on the power series expansion of the ratio of terms.

Theorem 3.5.2 (Gauss's Test). Let $\sum u_n$ be a series of positive terms. Suppose the ratio of successive terms can be expressed in the form:

$$\frac{u_n}{u_{n+1}} = 1 + \frac{\lambda}{n} + \frac{\gamma_n}{n^{1+\delta}},$$

where λ and $\delta > 0$ are constants, and $\{\gamma_n\}$ is a bounded sequence.

1. If $\lambda > 1$, the series converges.
2. If $\lambda \leq 1$, the series diverges.

Proof. Case 1: $\lambda \neq 1$

Consider the limit used in Raabe's Test:

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{\lambda}{n} + \frac{\gamma_n}{n^{1+\delta}} \right) = \lim_{n \rightarrow \infty} \left(\lambda + \frac{\gamma_n}{n^\delta} \right).$$

Since $\{\gamma_n\}$ is bounded and $\delta > 0$, the term $\frac{\gamma_n}{n^\delta}$ tends to 0. Thus, the limit is λ . By Raabe's Test, the series converges if $\lambda > 1$ and diverges if $\lambda < 1$.

Case 2: $\lambda = 1$

In this scenario, $\frac{u_n}{u_{n+1}} = 1 + \frac{1}{n} + \frac{\gamma_n}{n^{1+\delta}}$. Raabe's test yields a limit of 1 and is inconclusive. We proceed by comparing $\sum u_n$ with the harmonic series $\sum v_n = \sum \frac{1}{n}$, which diverges. Consider the ratio for the harmonic series:

$$\frac{v_n}{v_{n+1}} = \frac{n+1}{n} = 1 + \frac{1}{n}.$$

Subtracting the ratios of the two series:

$$\frac{u_n}{u_{n+1}} - \frac{v_n}{v_{n+1}} = \frac{\gamma_n}{n^{1+\delta}}.$$

Let us define a sequence $a_n = \frac{u_n}{v_n} = nu_n$. The behavior of this ratio determines the relationship between the series. We examine the ratio of successive terms of a_n :

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)u_{n+1}}{nu_n} = \left(1 + \frac{1}{n}\right) \left(\frac{u_{n+1}}{u_n}\right).$$

Substituting $\frac{u_{n+1}}{u_n} = \left(1 + \frac{1}{n} + \frac{\gamma_n}{n^{1+\delta}}\right)^{-1}$:

$$\frac{a_{n+1}}{a_n} = \left(1 + \frac{1}{n}\right) \left(1 - \frac{1}{n} + O\left(\frac{1}{n^2}\right)\right) = 1 - \frac{1}{n^2} + O\left(\frac{1}{n^2}\right).$$

This implies that for sufficiently large n , $\frac{a_{n+1}}{a_n} > 1 - \frac{k}{n^2}$ for some constant k . It can be shown (via infinite products or Weierstrass's criterion) that $\prod \frac{a_{n+1}}{a_n}$ converges to a non-zero finite limit. Therefore, $\lim_{n \rightarrow \infty} a_n = \lim(nu_n) = C > 0$. Consequently, $u_n \approx \frac{C}{n}$. Since $\sum \frac{1}{n}$ diverges, $\sum u_n$ diverges. ■

3.5.3 Cauchy's Integral Test

This test bridges the gap between the summation of series and the integration of functions, utilizing the geometric interpretation of the definite integral as the area under a curve.

Theorem 3.5.3 (Cauchy's Integral Test). Let $f(x)$ be a non-negative, monotonic decreasing, continuous function defined for $x \geq 1$, such that $f(n) = u_n$ for all $n \in \mathbb{N}$. Then the infinite series $\sum_{n=1}^{\infty} u_n$ and the improper integral $\int_1^{\infty} f(x) dx$ converge or diverge together.

Proof. Consider the area under the curve $y = f(x)$ over the interval $[1, n]$. We approximate this area using rectangles of width 1.

Since $f(x)$ is decreasing, for any $x \in [k, k+1]$, we have the inequality:

$$f(k+1) \leq f(x) \leq f(k).$$

Integrating this inequality from k to $k + 1$:

$$\int_k^{k+1} f(k+1) dx \leq \int_k^{k+1} f(x) dx \leq \int_k^{k+1} f(k) dx,$$

which simplifies to:

$$u_{k+1} \leq \int_k^{k+1} f(x) dx \leq u_k.$$

We sum these inequalities from $k = 1$ to $n - 1$:

$$\sum_{k=1}^{n-1} u_{k+1} \leq \sum_{k=1}^{n-1} \int_k^{k+1} f(x) dx \leq \sum_{k=1}^{n-1} u_k.$$

This translates to:

$$S_n - u_1 \leq \int_1^n f(x) dx \leq S_{n-1},$$

where S_n is the n -th partial sum of the series.

Since $u_n > 0$, both the sequence of partial sums $\{S_n\}$ and the integral sequence $I_n = \int_1^n f(x) dx$ are monotonically increasing.

Convergence Implication: If the integral converges to a limit I , then $S_n - u_1 \leq I$ for all n , implying $S_n \leq I + u_1$. Thus, $\{S_n\}$ is bounded above and monotonic, so the series converges.

Divergence Implication: If the integral diverges to $+\infty$, the inequality $\int_1^n f(x) dx \leq S_{n-1}$ ensures that S_{n-1} must also tend to $+\infty$. Thus, the series diverges.

Therefore, the series and the integral share the same character of convergence. ■

Cauchy's Integral Test - Solved Examples

Example 3.5.4. Discuss the convergence of the series $\sum_{n=1}^{\infty} ne^{-n^2}$.

Solution. We consider the function $f(x) = xe^{-x^2}$ defined for $x \geq 1$. Since $f(n) = ne^{-n^2}$ represents the terms of the series, we analyze the behavior of the function. The derivative is $f'(x) = e^{-x^2}(1 - 2x^2)$. For $x \geq 1$, we have $1 - 2x^2 < 0$, implying that $f(x)$ is a monotonically decreasing function of positive terms.

We evaluate the improper integral:

$$I = \int_1^{\infty} xe^{-x^2} dx.$$

Using the substitution $u = x^2$, leading to $du = 2x dx$:

$$I = \frac{1}{2} \int_1^{\infty} e^{-u} du = \frac{1}{2} [-e^{-u}]_1^{\infty} = \frac{1}{2} (0 - (-e^{-1})) = \frac{1}{2e}.$$

Since the integral converges to a finite value, by Cauchy's Integral Test, the series $\sum_{n=1}^{\infty} ne^{-n^2}$ converges. ■

Example 3.5.5. Determine the convergence of the Bertrand Series $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p}$ for $p > 0$.

Solution. Let $f(x) = \frac{1}{x(\ln x)^p}$ for $x \geq 2$. Clearly, $f(x)$ is positive, continuous, and decreasing. We examine the integral:

$$I = \int_2^{\infty} \frac{dx}{x(\ln x)^p}.$$

Substitute $u = \ln x$, so $du = \frac{1}{x} dx$. The limits change from $\ln 2$ to ∞ :

$$I = \int_{\ln 2}^{\infty} \frac{du}{u^p}.$$

This is a standard p -integral.

- If $p > 1$, the integral converges to $\frac{(\ln 2)^{1-p}}{p-1}$.
- If $0 < p \leq 1$, the integral diverges to ∞ .

Consequently, the series converges if $p > 1$ and diverges if $0 < p \leq 1$. ■

Example 3.5.6. Test the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$.

Solution. Consider the function $f(x) = \frac{1}{x^2 + 1}$. It is positive and decreasing for $x \geq 1$. Evaluating the integral:

$$\int_1^{\infty} \frac{dx}{x^2 + 1} = [\arctan x]_1^{\infty} = \lim_{t \rightarrow \infty} \arctan t - \arctan 1 = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}.$$

Since the integral is finite, the series converges. ■

The Logarithmic Test - Solved Examples

Example 3.5.7. Test the convergence of the series $\sum_{n=2}^{\infty} \frac{1}{(\ln n)^{\ln n}}$.

Solution. Let the general term be $u_n = (\ln n)^{-\ln n}$. Applying the Logarithmic Test, we consider the limit of $L_n = n \ln \frac{u_n}{u_{n+1}}$. First, calculate $\ln u_n = -\ln n \cdot \ln(\ln n)$. Then:

$$\ln \frac{u_n}{u_{n+1}} = \ln u_n - \ln u_{n+1} = \ln(n+1) \ln(\ln(n+1)) - \ln n \ln(\ln n).$$

This analytical approach is cumbersome. Alternatively, we can rearrange the term:

$$u_n = e^{-\ln n \cdot \ln(\ln n)} = (e^{\ln n})^{-\ln(\ln n)} = n^{-\ln(\ln n)} = \frac{1}{n^{\ln(\ln n)}}.$$

For large n , we have $\ln(\ln n) > 2$ (specifically for $n > e^{e^2}$). Thus, for sufficiently large n , we have $u_n < \frac{1}{n^2}$. By comparison with the convergent p -series ($p = 2$), the series converges. ■

Example 3.5.8. Discuss the convergence of the series $\sum_{n=1}^{\infty} x^{\ln n}$ for $x > 0$.

Solution. Let $u_n = x^{\ln n}$. We compute the logarithmic test statistic:

$$\ln \frac{u_n}{u_{n+1}} = \ln(x^{\ln n}) - \ln(x^{\ln(n+1)}) = (\ln n - \ln(n+1)) \ln x = \ln \left(\frac{n}{n+1} \right) \ln x.$$

Using the expansion $\ln(1-t) \approx -t$:

$$\ln \left(1 - \frac{1}{n+1} \right) \ln x \approx -\frac{1}{n} \ln x.$$

Multiplying by n :

$$\lim_{n \rightarrow \infty} n \ln \frac{u_n}{u_{n+1}} = \lim_{n \rightarrow \infty} n \left(-\frac{1}{n} \ln x \right) = -\ln x = \ln(1/x).$$

According to the Logarithmic Test:

- Converges if $\ln(1/x) > 1 \implies 1/x > e \implies x < 1/e$.
- Diverges if $\ln(1/x) < 1 \implies x > 1/e$.

If $x = 1/e$, the term becomes $(1/e)^{\ln n} = n^{-1}$, the harmonic series, which diverges. ■

Example 3.5.9. Test the series $\sum_{n=1}^{\infty} \left(1 - \frac{1}{\sqrt{n}}\right)^n$.

Solution. Let $u_n = (1 - n^{-1/2})^n$. Consider $\ln u_n = n \ln(1 - n^{-1/2})$. Using the expansion $\ln(1 - x) \approx -x - x^2/2$:

$$\ln u_n \approx n \left(-n^{-1/2} - \frac{1}{2}n^{-1} \right) = -\sqrt{n} - \frac{1}{2}.$$

The term $n \ln(u_n/u_{n+1})$ involves the difference $\sqrt{n+1} - \sqrt{n}$:

$$\ln(u_n/u_{n+1}) \approx (-\sqrt{n}) - (-\sqrt{n+1}) = \sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \approx \frac{1}{2\sqrt{n}}.$$

Multiplying by n :

$$\lim_{n \rightarrow \infty} n \left(\frac{1}{2\sqrt{n}} \right) = \infty.$$

Since the limit is greater than 1, the series converges. ■

Gauss's Test - Solved Examples

Example 3.5.10. Test the convergence of the Hypergeometric series $\sum_{n=1}^{\infty} \frac{\alpha(\alpha+1)\dots(\alpha+n-1)\beta(\beta+1)\dots(\beta+n-1)}{1 \cdot 2 \dots n \cdot \gamma(\gamma+1)\dots(\gamma+n-1)}$ where $\alpha, \beta, \gamma > 0$.

Solution. Let u_n be the general term. The ratio of consecutive terms is:

$$\frac{u_n}{u_{n+1}} = \frac{n+1}{\alpha+n} \cdot \frac{\gamma+n}{\beta+n} = \frac{(1+1/n)(1+\gamma/n)}{(1+\alpha/n)(1+\beta/n)}.$$

We expand this expression in powers of $1/n$ using $(1+x)^{-1} = 1 - x + x^2 - \dots$:

$$\begin{aligned} \frac{u_n}{u_{n+1}} &= \left(1 + \frac{1+\gamma}{n} + O(n^{-2})\right) \left(1 - \frac{\alpha+\beta}{n} + O(n^{-2})\right) \\ &= 1 + \frac{1+\gamma-\alpha-\beta}{n} + O\left(\frac{1}{n^2}\right). \end{aligned}$$

Comparing this to the Gauss form $1 + \frac{\lambda}{n} + O(n^{-2})$, we identify $\lambda = 1 + \gamma - \alpha - \beta$. By Gauss's Test:

- Converges if $\lambda > 1 \implies \gamma > \alpha + \beta$.
 - Diverges if $\lambda \leq 1 \implies \gamma \leq \alpha + \beta$.
-

Example 3.5.11. Discuss the convergence of $\sum_{n=1}^{\infty} \left[\frac{1 \cdot 3 \dots (2n-1)}{2 \cdot 4 \dots (2n)} \right]^2$.

Solution. Here $u_n = \left[\frac{(2n-1)!!}{(2n)!!} \right]^2$. The ratio is:

$$\frac{u_n}{u_{n+1}} = \left(\frac{2n+2}{2n+1} \right)^2 = \left(\frac{1+1/n}{1+1/(2n)} \right)^2.$$

Using binomial expansions $(1+x)^2 \approx 1+2x$ and $(1+x)^{-2} \approx 1-2x$:

$$\begin{aligned}\frac{u_n}{u_{n+1}} &= \left(1 + \frac{1}{n}\right)^2 \left(1 + \frac{1}{2n}\right)^{-2} \\ &= \left(1 + \frac{2}{n} + \frac{1}{n^2}\right) \left(1 - \frac{1}{n} + \frac{3}{4n^2}\right) \\ &= 1 + \left(\frac{2}{n} - \frac{1}{n}\right) + O\left(\frac{1}{n^2}\right) \\ &= 1 + \frac{1}{n} + O\left(\frac{1}{n^2}\right).\end{aligned}$$

Here $\lambda = 1$. Since $\lambda \leq 1$, the series diverges. ■

Example 3.5.12. Test the series $\sum_{n=1}^{\infty} \frac{2 \cdot 4 \cdots (2n+2)}{3 \cdot 5 \cdots (2n+3)}$.

Solution. The ratio of consecutive terms is:

$$\frac{u_n}{u_{n+1}} = \frac{3 \cdot 5 \cdots (2n+5)}{2 \cdot 4 \cdots (2n+4)} \cdot \frac{2 \cdot 4 \cdots (2n+2)}{3 \cdot 5 \cdots (2n+3)} = \frac{2n+5}{2n+4}.$$

Expanding the ratio:

$$\frac{u_n}{u_{n+1}} = \frac{2n+5}{2n+4} = \frac{2n+4+1}{2n+4} = 1 + \frac{1}{2n+4} = 1 + \frac{1}{2n(1+2/n)}.$$

Using $(1+x)^{-1} \approx 1-x$:

$$\frac{u_n}{u_{n+1}} = 1 + \frac{1}{2n} \left(1 - \frac{2}{n} + \dots\right) = 1 + \frac{1/2}{n} + O\left(\frac{1}{n^2}\right).$$

Here $\lambda = 1/2$. Since $\lambda < 1$, the series diverges. ■

Example 3.5.13. Determine the convergence of $\sum_{n=1}^{\infty} \frac{n!}{(x+1)(x+2) \cdots (x+n)}$ for $x > 0$ using Gauss's Test.

Solution. The reciprocal ratio is $\frac{u_n}{u_{n+1}} = \frac{x+n+1}{n+1} = \frac{n(1+(x+1)/n)}{n(1+1/n)}$. Expanding:

$$\begin{aligned}\frac{u_n}{u_{n+1}} &= \left(1 + \frac{x+1}{n}\right) \left(1 - \frac{1}{n} + O(n^{-2})\right) \\ &= 1 + \frac{x+1-1}{n} + O(n^{-2}) \\ &= 1 + \frac{x}{n} + O(n^{-2}).\end{aligned}$$

We identify $\lambda = x$.

- If $x > 1$, the series converges.
- If $x \leq 1$, the series diverges.

This confirms the result obtained via Raabe's test but resolves the $x = 1$ case definitively as divergence (since $\lambda = 1$ implies divergence in Gauss's test). ■

Chapter 4

Tutorial

This chapter establishes the topological foundations of real analysis, introducing neighbourhoods, open and closed sets, and limit points to analyze the structure of the real line. We explore the accumulation properties of infinite sets through the Bolzano-Weierstrass Theorem and the concept of closure. Furthermore, we investigate the theory of cardinality, distinguishing between countable and uncountable infinities to rigorously demonstrate the countability of rational numbers and the uncountability of the real continuum.

4.1 Neighbourhoods, Open Sets, and Limit Points

In this section, we establish the fundamental topological properties of the real number system $\mathbb{R}$. We begin with the concept of proximity using neighbourhoods.

4.1.1 Neighbourhood of a Point

Definition 4.1.1 (ϵ -Neighbourhood). Let $a \in \mathbb{R}$ and $\epsilon > 0$. The ϵ -neighbourhood of a , denoted by $N_\epsilon(a)$, is the set of all real numbers x such that $|x - a| < \epsilon$. In interval notation: $N_\epsilon(a) = (a - \epsilon, a + \epsilon)$.

Definition 4.1.2 (Neighbourhood). A set $S \subseteq \mathbb{R}$ is called a neighbourhood of a point $p \in \mathbb{R}$ if there exists an $\epsilon > 0$ such that $N_\epsilon(p) \subseteq S$.

Example 4.1.3. The open interval $(0, 1)$ is a neighbourhood of every point $p \in (0, 1)$. However, the closed interval $[0, 1]$ is a neighbourhood of every point in $(0, 1)$, but it is *not* a neighbourhood of 0 or 1, because any open interval centred at 0 must contain negative numbers, which are not in $[0, 1]$.

Example 4.1.4. Construct the ϵ -neighbourhood of the point $a = 3$ corresponding to $\epsilon = 0.5$.

Solution. Using the definition of an ϵ -neighbourhood, we have:

$$N_{0.5}(3) = \{x \in \mathbb{R} : |x - 3| < 0.5\}.$$

By expanding the absolute value inequality, we obtain:

$$-0.5 < x - 3 < 0.5 \implies 2.5 < x < 3.5.$$

Therefore, the neighbourhood is the open interval $N_{0.5}(3) = (2.5, 3.5)$. ■

Example 4.1.5. Show that the open interval $S = (1, 5)$ is a neighbourhood of the point $p = 2$.

Solution. We must find a positive real number $\epsilon > 0$ such that $(2 - \epsilon, 2 + \epsilon) \subseteq (1, 5)$. The distance from $p = 2$ to the left endpoint 1 is 1, and the distance to the right endpoint 5 is 3. We choose ϵ to be smaller than the minimum of these distances. Let $\epsilon = 0.5$. Then:

$$N_{0.5}(2) = (2 - 0.5, 2 + 0.5) = (1.5, 2.5).$$

Since $1 < 1.5$ and $2.5 < 5$, it follows that $(1.5, 2.5) \subset (1, 5)$. Thus, S is a neighbourhood of 2. ■

Example 4.1.6. Determine whether the closed interval $S = [0, 4]$ is a neighbourhood of the point $p = 4$.

Solution. For S to be a neighbourhood of 4, there must exist an $\epsilon > 0$ such that $(4 - \epsilon, 4 + \epsilon) \subseteq [0, 4]$. However, for any $\epsilon > 0$, the interval $(4 - \epsilon, 4 + \epsilon)$ contains real numbers strictly greater than 4. Specifically, the number $x = 4 + \frac{\epsilon}{2}$ satisfies:

$$4 < x < 4 + \epsilon.$$

Since $x > 4$, $x \notin [0, 4]$. Therefore, no such ϵ -neighbourhood is contained in S . Hence, $[0, 4]$ is not a neighbourhood of 4. ■

Example 4.1.7. Let $S = \mathbb{R} \setminus \{5\}$. Show that S is a neighbourhood of the point $p = 3$.

Solution. The set S consists of the union of two open intervals: $(-\infty, 5) \cup (5, \infty)$. The point $p = 3$ lies in the interval $(-\infty, 5)$. We can choose $\epsilon = 1$. The resulting neighbourhood is:

$$N_1(3) = (2, 4).$$

Since $5 \notin (2, 4)$, the entire interval $(2, 4)$ is contained within $\mathbb{R} \setminus \{5\}$. Thus, S is a neighbourhood of 3. ■

Example 4.1.8. Consider the finite set $S = \{1, 2, 3\}$. Is S a neighbourhood of the point 2?

Solution. By definition, for S to be a neighbourhood of 2, it must contain an open interval $(2 - \epsilon, 2 + \epsilon)$ for some $\epsilon > 0$. However, any open interval $(2 - \epsilon, 2 + \epsilon)$ contains infinitely many real numbers (specifically irrational numbers). The set S contains only three elements. Therefore, $(2 - \epsilon, 2 + \epsilon) \not\subseteq S$. Thus, a finite set cannot be a neighbourhood of any of its points. ■

Example 4.1.9. Prove that the set of rational numbers $\mathbb{Q}$ is not a neighbourhood of the point $p = 0$.

Solution. Assume, for the sake of contradiction, that $\mathbb{Q}$ is a neighbourhood of 0. Then there exists an $\epsilon > 0$ such that $(-\epsilon, \epsilon) \subseteq \mathbb{Q}$. However, between any two distinct real numbers, there exists an irrational number. This means that the interval $(-\epsilon, \epsilon)$ contains irrational numbers. Let ξ be an irrational number in $(-\epsilon, \epsilon)$. Then $\xi \notin \mathbb{Q}$, which implies $(-\epsilon, \epsilon) \not\subseteq \mathbb{Q}$. This contradiction shows that $\mathbb{Q}$ cannot contain any ϵ -neighbourhood. Therefore, $\mathbb{Q}$ is not a neighbourhood of 0 (or indeed, of any point). ■

4.1.2 Interior of a Set and Open Sets

Definition 4.1.10 (Interior Point). A point p is called an interior point of a set S if S is a neighbourhood of p . That is, there exists $\epsilon > 0$ such that $(p - \epsilon, p + \epsilon) \subseteq S$.

Definition 4.1.11 (Interior of a Set). The set of all interior points of a set S is called the interior of S and is denoted by S° or $\text{Int}(S)$.

Definition 4.1.12 (Open Set). A set $S \subseteq \mathbb{R}$ is said to be open if every point of S is an interior point of S . Equivalently, S is open if $S = S^\circ$.

Theorem 4.1.13 (Properties of Open Sets).

1. The union of an arbitrary family of open sets is open.
2. The intersection of a finite family of open sets is open.

Example 4.1.14. Consider the intersection of an infinite family of open sets: $\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$. The resulting set $\{0\}$ is not open. This illustrates why the intersection property holds only for finite families.

Example 4.1.15. Find the interior of the open interval $S = (2, 5)$ and determine if the set is open.

Solution. Let p be any element of S . Then $2 < p < 5$. We need to find an $\epsilon > 0$ such that the neighbourhood $(p - \epsilon, p + \epsilon)$ is contained entirely within S . Let $d_1 = p - 2$ (the distance from p to the left endpoint) and $d_2 = 5 - p$ (the distance from p to the right endpoint). Choose $\epsilon = \min\{d_1, d_2\}$. Since $p \neq 2$ and $p \neq 5$, we have $\epsilon > 0$. Consider any $x \in (p - \epsilon, p + \epsilon)$. Then:

$$p - \epsilon < x < p + \epsilon.$$

Since $\epsilon \leq p - 2$, we have $p - \epsilon \geq 2$. Since $\epsilon \leq 5 - p$, we have $p + \epsilon \leq 5$. Thus, $2 \leq p - \epsilon < x < p + \epsilon \leq 5$, which implies $x \in (2, 5)$. Therefore, $(p - \epsilon, p + \epsilon) \subseteq S$. Since p was arbitrary, every point in S is an interior point. Thus, $S^\circ = (2, 5)$. Since $S = S^\circ$, the set S is an open set. ■

Example 4.1.16. Find the interior of the closed interval $S = [1, 3]$ and determine if the set is open.

Solution. We test the points in S :

- Let $p \in (1, 3)$. As shown in the previous example, there exists an ϵ such that $(p - \epsilon, p + \epsilon) \subseteq (1, 3) \subseteq [1, 3]$. So, all points strictly between 1 and 3 are interior points.
- Consider the left endpoint $p = 1$. For any $\epsilon > 0$, the interval $(1 - \epsilon, 1 + \epsilon)$ contains numbers strictly less than 1 (for example, $1 - \frac{\epsilon}{2}$). These numbers are not in S . Therefore, $(1 - \epsilon, 1 + \epsilon) \not\subseteq S$, so 1 is not an interior point.
- Similarly, for the right endpoint $p = 3$, any neighbourhood $(3 - \epsilon, 3 + \epsilon)$ contains numbers greater than 3, which are not in S . Thus, 3 is not an interior point.

The set of interior points is $S^\circ = (1, 3)$. Since $S = [1, 3]$ but $S^\circ = (1, 3)$, we have $S \neq S^\circ$. Therefore, the set S is not open. ■

Example 4.1.17. Let $S = \mathbb{N} = \{1, 2, 3, \dots\}$. Find S° and determine if $\mathbb{N}$ is open.

Solution. Let $n \in \mathbb{N}$. For n to be an interior point, there must exist an $\epsilon > 0$ such that $(n - \epsilon, n + \epsilon) \subseteq \mathbb{N}$. However, the interval $(n - \epsilon, n + \epsilon)$ contains non-integer real numbers. For instance, if we choose $\epsilon < 1$, the number $n + \frac{\epsilon}{2}$ is in the interval but is not an integer. Therefore, $(n - \epsilon, n + \epsilon) \not\subseteq \mathbb{N}$. Since no point in $\mathbb{N}$ satisfies the condition, the set of interior points is empty: $S^\circ = \emptyset$. Since $S \neq \emptyset$ and $S^\circ = \emptyset$, S is not open. ■

Example 4.1.18. Find the interior of the set of rational numbers $S = \mathbb{Q}$. Is $\mathbb{Q}$ open?

Solution. Let $q \in \mathbb{Q}$. Assume, for contradiction, that q is an interior point. Then there exists $\epsilon > 0$ such that $(q - \epsilon, q + \epsilon) \subseteq \mathbb{Q}$. However, by the property of real numbers, any interval with positive length contains both rational and irrational numbers. Specifically, there exists an irrational number ξ such that $q - \epsilon < \xi < q + \epsilon$. Since $\xi \notin \mathbb{Q}$, the interval $(q - \epsilon, q + \epsilon)$ is not a subset of $\mathbb{Q}$. Thus, no point in $\mathbb{Q}$ is an interior point. $S^\circ = \emptyset$. Therefore, $\mathbb{Q}$ is not an open set. ■

Example 4.1.19. Consider the set $S = [0, \infty) = \{x \in \mathbb{R} : x \geq 0\}$. Determine S° and whether S is open.

Solution. We examine the points in S :

- Let $p > 0$. Choose $\epsilon = p$. Then $(p - \epsilon, p + \epsilon) = (0, 2p)$. Every number in this interval is positive, so $(0, 2p) \subseteq S$. Thus, every $p > 0$ is an interior point.
- Let $p = 0$. Consider any $\epsilon > 0$. The neighbourhood is $(-\epsilon, \epsilon)$. This interval contains negative numbers (e.g., $-\frac{\epsilon}{2}$). Since negative numbers are not in S , $(-\epsilon, \epsilon) \not\subseteq S$. Thus, 0 is not an interior point.

The interior of the set is $S^\circ = (0, \infty)$. Since $0 \in S$ but $0 \notin S^\circ$, the set is not open. ■

Example 4.1.20. Let $S = (1, 2) \cup (3, 4)$. Show that S is an open set.

Solution. Let $p \in S$. Then either $p \in (1, 2)$ or $p \in (3, 4)$.

- Case 1: If $p \in (1, 2)$, let $\epsilon_1 = \min\{p - 1, 2 - p\}$. Then $(p - \epsilon_1, p + \epsilon_1) \subseteq (1, 2) \subseteq S$. Thus, p is an interior point.
- Case 2: If $p \in (3, 4)$, let $\epsilon_2 = \min\{p - 3, 4 - p\}$. Then $(p - \epsilon_2, p + \epsilon_2) \subseteq (3, 4) \subseteq S$. Thus, p is an interior point.

In both cases, p is an interior point of S . Therefore, $S^\circ = S$, which implies that S is an open set. ■

4.1.3 Limit Points and The Derived Set

Definition 4.1.21 (Limit Point / Accumulation Point). A real number p is a limit point of a set $S \subseteq \mathbb{R}$ if every neighbourhood of p contains a point of S other than p . Symbolically, p is a limit point of S if for all $\epsilon > 0$: $(N_\epsilon(p) \cap S) \setminus \{p\} \neq \emptyset$.

Definition 4.1.22 (Derived Set). The set of all limit points of a set S is called the derived set of S and is denoted by S' .

Example 4.1.23. Let $S = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\}$. The point 0 is a limit point of S because every interval $(-\epsilon, \epsilon)$ contains terms of the sequence $\frac{1}{n}$ for sufficiently large n . Note that $0 \notin S$. The derived set is $S' = \{0\}$.

Example 4.1.24. Let $\mathbb{Z}$ be the set of integers. For any $p \in \mathbb{R}$, we can find a neighbourhood of p of length less than 1 that contains no integer (if $p \notin \mathbb{Z}$) or no other integer (if $p \in \mathbb{Z}$). Thus, $\mathbb{Z}' = \emptyset$.

Example 4.1.25. Find the derived set of the finite set $S = \{1, 2, 3, 4\}$.

Solution. Let $p \in \mathbb{R}$. We examine whether any neighbourhood of p contains a point of S distinct from p . Let δ be the minimum distance between distinct points in $S \cup \{p\}$. Specifically, let $\epsilon = 0.5$. If we construct the neighbourhood $N_{0.5}(p) = (p - 0.5, p + 0.5)$:

- If $p \in S$ (e.g., $p = 1$), the interval $(0.5, 1.5)$ contains 1, but no other element of S . The condition $(N_\epsilon(p) \cap S) \setminus \{p\} \neq \emptyset$ fails.
- If $p \notin S$, we can find a sufficiently small ϵ such that $N_\epsilon(p)$ contains no points of S at all.

Therefore, a finite set has no limit points. The derived set is $S' = \emptyset$. ■

Example 4.1.26. Determine the limit points and the derived set of the set $S = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\}$.

Solution. We analyze the point 0. Let $\epsilon > 0$ be given. By the Archimedean property of real numbers, there exists a natural number k such that $\frac{1}{k} < \epsilon$. Since $k \in \mathbb{N}$, $\frac{1}{k} > 0$. Thus, $\frac{1}{k} \in (-\epsilon, \epsilon)$. Since $\frac{1}{k} \in S$ and $\frac{1}{k} \neq 0$, the neighbourhood $(-\epsilon, \epsilon)$ contains a point of S distinct from 0. Hence, 0 is a limit point. Now consider any non-zero point p . If $p \notin S$ and $p \neq 0$, we can choose ϵ such that the neighbourhood excludes all points of S . If $p \in S$, say $p = \frac{1}{n}$, we can choose ϵ smaller than the distance to its neighbors $\frac{1}{n-1}$ and $\frac{1}{n+1}$. This neighbourhood will contain only $\frac{1}{n}$ and no other point of S . Thus, 0 is the only limit point. The derived set is $S' = \{0\}$. ■

Example 4.1.27. Find the derived set of the open interval $S = (0, 1)$.

Solution. Consider any point $p \in \mathbb{R}$.

- If $0 < p < 1$, then for any $\epsilon > 0$, the interval $(p - \epsilon, p + \epsilon)$ overlaps with S . Even if we remove p , the density of real numbers ensures there are other points of S in this neighbourhood. Thus, every point in $(0, 1)$ is a limit point.
- If $p = 0$, consider $(-\epsilon, \epsilon)$. The right half $(0, \epsilon)$ has a non-empty intersection with $(0, 1)$ (assuming $\epsilon < 1$, otherwise it fully contains it). Thus, 0 is a limit point.
- Similarly, if $p = 1$, the interval $(1 - \epsilon, 1 + \epsilon)$ intersects S in $(1 - \epsilon, 1)$. Thus, 1 is a limit point.
- If $p > 1$ or $p < 0$, we can find a neighbourhood disjoint from S .

Therefore, the set of limit points is the closed interval $S' = [0, 1]$. ■

Example 4.1.28. Find the limit points of the set of integers $\mathbb{Z}$.

Solution. Let $p \in \mathbb{R}$. We test if p is a limit point. Choose $\epsilon = 0.5$. The neighbourhood is $N_{0.5}(p) = (p - 0.5, p + 0.5)$. The length of this interval is 1. Since the minimum distance between any two distinct integers is 1, this interval can contain at most one integer.

- If $p \in \mathbb{Z}$, the interval contains only p . The set $(N_{0.5}(p) \cap \mathbb{Z}) \setminus \{p\}$ is empty.
- If $p \notin \mathbb{Z}$, and we choose ϵ sufficiently small (specifically $\epsilon < \min(p - \lfloor p \rfloor, \lceil p \rceil - p)$), the interval contains no integers.

Since no neighbourhood contains a point of $\mathbb{Z}$ distinct from p , the set has no limit points. The derived set is $\mathbb{Z}' = \emptyset$. ■

Example 4.1.29. Determine the derived set of the set of rational numbers $\mathbb{Q}$.

Solution. Let p be any real number ($p \in \mathbb{R}$). Let $\epsilon > 0$ be any positive number. Consider the neighbourhood $(p - \epsilon, p + \epsilon)$. By the density property of rational numbers in real numbers, there exist infinitely many rational numbers within any open interval. Therefore, the interval $(p - \epsilon, p + \epsilon)$ contains a rational number q such that $q \neq p$ (in fact, it contains infinitely many). This implies that every real number is a limit point of $\mathbb{Q}$. Consequently, the derived set is $\mathbb{Q}' = \mathbb{R}$. ■

Example 4.1.30. Let $S = [0, 1) \cup \{2\}$. Find the derived set S' .

Solution. We analyze the two components of the set S : the interval $[0, 1)$ and the isolated point $\{2\}$.

- For the part $[0, 1)$, as established in previous examples, every point in the closed interval $[0, 1]$ is a limit point. This includes 0 and 1.
- Now consider the point 2. Let $\epsilon = 0.5$. The neighbourhood $(1.5, 2.5)$ intersects S only at the point 2. The condition for a limit point requires the intersection to contain a point *other than* 2. Since $((1.5, 2.5) \cap S) \setminus \{2\} = \{2\} \setminus \{2\} = \emptyset$, the point 2 is not a limit point. It is an isolated point.

Thus, the limit points are generated solely by the interval portion. The derived set is $S' = [0, 1]$. ■

Problem 4.1.31. Find the derived set of $S = (0, 1)$.

Solution. Every point $x \in (0, 1)$ is a limit point. Furthermore, 0 is a limit point because any interval $(-\epsilon, \epsilon)$ intersects $(0, 1)$. Similarly, 1 is a limit point. Thus, $S' = [0, 1]$. ■

Problem 4.1.32. Determine the limit points of $S = \mathbb{Q}$, the set of rational numbers.

Solution. By the density property of real numbers, every interval (a, b) contains infinitely many rational numbers. Therefore, for any $x \in \mathbb{R}$ and any $\epsilon > 0$, the neighbourhood $(x - \epsilon, x + \epsilon)$ contains rational numbers distinct from x . Hence, $\mathbb{Q}' = \mathbb{R}$. ■

4.2 The Bolzano-Weierstrass Theorem and Closed Sets

4.2.1 The Bolzano-Weierstrass Theorem

Theorem 4.2.1 (Bolzano-Weierstrass Theorem). *Every bounded, infinite subset of $\mathbb{R}$ has at least one limit point in $\mathbb{R}$.*

Remark 4.2.2. Both conditions (boundedness and infiniteness) are necessary.

1. A finite set has no limit points.
2. The set $\mathbb{N}$ is infinite but unbounded, and it has no limit points.

Definition 4.2.3 (Closed Set). A set $S \subseteq \mathbb{R}$ is said to be closed if it contains all its limit points. That is, S is closed if $S' \subseteq S$.

Theorem 4.2.4 (Open and Closed Duality). *A set S is closed if and only if its complement $S^c = \mathbb{R} \setminus S$ is open.*

Definition 4.2.5 (Closure of a Set). The closure of a set S , denoted by $\bar{S}$ (or $\text{cl}(S)$), is the union of S and its derived set S' . $\bar{S} = S \cup S'$.

Theorem 4.2.6 (Results on Closure). *Let $S \subseteq \mathbb{R}$.*

1. $\bar{S}$ is the smallest closed set containing S .
2. S is closed if and only if $S = \bar{S}$.
3. $\bar{\bar{S}} = \bar{S}$.

Example 4.2.7. Consider $S = (0, 1)$.

- The derived set is $S' = [0, 1]$.
- The closure is $\bar{S} = (0, 1) \cup [0, 1] = [0, 1]$.

Example 4.2.8. Verify the Bolzano-Weierstrass Theorem for the set $S = \left\{ (-1)^n \left(1 + \frac{1}{n} \right) : n \in \mathbb{N} \right\}$.

Solution. First, we determine if the set S satisfies the hypotheses of the Bolzano-Weierstrass Theorem. The set is infinite because it contains a distinct term for every $n \in \mathbb{N}$ (except potentially where odd/even terms collide, but the terms approach different limits, ensuring infinitely many distinct points). We examine boundedness. For any $n \in \mathbb{N}$, $0 < \frac{1}{n} \leq 1$. Thus, $1 < 1 + \frac{1}{n} \leq 2$. The term $(-1)^n$ alternates between -1 and 1 . Therefore, every element $x \in S$ satisfies $-2 \leq x \leq 2$. Hence, S is bounded. Since S is an infinite bounded subset of $\mathbb{R}$, by the Bolzano-Weierstrass Theorem, it must have at least one limit point. We observe the subsequences: For even n , terms approach 1 . For odd n , terms approach -1 . Thus, the set has two limit points, 1 and -1 , confirming the theorem. ■

Example 4.2.9. Demonstrate why the boundedness condition is necessary in the Bolzano-Weierstrass Theorem using the set $S = \{2^n : n \in \mathbb{N}\}$.

Solution. The set is $S = \{2, 4, 8, 16, \dots\}$. This set is clearly infinite as the mapping $n \mapsto 2^n$ is a bijection from $\mathbb{N}$ to S . However, the set is unbounded above because for any real number M , there exists an n such that $2^n > M$. We calculate the derived set S' . Let $p \in \mathbb{R}$. We can choose a neighbourhood of p that is small enough (e.g., length 0.5) such that it contains at most one element of S (or none if p is large and between powers of 2). Thus, no point p is a limit point. $S' = \emptyset$. This example shows that an infinite set that is not bounded may fail to have a limit point. ■

Example 4.2.10. Determine whether the set $S = [0, 1)$ is closed using the definition of limit points.

Solution. A set S is closed if $S' \subseteq S$. We determine the derived set S' . As established in previous sections, the limit points of an interval (open, closed, or half-open) consist of the closed interval. Thus, $S' = [0, 1]$. We check the inclusion condition: Is $[0, 1] \subseteq [0, 1)$? The point $1 \in S'$ because every neighbourhood of 1 intersects $[0, 1)$. However, $1 \notin [0, 1) = S$. Since there is a limit point (1) that does not belong to the set S , the set is not closed. ■

Example 4.2.11. Show that the set $S = [2, 5]$ is a closed set.

Solution. The derived set of the closed interval $S = [2, 5]$ is $S' = [2, 5]$. We observe that $S' = S$. Since $S' \subseteq S$, the set contains all its limit points. Therefore, S is a closed set. ■

Example 4.2.12. Use the Open and Closed Duality Theorem to prove that the set $S = \mathbb{R} \setminus \{3\}$ is open.

Solution. The duality theorem states that a set is open if and only if its complement is closed. Let $A = S = \mathbb{R} \setminus \{3\}$. The complement of A in $\mathbb{R}$ is $A^c = \{3\}$. We analyze if the finite set $\{3\}$ is closed. The derived set of any finite set is empty. Thus, $(\{3\})' = \emptyset$. Since $\emptyset \subseteq \{3\}$, the set $\{3\}$ is closed. Because A^c is closed, the original set $A = \mathbb{R} \setminus \{3\}$ must be open. ■

Example 4.2.13. Find the closure of the open interval $S = (a, b)$.

Solution. By definition, the closure is $\bar{S} = S \cup S'$. The set is $S = (a, b)$. The derived set (set of limit points) is $S' = [a, b]$. Therefore:

$$\bar{S} = (a, b) \cup [a, b] = [a, b].$$

This confirms that the closure of an open interval is the corresponding closed interval. ■

Example 4.2.14. Find the closure of the set of rational numbers, $\mathbb{Q}$.

Solution. We have $S = \mathbb{Q}$. We previously established that due to the density of rationals in reals, every real number is a limit point of $\mathbb{Q}$. Thus, the derived set is $S' = \mathbb{R}$. The closure is:

$$\bar{S} = S \cup S' = \mathbb{Q} \cup \mathbb{R} = \mathbb{R}.$$

Since $\bar{\mathbb{Q}} = \mathbb{R}$, we say that $\mathbb{Q}$ is dense in $\mathbb{R}$. ■

Example 4.2.15. Find the closure of the set $S = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$.

Solution. The set is $S = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$. We first identify the derived set S' . The only limit point of S is 0. Thus, $S' = \{0\}$. The closure is defined as $\bar{S} = S \cup S'$.

$$\bar{S} = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\} \cup \{0\} = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup \{0\}.$$

We observe that $\bar{S}$ is a closed set because its complement $\mathbb{R} \setminus \bar{S} = (-\infty, 0) \cup \bigcup_{n=1}^{\infty} \left(\frac{1}{n+1}, \frac{1}{n}\right) \cup (1, \infty)$ is a union of open intervals, which is open. ■

4.3 Cardinality of Sets

In this section, we extend the concept of "size" from finite sets to infinite sets. We introduce the notion of one-to-one correspondence to compare the magnitude of infinite collections.

4.3.1 Equipotence and Cardinality

Definition 4.3.1 (Equipotent Sets). Two sets A and B are said to be equipotent (or have the same cardinality), written as $A \sim B$ or $|A| = |B|$, if there exists a bijection (a function that is both one-to-one and onto) $f : A \rightarrow B$.

Definition 4.3.2 (Finite and Infinite Sets). A set S is finite if it is empty or if $S \sim \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$. If a set is not finite, it is called infinite.

4.3.2 Countable Sets

Definition 4.3.3 (Countable Set). A set S is said to be countable (or at most countable) if it is either finite or countably infinite (denumerable). A set is countably infinite if $S \sim \mathbb{N}$.

Remark 4.3.4. If a set S is countably infinite, its elements can be arranged in a sequence $x_1, x_2, x_3, \dots$ without repetition, such that every element of S appears exactly once.

Example 4.3.5. The set of integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is countable. We can define a bijection $f: \mathbb{N} \rightarrow \mathbb{Z}$ by arranging the integers as $0, 1, -1, 2, -2, \dots$. Explicitly:

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\frac{n-1}{2} & \text{if } n \text{ is odd} \end{cases}$$

Theorem 4.3.6 (Subset of a Countable Set). *Every subset of a countable set is countable.*

Proof. Let A be a countable set and $B \subseteq A$. If B is finite, it is countable by definition. If B is infinite, the elements of A can be listed as a sequence. By deleting elements from this sequence that are not in B , we obtain a sequence listing the elements of B . Thus, B is countable. ■

Theorem 4.3.7 (Countable Union). *The union of a countable family of countable sets is countable. If $\{A_n\}_{n \in \mathbb{N}}$ is a collection of countable sets, then $S = \bigcup_{n=1}^{\infty} A_n$ is countable.*

4.3.3 Countability of Rational Numbers

One might suspect that the set of rational numbers $\mathbb{Q}$ is "larger" than the natural numbers because rational numbers are dense on the real line. However, they share the same cardinality.

Theorem 4.3.8. *The set $\mathbb{Q}$ of rational numbers is countable.*

Proof. We denote $\mathbb{Q}^+$ as the set of positive rational numbers. Consider the array of positive rationals p/q where p is the row number and q is the column number. We can traverse this array diagonally (Cantor's Zigzag argument) to list them in a sequence: $\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{3}{1}, \frac{2}{2}, \frac{1}{3}, \dots$. By skipping fractions that are not in lowest terms (e.g., $\frac{2}{2} = 1$), we establish a bijection between $\mathbb{N}$ and $\mathbb{Q}^+$. Since $\mathbb{Q} = \mathbb{Q}^+ \cup \mathbb{Q}^- \cup \{0\}$, and the union of countable sets is countable, $\mathbb{Q}$ is countable. ■

4.3.4 Uncountable Sets and the Reals

Definition 4.3.9 (Uncountable Set). A set S is said to be uncountable if it is infinite and not countable. That is, there exists no bijection between S and $\mathbb{N}$.

Theorem 4.3.10 (Cantor's Theorem / Uncountability of $\mathbb{R}$). *The set of real numbers $\mathbb{R}$ is uncountable.*

Proof. It suffices to show that the open interval $I = (0, 1)$ is uncountable. We use Cantor's Diagonal Argument. Assume, for the sake of contradiction, that $(0, 1)$ is countable. Then its elements can be enumerated in a sequence $x_1, x_2, x_3, \dots$. We represent each x_i by its unique infinite decimal expansion (avoiding sequences ending in infinite 9s to ensure uniqueness):

$$\begin{aligned} x_1 &= 0.d_{11}d_{12}d_{13}\dots \\ x_2 &= 0.d_{21}d_{22}d_{23}\dots \\ x_3 &= 0.d_{31}d_{32}d_{33}\dots \\ &\vdots \end{aligned}$$

where $d_{ij} \in \{0, 1, \dots, 9\}$. We construct a number $y = 0.y_1y_2y_3 \dots \in (0, 1)$ such that y differs from x_n in the n -th decimal place. Define:

$$y_n = \begin{cases} 4 & \text{if } d_{nn} \neq 4 \\ 5 & \text{if } d_{nn} = 4 \end{cases}$$

Then $y_n \neq d_{nn}$, which implies $y \neq x_n$ for any $n \in \mathbb{N}$. Thus, y is a real number in $(0, 1)$ that is not contained in the enumeration. This contradicts the assumption that the set is countable. Therefore, $(0, 1)$ is uncountable, and consequently, $\mathbb{R}$ is uncountable. ■

Corollary 4.3.11. *The set of irrational numbers $\mathbb{P} = \mathbb{R} \setminus \mathbb{Q}$ is uncountable.*

Proof. We know that $\mathbb{R} = \mathbb{Q} \cup \mathbb{P}$. If $\mathbb{P}$ were countable, then $\mathbb{R}$ would be the union of two countable sets, making $\mathbb{R}$ countable. This contradicts Cantor's Theorem. Thus, $\mathbb{P}$ must be uncountable. ■

4.3.5 Cantor's Power Set Theorem

Theorem 4.3.12. *For any set A , the cardinality of the power set $\mathcal{P}(A)$ (the set of all subsets of A) is strictly greater than the cardinality of A .*

Remark 4.3.13. This implies that there is no "largest" infinity. We have a hierarchy of infinities: $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| < \dots$. The cardinality of $\mathbb{N}$ is denoted $\aleph_0$ (aleph-null), and the cardinality of $\mathbb{R}$ is denoted $\mathfrak{c}$ (continuum). It is a known result that $\mathfrak{c} = 2^{\aleph_0} = |\mathcal{P}(\mathbb{N})|$.

Example 4.3.14. Show that the set of all even natural numbers $E = \{2, 4, 6, \dots\}$ has the same cardinality as $\mathbb{N}$.

Solution. We define a function $f : \mathbb{N} \rightarrow E$ by $f(n) = 2n$.

- **Injectivity:** Let $f(n) = f(m)$. Then $2n = 2m$, which implies $n = m$.
- **Surjectivity:** Let $y \in E$. By definition of even numbers, $y = 2k$ for some $k \in \mathbb{N}$. Thus $f(k) = y$.

Since f is a bijection, $|E| = |\mathbb{N}|$. This illustrates Galileo's Paradox: a proper subset of an infinite set can be equipotent to the whole set. ■

Example 4.3.15. Prove that the interval $(0, 1)$ is equipotent to the interval $(0, \infty)$.

Solution. We seek a bijection $f : (0, 1) \rightarrow (0, \infty)$. Consider the function $f(x) = \frac{1}{x} - 1$. Let $x \in (0, 1)$. Then $\frac{1}{x} \in (1, \infty)$, so $\frac{1}{x} - 1 \in (0, \infty)$.

- **One-to-one:** If $\frac{1}{x_1} - 1 = \frac{1}{x_2} - 1$, then $\frac{1}{x_1} = \frac{1}{x_2} \implies x_1 = x_2$.
- **Onto:** Let $y \in (0, \infty)$. We solve $y = \frac{1}{x} - 1$ for x . $y + 1 = \frac{1}{x} \implies x = \frac{1}{y + 1}$. Since $y > 0$, $y + 1 > 1$, so $0 < \frac{1}{y + 1} < 1$. Thus $x \in (0, 1)$.

Therefore, $|(0, 1)| = |(0, \infty)| = \mathfrak{c}$. ■

Example 4.3.16. Let A be the set of all sequences of length finite length consisting of 0s and 1s. Is A countable?

Solution. Let A_n be the set of sequences of length n . For any fixed n , $|A_n| = 2^n$, which is finite. The set of all such finite sequences is $A = \bigcup_{n=1}^{\infty} A_n$. This is a countable union of finite sets. By the theorem regarding the union of countable sets, A is countable. ■

Example 4.3.17. Show that the set of algebraic numbers is countable. (An algebraic number is a root of a non-zero polynomial with integer coefficients).

Solution. Let P_n be the set of polynomials of degree n with integer coefficients. Since coefficients come from $\mathbb{Z} \times \cdots \times \mathbb{Z}$ ($n+1$ times), and $\mathbb{Z}$ is countable, the set of such polynomials is countable. A polynomial of degree n has at most n roots. Let S_n be the set of roots of polynomials in P_n . Since P_n is countable and each polynomial contributes a finite number of roots, S_n is countable. The set of all algebraic numbers is $\bigcup_{n=1}^{\infty} S_n$. Being a countable union of countable sets, the set of algebraic numbers is countable. ■

Example 4.3.18. Deduce the existence of transcendental numbers.

Solution. Let $\mathbb{A}$ be the set of algebraic numbers and $\mathbb{T}$ be the set of transcendental numbers. We have $\mathbb{R} = \mathbb{A} \cup \mathbb{T}$. From the previous example, $\mathbb{A}$ is countable. From Cantor's Theorem, $\mathbb{R}$ is uncountable. If $\mathbb{T}$ were countable (or empty), then $\mathbb{R}$ would be the union of two countable sets, which implies $\mathbb{R}$ is countable. This is a contradiction. Therefore, $\mathbb{T}$ must be uncountable. In fact, almost all real numbers are transcendental. ■

Example 4.3.19. Prove that the interval $[0, 1]$ has the same cardinality as $(0, 1)$.

Solution. We construct a bijection $f : [0, 1] \rightarrow (0, 1)$. We need to "absorb" the endpoints 0 and 1 into the infinite set. Select a countable sequence inside $(0, 1)$, say $S = \{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$. Define the mapping f as follows:

$$f(x) = \begin{cases} \frac{1}{2} & \text{if } x = 0 \\ \frac{1}{3} & \text{if } x = 1 \\ \frac{1}{n+2} & \text{if } x = \frac{1}{n} \in S \quad (n \geq 2) \\ x & \text{if } x \in (0, 1) \setminus S \end{cases}$$

This map shifts the sequence S and places 0 and 1 at the start of the sequence image. This is a bijection, proving $|[0, 1]| = |(0, 1)| = \mathfrak{c}$. ■

Exercise 4.3.20. 1. Find the interior of the set $S = [0, 5) \cup \{6\}$.

2. Determine the interior of the set of integers $\mathbb{Z}$.

3. Can a finite non-empty set $S \subset \mathbb{R}$ be a neighbourhood of any of its points? Justify.

4. Let $S = \bigcup_{n \in \mathbb{Z}} (n, n+1)$. Is S an open set?

5. Consider the sets $A_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$ for $n \in \mathbb{N}$. Is the intersection $\bigcap_{n=1}^{\infty} A_n$ an open set?

6. Find the derived set (set of limit points) of the interval $S = (3, 7)$.

7. Find the derived set of $S = \left\{1 + \frac{(-1)^n}{n} : n \in \mathbb{N}\right\}$.

8. Determine the limit points of the set $\mathbb{N}$.

9. Does the Bolzano-Weierstrass Theorem apply to the set $S = \{\sin(n) : n \in \mathbb{N}\}$? Explain briefly.

10. Why does the set $S = \{n^2 : n \in \mathbb{N}\}$ fail to have a limit point despite being infinite?

11. Find the closure of the set $S = (0, \sqrt{2}) \cap \mathbb{Q}$.

12. Is the set $S = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup \{0\}$ closed?

13. Determine the closure of the set of irrational numbers, $\bar{\mathbb{P}}$, where $\mathbb{P} = \mathbb{R} \setminus \mathbb{Q}$.

14. Find the smallest closed set containing the open interval $(2, 3)$.

15. Is the set of Prime Numbers countable or uncountable?
16. Is the Cartesian product $\mathbb{Q} \times \mathbb{Q}$ countable?
17. What is the cardinality of the set of all infinite sequences formed by digits 0 and 1?
18. Let A be the set of all circles in the plane centered at rational coordinates with rational radii. Is A countable?
19. Is the set $[0, 1] \setminus \mathbb{Q}$ countable or uncountable?
20. Compare the cardinality of the power set of naturals, $\mathcal{P}(\mathbb{N})$, and the set of real numbers $\mathbb{R}$.

Answers

1. $S^\circ = (0, 5)$. (The point 6 is isolated and 0 is a boundary point not included in the interior).
2. $\mathbb{Z}^\circ = \emptyset$.
3. No. Any neighbourhood must contain an open interval, which contains infinitely many points. A finite set cannot contain an open interval.
4. Yes. It is a union of open intervals, and the union of any family of open sets is open.
5. No. The intersection is $\{0\}$, which is a finite set (closed) and not open.
6. $S' = [3, 7]$.
7. $S' = \{1\}$.
8. $\emptyset$. (The set is unbounded and elements are spaced by at least 1).
9. Yes. The set is infinite and bounded (since $-1 \leq \sin(n) \leq 1$), so it must have at least one limit point.
10. The set is unbounded. BWT requires the set to be bounded.
11. $\bar{S} = [0, \sqrt{2}]$. (Rationals are dense).
12. Yes. The limit point of the sequence is 0, which is included in the set.
13. $\bar{\mathbb{P}} = \mathbb{R}$. (Irrationals are dense in $\mathbb{R}$).
14. $[2, 3]$.
15. Countable. (It is an infinite subset of the countable set $\mathbb{N}$).
16. Countable. (Finite Cartesian product of countable sets is countable).
17. Uncountable. (Has cardinality $2^{\aleph_0} = \mathfrak{c}$).
18. Countable. (Corresponds to $\mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}$, which is countable).
19. Uncountable. (Removing a countable set $\mathbb{Q}$ from an uncountable set $[0, 1]$ leaves an uncountable set).
20. They are equal. $|\mathcal{P}(\mathbb{N})| = |\mathbb{R}| = \mathfrak{c}$.

Concluding Remarks: A Note to the Student

Congratulations on completing the course **MMT422J1: Real Analysis-I**.

Real Analysis is often described as the soul of rigorous mathematics. Over the course of this semester, you have journeyed far beyond the mechanical computations of calculus into the logical foundations that hold the number system together.

Reflecting on the Journey

You began in **Unit I** by questioning the very nature of numbers, understanding the completeness of $\mathbb{R}$ and the gaps in $\mathbb{Q}$. In **Unit II** and **Unit III**, you mastered the delicate $\epsilon - \delta$ arguments, taming the behavior of infinite sequences and series. Finally, in **Unit IV**, you stepped into the abstract world of topology and set theory, learning that some infinities are "larger" than others—a concept that challenged the greatest minds of the 19th century.

The Value of this Course

Completing this module proves that you possess not just mathematical knowledge, but mathematical *maturity*. You have learned that:

- **Precision is power:** A definition is not just text; it is a tool.
- **Intuition requires proof:** What seems obvious (like the Intermediate Value Theorem) requires rigorous justification.
- **The finite informs the infinite:** We understand the continuum by analyzing finite approximations.

Looking Ahead

This course was not a destination, but a gateway. The tools you have sharpened here—supremums, limit points, and convergence tests—are the prerequisites for advanced landscapes such as *Metric Spaces*, *Complex Analysis*, *Measure Theory*, and *Functional Analysis*.

As you move forward to future semesters, remember the discipline you developed here. The rigor of Real Analysis is the bedrock upon which modern mathematics stands. Carry this clarity of thought with you, and continue to explore the profound beauty of the mathematical universe.

"Mathematics is the most beautiful and most powerful creation of the human spirit."

— Stefan Banach

Best wishes for your future studies.